Engineering Applications of Artificial Intelligence 156 (2025) 111034

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Promoting sustainable urban mobility: An integrated fuzzy decision-making model for assessing autonomous bus alternatives

Ömer Faruk Görçün [a], Mehmet Özçalıcı [b], Hasan Emin Gurler [b], Dragan Pamucar [c,d,e], Vladimir Simic [f,g,*]

[a] Department of Business Administration, Faculty of Economics, Administrative and Social Sciences at Kadir Has University, Cibali Av. Kadir Has St. Fatih, Istanbul, 34083, Turkey
[b] Department of International Trade and Logistics, Faculty of Economics and Administrative Sciences, Kilis 7 Aralık University, Kilis, Turkey
[c] Department of Operations Research and Statistics, Faculty of Organizational Sciences, University of Belgrade, Belgrade, Serbia
[d] Department of Industrial Engineering & Management, Yuan Ze University, Taoyuan City, 320315, Taiwan
[e] İstinye University, Faculty of Engineering and Natural Sciences, Industrial Engineering Department, Sarıyer, 34396, İstanbul, Türkiye
[f] University of Belgrade, Faculty of Transport and Traffic Engineering, Vojvode Stepe 305, 11010, Belgrade, Serbia
[g] Department of Computer Science and Engineering, College of Informatics, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea

ARTICLE INFO

Keywords:
Urban transportation
Autonomous buses
Sustainability
Fuzzy Dombi Bonferroni method
Fuzzy logarithm methodology of additive weights

ABSTRACT

Today, in addition to the increasing pressures on urban transportation authorities to achieve sustainability goals, it has become necessary to comprehensively evaluate innovative transportation technologies such as autonomous (driverless) buses due to the increasing demand for public transportation applications that will contribute to making urban transportation more sustainable with its environmental, social, economic and social dimensions. In addition, the reservations and hesitations of decision-makers about integrating autonomous buses into urban transportation systems have not been eliminated. These hesitations and reservations are mainly due to critical research, theoretical gaps, and limitations in practice. Considering these gaps, this study presents an innovative model that integrates the fuzzy logarithm methodology of additive weights (LMAW) method and the fuzzy Dombi Bonferroni (DOBI) method to evaluate and rank 20 different autonomous bus alternatives with 33 sustainability criteria. The proposed integrated decision-making procedure can effectively manage complex uncertainties while examining whether autonomous bus alternatives can be integrated into urban transportation systems based on sustainability, considering four-dimensional sustainability criteria. This finding indicates that urban transportation's user-oriented and reliable nature is critical to achieving sustainability goals. In addition, the Proterra Catalyst (A9) is the autonomous bus with the highest sustainability performance for use in urban transport, followed by the Mercedes-Benz Future Bus (A18) and Mercedes-Benz eCitaro (A8). These results regarding alternatives underline the importance of advances in autonomous vehicle technology and making these vehicles more sustainable in evaluation processes.

1. Introduction

Today, especially in big cities, non-autonomous driver-based rubber-wheeled buses are still the dominant urban transportation system in many world cities and have the highest share compared to other modes of transportation. Although this share varies depending on the rail system infrastructures of countries and cities, 30–40 % of transportation is provided by buses in cities with a developed and widespread rail transportation system, while this rate can exceed 50 % in developing countries (Correa et al., 2017).

Some of the advantages of these urban transportation systems explain why bus-based urban transportation is still the most dominant type. First, buses are a highly flexible type of transport and can use any road in the city as a route (Kuo et al., 2023; Basuri et al., 2025). In addition, existing routes can be quickly and easily modified according to changing requirements, while adapting to new residential areas is

* Corresponding author. Department of Computer Science and Engineering, College of Informatics, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul, 02841, Republic of Korea.
E-mail addresses: omer.gorcun@khas.edu.tr (Ö.F. Görçün), mozcalici@kilis.edu.tr (M. Özçalıcı), hasan.gurler@kilis.edu.tr (H.E. Gurler), vsima@korea.ac.kr (V. Simic).

https://doi.org/10.1016/j.engappai.2025.111034
Received 31 January 2025; Received in revised form 24 March 2025; Accepted 28 April 2025
Available online 21 May 2025
0952-1976/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

extremely easy (Tütüncü et al., 2023).

In addition, bus-based transportation systems have extremely reasonable installation costs compared to rail transportation systems such as subways, trams, etc. Low installation costs make choosing and adopting a bus transportation system in many developing countries easier. Urban public transportation buses are a more accessible transportation system than other alternative modes of transportation. Unlike rail transportation systems, buses can be accessed from almost any point in the city (Cui et al., 2023). At the same time, many urban transportation alternatives besides buses need infrastructure construction, which can take a long time. Buses can be planned and commissioned quickly, overcoming this challenge. Thanks to its flexibility, the buses used according to the requirements can be changed, and buses that meet every requirement can be integrated into the system. For example, on lines with a high number of passengers, buses with a higher passenger capacity and double-decker ones can be added, or in cases where the traffic volume is high, and the number of passengers carried by bus is low, smaller sized and capacity buses can be preferred.

In addition, urban transport by bus provides critical flexibility advantages in terms of operation (Moslem et al., 2023; Imran and Ullah, 2025). Accordingly, trips can be reduced or increased depending on traffic and other conditions. However, with the development of technology in recent years, electric and hybrid buses have started to be used in a limited framework and at the starting point. Accordingly, new bus technologies have proven promising in reducing carbon emissions and sustainability. Finally, bus systems for the city with drivers can be easily integrated with all other urban transportation systems, and multimodal transportation systems can be configured in cities.

Despite these advantages, driver-driven urban bus transportation is not a promising transportation system for a sustainable future for rapidly growing cities. One of the main reasons for this is the unresolved limitations and structural problems related to driver-driven buses. Urban bus systems pose critical issues and some risks related to safety. These structural problems range from traffic accidents to environmental impacts, operational inefficiencies, and external costs. Accordingly, the fact that autonomous buses have the potential to reduce or eliminate accidents caused by conventional buses with drivers shows that driverless vehicle technology can be a serious and critical alternative to conventional driver-driven buses in urban transportation from a safety, security, and efficiency perspective (Deveci et al., 2021).

However, despite promising advances in autonomous vehicle technology, many cities worldwide still need to use wheel-driven buses as their primary mode of transportation (Younis Al-Zibaree and Konur, 2023). When the critical problems caused by this situation are considered, it is seen that the most vital issues are directly related to security risks. In addition, the problems and risks that arise negatively affect various perspectives, such as traffic accidents, environmental pollution, and operational inefficiency.

One of the most critical problems of driver-driven urban buses is traffic accidents caused by drivers due to distraction, driving errors, violations, etc. While the number of collision-type accidents caused by driver error that occurred recently on the Bus Rapid Transit (BRT) line used only in the city of Istanbul was determined as 17, the number of accidents such as buses used in the BRT system hitting the passenger for the same reason was defined as 15 (Akbulut et al., 2017). Rubber-tired buses had 2936 accidents last year, resulting in two deaths and 287 injuries (I.E.T.T., 2024). As a result, driver-driven buses used in urban transportation pose significant risks to passengers and other stakeholders. Contributing to the increase in traffic density is one of the problems caused by such urban transportation systems. Buses adversely affect the traffic flow by moving extraordinarily slowly at traffic lights and entering and exiting stops in urban transportation. The effect of this problem increases, especially during peak hours.

While the risks that occur while driving due to human factors endanger the safety of passengers, they significantly reduce user satisfaction. In addition, the fact that fossil-based fuels are used in almost all buses is mainly responsible for increased environmental pollution and emissions. While Połom and Wiśniewski (2021) claim that buses used in urban transport are responsible for 16.9 % of the total carbon emissions of cities, Nanaki et al. (2017) suggest that this rate reaches 40 % in many major European cities.

Autonomous buses deserve further discussion and evaluation, as they represent an essential opportunity to reduce the problems and risks caused by driver-driven buses used in urban transportation and to create a more reliable transportation environment. Considering these tools' advantages, one of the most essential advantages is their ability to minimize human errors. Especially considering that most accidents involving buses with conventional drivers are caused by human mistakes, decision-making processes based on precise algorithms are created in a very subtle way, and a driving system with continuous learning capability can eliminate such problems and make driving perfect.

Although the ability of autonomous buses to adapt to urban traffic conditions, which are extraordinarily complex and affected by numerous variables, is still limited, autonomous buses equipped with advanced detection systems can react instantly to the risks that occur through continuous and real-time monitoring. They have remarkable potential to prevent possible accidents. In this context, driverless buses can increase the safety of other vehicles depending on their technological advantages in mixed traffic, where different types of vehicles travel. They can improve the quality of urban transportation services produced in metropolitan areas with high population and user density (Deveci et al., 2022b).

Autonomous vehicles have the potential not only to reduce accidents but also to lower carbon emissions in urban transportation. These vehicles can plan routes in real time to alleviate the adverse effects of traffic congestion and significantly optimize fuel consumption (Tanveer et al., 2020), thereby offering potential economic benefits. Additionally, many autonomous buses utilize alternative renewable energy sources, such as hydrogen fuel cells or electricity, resulting in already low emissions. In this context, Fidanoglu et al. (2023) highlighted that autonomous public vehicles can significantly enhance service quality, especially in last-mile mobility, and improve energy efficiency. Moreover, driverless vehicle driving and managing algorithms enable more firm driving alternatives, which can help reduce final energy usage and other negative externalities by maintaining a consistent speed under normal conditions.

Additionally, it eliminates safety risks due to standardized autonomous buses and instantaneous decision-making capabilities. Managing autonomous buses used in urban transportation from a single centre and using intelligent transportation systems can significantly increase traffic safety and improve efficiency and operational efficiency (Deveci et al., 2022b). In this context, sensors, improved algorithms, and other autonomous vehicle technologies adopt a more controlled and different driving style, enabling better management of the risks encountered in urban transportation. Furthermore, autonomous buses, which can operate at more controllable and stable speeds in urban traffic, play a crucial role in reducing the severity and frequency of possible accidents, paving the way for a safer urban transportation system.

However, in addition to urban transport authorities, users must be sufficiently convinced about using and disseminating autonomous (driverless) buses in urban transportation. The reasons why autonomous buses have yet to be adopted and widespread are multidimensional, including problems such as technological, economic, social, and legal regulations and the autonomous bus infrastructure needed. Technological advancement is a crucial factor in the successful implementation of autonomous buses. Autonomous vehicle technology must still be advanced enough to respond to highly variable and complex conditions. On the other hand, while the integration of autonomous vehicles into real-time traffic management systems facilitates the adaptation of these vehicles to urban transportation systems (Gokasar et al., 2024), it has the potential to significantly increase traffic safety and traffic

management efficiency in the city by reducing accidents that occur primarily because of driver errors (Chougule et al., 2024).

Therefore, for adopting autonomous buses in urban transportation, it is necessary to focus on models in which sensing systems, artificial intelligence deep learning, and data processing systems are further developed. Another important problem that increases decision-makers' hesitation in urban transportation is the purchase, maintenance, and operating costs of these vehicles and other required expenses. In addition, how urban transport authorities can transform their existing vehicle fleet and the additional budget pressure caused by the high infrastructure investments required for autonomous buses force decision-makers to be more hesitant.

The lack of social acceptance of driverless autonomous vehicles in many cities is one of the critical barriers to their adoption (Nastjuk et al., 2020). Increasing awareness and education is a critical priority for policymakers to improve public acceptance of autonomous vehicles. In addition to end-users' concerns about the safety provided by autonomous buses, the resistance of city residents to technological developments slows down the integration of autonomous vehicles into urban transportation systems and makes it challenging to adopt. At the same time, legal regulations regarding the use of autonomous buses in urban transportation have yet to be created. The transportation infrastructure must have technological elements such as intelligent transportation systems and 5G networks to use autonomous (driverless) buses in urban transportation. The fact that the existing infrastructure needs to be improved is among the obstacles to this transformation.

Another reason autonomous buses have not been sufficiently adopted in urban transportation is that decision-makers need to get more support from the literature, and the literature needs to develop on this subject sufficiently. In this context, the findings of a preliminary study on the use of autonomous buses in urban transportation show critical research gaps in the relevant literature. First of all, studies focusing on the selection of autonomous vehicles in the literature are based on the usability of shared autonomous vehicles in urban transportation (Fidanoglu et al., 2023), operating models for autonomous buses (Deveci et al., 2022a), the use of driverless vehicles in the metaverse environment (Deveci et al., 2022b), traffic management and optimization for autonomous vehicles (Gokasar et al., 2024) and autonomous in the logistics industry they focused on topics such as the use of vehicles, simulating the use of autonomous vehicles (Dwipayana et al., 2024). In addition, in some studies, issues such as social acceptance and passenger experiences of driverless buses (Ariza-Álvarez et al., 2023) and the environmental impact of autonomous buses (Hasan et al., 2022) were discussed. On the other hand, a study to determine the most suitable autonomous buses that can be used in urban transportation with social, economic, environmental, technical, and operational sustainability dimensions is not available in the relevant literature to the authors' knowledge.

In addition, since the issues discussed in the relevant literature are not directly related to the focus of this study, it may not be possible to generalize the findings and inferences obtained in terms of the decision problem discussed in this study. Likewise, whether the methodologies proposed in these studies will effectively solve the decision problem addressed in this study is still being determined. At the same time, the approaches proposed in previous studies need more comprehensibility and an easy-to-follow simple basic algorithm to deal with the decision problem and produce reasonable and rational solutions. However, there is a significant disconnect between research and practice, as decision-makers in the urban transportation industry have not adequately adopted these approaches in real-life conditions. Ultimately, most previous studies have not focused on a decision-making problem encountered in real-world conditions but instead have tried to simulate a possible decision-making problem.

In this context, this current investigation seeks to answer some of the research questions that have arisen, focusing on the novelty and potential impact of the study. The identified research problems can be defined as follows: (RQ1) What are the critical factors and criteria affecting the sequencing performance of driverless buses to be evaluated for integration into urban transportation systems? This question aims to reveal the relative importance of the sustainability criteria and their interactions in a novel approach that could significantly impact urban transportation planning. (RQ2) What decision-making model should be developed to overcome existing research and theoretical gaps in the literature and practice? This question seeks to obtain logical and rational answers related to the determination of the structure of the decision-making model that needs to be developed to address the decision-making problem and to produce logical solutions, introducing a novel model that could revolutionize decision-making in this field. (RQ3). How can the sustainability performance of self-driving autonomous vehicles be improved to be more effectively adapted to urban transportation systems? The answers to this question aim to strengthen the application aspect of the research and its relevance to real-life problems, introducing novel solutions for improving sustainability in urban transportation. (RQ4) How do the interactions and relationships between autonomous buses' technical, social, economic, environmental, and operational sustainability performances affect evaluation? The main objective of this question is to identify how decision-makers can balance different sustainability criteria in an actual decision-making process, introducing a novel perspective on sustainable decision-making.

In this study, the authors aim to examine the autonomous bus alternatives that can be used in urban transportation. The study conducts a comprehensive and sensitive evaluation, considering multidimensional sustainability, and addresses research gaps and theoretical evaluation inadequacies. The current work aims to rank these driverless buses based on their sustainability performance, identifying their potential for public transportation. This study represents a critical innovation in the relevant industry and significantly contributes to the literature. It provides a practical roadmap that can guide decision-makers in urban transportation authorities, evaluating and selecting the most suitable autonomous buses from a sustainability perspective in real-life conditions.

At the same time, this study meticulously considers multidimensional sustainability criteria. It identifies the most effective and critical ones and thoroughly discusses their impact on the evaluation processes and choices of suitable autonomous buses for urban transportation. The study also reveals the intricate interaction and relationships between the social, economic, environmental, and technical sustainability criteria included in the evaluation process.

Moreover, Fuzzy Aczel-Alsina presents a practical decision-making model that integrates LMAW and DOBI methods, effectively addressing the decision problem under consideration. This model, known for its consistency, stability, and robustness, offers a flexible framework for decision-making. It can evaluate linear and nonlinear interactions and relationships between the criteria, and its basic algorithm can be easily understood and applied by decision-makers without advanced mathematical knowledge. Due to fewer calculations and implementation steps, this model provides a practical and reliable evaluation tool for decision-makers.

The integration of fuzzy LMAW and fuzzy DOBI methods presents an innovative advancement over existing decision-making models, particularly in evaluating autonomous bus alternatives, due to several key factors. The integration of fuzzy LMAW precisely captures ambiguity by assigning criteria weights through expert-derived fuzzy numbers, while the fuzzy DOBI method excels in ranking alternatives through advanced fuzzy aggregation operators. The resulting framework thus robustly addresses real-world ambiguity. The combined use of fuzzy LMAW and fuzzy DOBI forms a sequential analytical process: LMAW accurately determines the weights of evaluation criteria, and DOBI subsequently uses these weights for nuanced, interactive alternative rankings. This stepwise synergy ensures a more precise and systematic evaluation of autonomous bus alternatives.

Following a comprehensive introduction, the rest of the investigation

is structured as follows. Section 2 demonstrates the conclusions of a comprehensive and detailed literature investigation, and the contributions of earlier studies and the research and theoretical gaps that still need to be filled are discussed extensively. Section 3 illustrates the proposed integrated decision-making model's basic algorithm, implementation steps, and mathematical applications. Section 4 presents the findings obtained by applying the developed decision-making tool to evaluate autonomous bus alternatives for urban transportation. In Section 5, a comprehensive robustness check, designed to test the validity and robustness of the model thoroughly, has been carried out, providing reassurance about its reliability. Section 6 summarizes and discusses the results obtained, while Section 7 concludes the study and shows the limitations of the study and research guidelines for future studies.

2. Research background

This chapter provides a detailed overview of the results from a comprehensive literature review. During this process, we carefully examined prominent citation databases. We collected previous studies using keywords such as autonomous vehicles, driverless buses, urban transportation, public transportation, urban passenger transportation, multi-criteria decision-making, and their combinations. This thorough examination allowed us to identify the key themes of these studies, the methodological frameworks used, the findings reported, and the criteria and alternatives considered in the research.

Considering this information, the researchers evaluated the current trends in the literature, the developments in the process, and the authors' findings in previous studies that they had in common and diverged. Autonomous vehicle technologies have not only attracted more and more attention from the research community members and urban transportation authorities in recent years but have also become a growing and important field due to their potential to improve the efficiency and effectiveness of urban transportation and make technical, environmental, social, and economic critical contributions to sustainability.

In the studies carried out from the early period of the relevant literature to the present, researchers have discussed autonomous vehicle technologies from different perspectives: driverless autonomous vehicles, social acceptance of these vehicles, developments in autonomous vehicle technology, intelligent transportation systems required by autonomous vehicles, remote sensing, and communication technologies have been examined in a wide range of infrastructure requirements. Owczarzak and Zak (2015) examined driverless vehicles' advantages by comparing them with conventional urban transportation systems by applying the ELECTREE III method in light of criteria such as average travel time, transportation cost, and reliability. The authors revealed that autonomous vehicles have a lower environmental impact and are a more accessible transportation system. They underlined that the most critical obstacles to the development of driverless vehicle technology are the high technological infrastructure costs required by these vehicles and the lack of inclusive legal regulations, yet it is an urgent issue that needs to be addressed.

In the experimental study of Papadima et al. (2020), the authors developed and applied a Conjoint Analysis model to evaluate driverless buses' social acceptance levels. While the authors emphasize that smart cities are more prone to adopt autonomous vehicle technologies and applications, they underline that the security concerns of individuals living in these cities regarding autonomous vehicles are extremely low. According to the authors, ticket prices are the most critical factor in adopting autonomous buses. In addition, increasing the frequency of service of autonomous buses and improving user experiences may enable faster and easier adoption of autonomous vehicle technologies in urban transportation.

Yu et al. (2017) developed a deep Q-learning-based approach to reduce possible accidents by developing more effective decisions while driving autonomous buses. Their work, which aimed to put the overtaking decisions of these vehicles on a more rational basis, has led to significant safety improvements. By combining Deep Neural Networks (DNN) and Q-learning methodology, they designed a model that can produce a comprehensive algorithm. The researchers ran these algorithms in actual traffic conditions by conducting experimental applications and noted that autonomous buses developed safer driving behaviour. This research not only proves the potential of the proposed model to facilitate the adaptation of autonomous buses to urban transportation systems but also provides reassurance about the safety of this driverless vehicle technology. Also, Badi et al. (2024, 2025) proposed innovative optimization techniques for minimizing transportation and inventory costs. The effectiveness of the proposed optimization model has been showed through two real-world case studies.

Salonen and Haavisto (2019) evaluated the social acceptance of driverless vehicles by examining passenger experiences of autonomous urban transport vehicles in the Finnish context using an Interbehavioral Theory (TIB) approach. The findings show that users' safety concerns about autonomous vehicles are decisive in adopting these vehicles in urban transportation. According to the authors, users' confidence in autonomous buses is similar to rail system vehicles. The authors emphasized that autonomous buses should be further developed in terms of safety, and user experiences should be increased to become widespread in urban transportation.

López-Lambas and Alonso (2019) examined the barriers associated with the adoption of autonomous buses in the context of urban transport and the psychological attitudes of users. The authors took the problem in the context of Spain and formed focus groups, and discussed the issue in these groups. The results show that increasing the adoption of driverless buses is necessary to add new technological equipment that will improve the safety perceptions of users, create awareness in society, and take into account socio-cultural differences when determining policies regarding autonomous buses.

Rosell and Allen (2020) conducted a comprehensive analysis using the three-stage SEM-MIMIC model to evaluate the factors affecting the preference for driverless buses and users' satisfaction. The main finding of the research is that users' perception of security has critical and significant effects on user satisfaction. In that regard, the investigators highlighted the need for further advancement of safety-related technologies for users' adoption of driverless buses, highlighting their research's impact on the transportation technology development field.

Huang et al. (2020) conducted a comprehensive review using the Adaptive Network-Based Fuzzy Inference System (ANFIS) to assess users' social acceptance of autonomous vehicles. The study findings indicate that users' ease of use and risk perceptions are the factors that most affect the adoption of autonomous vehicles. They stressed the crucial need to inform users more about the contributions, benefits, and risks of autonomous vehicles, as this would significantly increase the acceptance of autonomous vehicles.

Using the planned behaviour theory framework, Launonen et al. (2021) delved into users' experiences with autonomous buses. The research aimed to understand passengers' attitudes, perceptions, and emotions using driverless buses in real-life conditions and the key factors influencing them. The study's most crucial finding is that users' most important expectation regarding autonomous vehicle technologies is security. This insight has significant practical implications, underscoring the need for manufacturers to improve and enhance perceived safety and reliability for the widespread adoption of autonomous vehicles.

Deveci et al. (2021) conducted a comprehensive study on the usability of autonomous vehicles in urban transportation and the effectiveness, efficiency, and benefits of real-time traffic management systems. They introduced the extended fuzzy CoCoSo method with the Power Heronian function and the logarithmic method, a robust approach for evaluating and ranking these systems. In their model application, the authors identified four main and 13 sub-criteria and thoroughly examined various strategies and practices related to

autonomous vehicles. The study's findings suggest that integrating autonomous vehicles into urban traffic management systems could have the most significant impact.

In their study tailored to Oslo, Mouratidis and Cobeña Serrano (2021) explored passenger experiences with driverless buses through questionnaires and bilateral interviews. They also compared users' opinions before and after they were informed about these vehicles. The study's results reveal a reassuringly high user perception of safety and a desire for increased vehicle speed and reduced sudden braking, indicating a positive user attitude towards these vehicles.

Korkmaz et al. (2021) delved into the social acceptance of autonomous public transportation systems in Istanbul and the factors influencing users' preference for these vehicles using structural equation modelling (SEM). Their findings underscore the critical role of performance, safety, and reliability in adopting autonomous public vehicles. Moreover, the authors' research suggests that autonomous urban transport vehicles have the potential to significantly enhance the performance and efficiency of public transport, offering a promising outlook for the future.

Wicki (2021) conducted an experimental study and evaluated the results of the panel survey conducted during the driverless bus trial in Schaffhausen, Switzerland, and examined how disruptive technological developments are perceived in society, especially autonomous vehicle technologies. The author reviews how experiences using autonomous vehicles affect concerns about risks such as safety, etc. The results reveal that while there is an acceptable acceptance of autonomous vehicles in society, users' autonomous bus experiences do not significantly increase preferences. The study emphasizes that there may be changes in individuals' perceptions of autonomous vehicles, especially accidents related to autonomous vehicles, significantly affecting these perceptions. It highlights the evolving nature of technology and the need for ongoing research and adaptation.

Deveci et al. (2022a) used an integrated model consisting of the Sequential Priority Approach (OPA) method, which prioritizes tasks based on their importance and urgency, and fuzzy Dombi Bonferroni (DOBI) methods, which are used to handle uncertainty in decision-making, to guide policymakers regarding the adaptation of driverless buses to urban transportation systems. They discussed four different operational scenarios: private uses, last-mile transport, mixed traffic, the operation of autonomous buses in closed systems, and sustainability. They evaluated these options to determine the most appropriate one based on critical factors such as safety and efficiency. The study's main finding is that the mixed traffic option is the most suitable alternative because it can be integrated into urban transportation systems more easily and quickly compared to other options, reducing environmental impacts relatively more and improving traffic flows in urban areas.

Deveci et al. (2022b) examined the potential for improving urban transportation by integrating autonomous vehicle and metaverse technologies, a virtual reality space where users can interact with a computer-generated environment and other users, with a hybrid decision-making model integrating qROF clusters and extended OPA and RAFSI methods. In this context, the authors evaluated the potential of three alternative applications for autonomous vehicles and metaverse integration within the framework of fourteen criteria. They found that the alternative of managing autonomous vehicles within the framework of a central transportation system is the most appropriate option, depending on the potential to make urban traffic safer, further develop autonomous vehicle technology, and improve urban transportation from a sustainability perspective.

Hasan et al. (2022) conducted an experimental study in real-life conditions to compare driverless autonomous vehicle-based public transportation with conventional urban transportation systems and evaluate the environmental impact of urban transportation systems. They developed a micro-simulation-based model and examined four scenarios they created by considering fuel energy efficiency, fleet control, vehicle platooning, and infrequent vehicle transit times. Their findings underscore the potential of autonomous urban transportation types to significantly reduce environmental pollution and carbon emissions from urban transportation, offering a promising outlook for the future.

Biswas et al. (2022) delved into the enhancement of the lane-changing behaviour of autonomous vehicles, considering a comprehensive approach that includes artificial intelligence, cyber-physical systems (CPS), and Internet of Things (IoT)-based infrastructures. This approach, which considers the interconnected nature of these technologies, is crucial for further improving the safety of autonomous vehicles. The study evaluated how autonomous vehicles can develop attitudes similar to human behaviours in lane change situations, going beyond ordinary driving behaviours. The results highlighted the significant role of lateral control systems in increasing autonomous vehicles' safety while acknowledging data flow interruptions as a critical obstacle.

Li et al. (2023) conducted a comparative study of autonomous demand-driven public transportation types and conventional buses, focusing on user behaviours and preferences. The study aimed to understand users' autonomous public transportation preferences, the psychological motivations affecting these preferences, and how these findings can be integrated into policy and strategy development for urban transportation. The researchers collected data through questionnaires and bilateral user interviews and analyzed them using the Integrated Choice and Latent Variable model. The study's findings show that socio-demographic factors and travel-related criteria directly influence user preferences for autonomous public vehicles and have practical implications for urban transportation policy and strategy development.

Karami et al. (2023) studied users' attitudes towards fully autonomous vehicles, particularly in long-distance journeys. They investigated the impact of habits and satisfaction levels and compared the preferences for autonomous cars, aeroplanes, trains, and driverless buses. The study's findings revealed that safety is the primary concern for users in transportation, leading them to favour autonomous trains. Interestingly, the study also found that attitudes towards users play a more significant role in determining their transportation choices than their travel habits. The authors stressed the importance of policymakers developing tailored policies and strategies to various socio-cultural and demographic factors to promote the adoption of autonomous vehicles.

Fidanoglu et al. (2023) delved into the potential of integrating shared autonomous vehicles into urban transportation systems, particularly in addressing the last-mile problem of public transportation systems. Their investigation, which focused on the Bursa province, assessed the feasibility of this integration using real-time data. The authors' analysis of bus routes and the usability of autonomous vehicles on these routes using the dynamic vehicle routing problem method revealed a promising solution. Shared autonomous vehicles outperformed conventional buses in optimizing energy consumption and improving service quality, and they showed the potential to significantly reduce carbon emissions and energy consumption.

Rahnamay Bonab et al. (2023) conducted a comprehensive analysis using the Spherical fuzzy set (SFS)–based Choquet integral (CI) method. Their thorough evaluation and selection of autonomous vehicles for the logistics industry, which considered cost, efficiency, safety, and environmental impact, provides a solid foundation for future research and development in the field, reassuring the audience about the validity of the research.

Gokasar et al. (2023) evaluated how to reduce traffic accidents and the externality of urban transportation and improve traffic safety by increasing the effectiveness and performance of autonomous vehicles in mixed traffic in urban areas in real-life conditions. To do this, they used the type-2 neutrosophic number-based (T2NN) Entropy-COPRAS integrated decision-making model to multidimensionally compare the options for four different autonomous vehicle applications, such as autonomous vehicles that behave like human-driven vehicles, including

traditional vehicles, connected vehicles, autonomous vehicles using an algorithm for event management, and traditional event management methodology. In this context, the authors suggested that autonomous vehicles using algorithms for event management would be the most optimal choice as the critical finding of their study. They suggested that the advantages of driverless vehicles that behave like human-driven vehicles are pretty low compared to the other three alternatives. The authors' finding shows that autonomous vehicles need more advanced algorithms to make faster, more accurate, and rational decisions in a complex traffic environment.

Gokasar et al. (2023) examined how metaverse technology and advanced sensors can be integrated into autonomous vehicles with the help of the fuzzy full consistency method (fuzzy FUCOM) to improve traffic safety and efficiency. While integrating metaverse technology into autonomous vehicles offers significant benefits, such as enhanced sensing capabilities and a more comprehensive view of the surrounding environment, it also presents challenges. These challenges may include the complexity of the technology, the need for specialized skills to operate and maintain it, and potential privacy and security concerns. In this context, the authors identified and evaluated four alternatives to the proposed model. According to the results, using self-generating remote sensing systems (sensors) in metaverse technology to be integrated into autonomous vehicles has been determined as the most appropriate solution, and the importance of uninterrupted data collection with metaverse technology has been emphasized. The study provides critical and valuable insights for decision-makers considering integrating metaverse technology into autonomous vehicles.

Ariza-Álvarez et al. (2023) examined the social acceptance of autonomous buses used in passenger transportation in Spain for users and the experiences of users in an empirical study under real-life conditions. With the help of discrete choice analysis and logit modelling, the authors reviewed the relationships between users' characteristics and preferences. The study reveals that users with high environmental awareness and technology use experience can prefer autonomous buses more. At the same time, autonomous buses provide customer satisfaction at an acceptable level. In addition, they emphasized that socio-cultural and demographic factors significantly affect user preferences. In this context, the study underscores the importance of policymakers considering these factors when developing strategies, thereby involving the audience in decision-making.

Sitinjak et al. (2023) researched to evaluate the social acceptance of autonomous public vehicles in the Indonesian context. Their study, which utilized a questionnaire, focused on the contributions and advantages of these vehicles. The authors identified factors that affect users' attitudes and perceptions toward autonomous urban vehicles by analyzing demographic characteristics, concerns, and preference differences. The study's findings, which suggest that autonomous public vehicles could significantly improve the quality of urban transportation and eliminate existing problems, have important implications for future adoption as individual awareness of their benefits increases.

Lee et al. (2024) conducted extensive research with the help of a survey to identify and better understand how autonomous public transport vehicles are perceived by users in the context of driverless buses and the risks arising from these vehicles. The authors analyzed the user evaluations collected by questionnaires with a sequential probit model with three random parameters. They reported that users of urban transportation systems see autonomous vehicles as relatively more risky than conventional public transport vehicles. This perception could potentially impact the adoption rate of autonomous vehicles in urban areas. However, the authors found that socio-economic factors influence users' tendencies to use autonomous vehicles in urban transportation. Accordingly, users' security perceptions of autonomous vehicles and socio-economic factors are the factors that urban transportation authorities should consider when they produce policies regarding autonomous vehicles.

Gokasar et al. (2024) aimed to develop an approach that allows urban traffic to be managed in real-time using a small number of connected autonomous vehicles, significantly optimizing urban traffic and designing a system for rapid response in emergencies. The authors based the principles of the study on the tests performed in the SUMO (Simulation of Urban Mobility) simulation environment. The study's results show that using an appropriate number of autonomous vehicles increases the flow speed while reducing traffic density. It improves the current traffic conditions and offers a promising future for urban transportation. When disruptive situations such as traffic accidents occur, effective management of these extraordinary situations can further improve the traffic in the city. In addition, it emphasizes the potential of these vehicles' technological features, such as connectivity and autonomy, in the context of improving traffic conditions.

Dwipayana et al. (2024) aimed to produce a prototype of an autonomous bus that is suitable and responsive to the requirements by simulating the use of autonomous buses in tourism activities in Bali and reviewed how to reduce the externalities (such as traffic congestion, environmental pollution, accidents, etc.) that these vehicles will provide by using autonomous buses in the tourism industry. The authors designed this research as a research and development activity and revealed how an autonomous bus could be produced step by step. At the end of the study, they successfully managed to run the produced autonomous bus prototype, marking a significant milestone in the development of autonomous vehicles. The results reveal that using autonomous buses, especially in tourism regions, can significantly reduce the externality of transportation, including carbon emissions.

Zakeri et al. (2024) developed a hybrid procedure covering a machine learning (ML) algorithm, a random forest (RF) algorithm, and three innovative multi-criteria decision-making (MCDM) algorithms within the framework of a decision support system for the adaptation of autonomous vehicles to urban transportation systems. In this context, it examined four scenarios related to the integration of autonomous vehicles, addressed the obstacles and challenges, and analyzed all of them comprehensively. This thorough analysis provides a comprehensive understanding of the complexities of integrating autonomous vehicles, ensuring that decision-makers are well-informed.

Akram et al. (2024) undertook a pioneering analysis, extending the ELECTRE IV approach with global fuzzy sets (SFS). They aimed to clear mixed traffic for autonomous vehicles in Istanbul, identify transportation corridors, and tackle the challenges of integrating autonomous vehicles into urban transportation systems. Their work is particularly significant in identifying suitable corridors for driverless vehicles.

Zarbakhshnia and Ma (2024) identified critical success factors for integrating driverless autonomous vehicles into sustainable urban transportation systems. Their use of fuzzy DEMATEL and fuzzy Analytic Network Process (ANP) methods revealed that users' social acceptance, financial costs, and environmental impacts are the most influential criteria. This insight is crucial for understanding the adoption of autonomous vehicles, with the authors also highlighting the importance of the energy source in reducing their environmental impact.

Hasan et al. (2024) developed a multi-criteria decision support system to evaluate autonomous vehicles, an innovative transportation technology, in the context of sustainability and to optimize transportation systems. The methodology involved in this development included a comprehensive review of existing research, identifying key criteria for evaluating autonomous vehicles and developing a model that could effectively assess these vehicles' sustainability and optimization potential. In this context, researchers tried to evaluate the life cycle of projects for technologies such as autonomous vehicles. The authors also conducted an experimental study by testing this model on a five-lane highway project in Abu Dhabi. The research findings show that it creates significant life cycle effects for users in terms of time savings, minimization of energy consumption, and operating costs. In addition, factors such as the increase in traffic volume and the lack of sufficient space in the parking lots significantly affect the sustainability performance of the options. In addition, innovative applications such as

autonomous BRT have reduced lifecycle costs and carbon emissions by more than half.

Babaei et al. (2024) studied object detection architectures for autonomous vehicles using object detection algorithms. They ranked these algorithms according to their level of impact and importance using the practical implications of their findings, which highlight the critical role of algorithms with a higher degree of accuracy in dynamic and variable situations, which are significant for developing autonomous vehicles.

2.1. General trends and developments in the literature

It is seen that the literature related to autonomous vehicles generally focuses on the adaptation and integration of driverless public vehicles, especially buses, used in public transportation to urban transportation systems, the social acceptance and adoption of users regarding autonomous public vehicles, and technical issues for improving the safety and other capabilities of these vehicles.

In this context, when the studies related to autonomous vehicles are evaluated by considering the research problems they focus on and the methodological frameworks used, the studies in the relevant literature can be examined under three main headings. It can be defined as early studies from 2015 to 2018 when the first study on autonomous vehicles was conducted. In this period, studies have primarily focused on comparing driverless and conventional transportation systems, examining the potential of autonomous vehicles, their advantages, and their difficulties adapting them to transportation systems.

Methodologically, it is seen that the authors prefer different approaches such as ELECTREE III (Owczarzak and Zak, 2015), Conjoint Analysis (Papadima et al., 2020), Deep Neural Networks (DNN) and Q-learning methodology (Yu et al., 2017) within the scope of different evaluations. In this period, it has been found that the environmental impacts of autonomous vehicles, especially carbon emissions, are significantly lower than conventional vehicles, thereby promoting a more environmentally conscious approach to transportation. In addition, it has been argued that factors such as the extraordinarily high technological infrastructure costs required for autonomous vehicles and the inadequacy of legal regulations are the most critical obstacles to the widespread adoption of autonomous vehicles in this period. During this period, researchers mainly focused on factors such as the safety and costs of autonomous vehicles, while the number of studies on social acceptance of autonomous vehicles is minimal.

In the second period, which covers 2019 to 2021, researchers focused more on social acceptance and user experiences of autonomous vehicles. Critical to transportation and technology, their findings show that users' perceptions of safety and satisfaction are key to adopting driverless public vehicles. Many authors, especially Salonen and Haavisto (2019), emphasized that improving the security perception of passengers is extremely important in disseminating autonomous public transportation systems. López-Lambas and Alonso (2019), Mouratidis and Cobeña Serrano (2021) and Rosell and Allen (2020) tried to measure users' perceptions and satisfaction with driverless autonomous vehicles within the framework of different approaches, while Deveci et al. (2021) analyzed the usability of autonomous vehicles in urban transportation and real-time traffic conditions, as a study that presented a multi-criteria decision-making tool for the first time, unlike other authors in this period. It has been. In this period, while the authors examined autonomous vehicles from a sustainability perspective, many authors evaluated the effects of autonomous vehicles on reducing carbon emissions.

In recent studies (2022–2024), it is seen that researchers are increasingly turning to technical and specific research problems and topics. Accordingly, regarding autonomous vehicles being more reliable, advanced, equipped, and performing, the authors are investigating the potential of integrating autonomous vehicles with metaverse, 5G, IoT, and cyber security systems. In this context, Deveci et al. (2022b) emphasized that autonomous vehicles powered by metaverse technology can make traffic in the city more sustainable, safe, and efficient. In addition, micro simulations and the literature have examined how autonomous vehicles can be used in traffic for the city (Gokasar et al., 2024).

Researchers have increasingly used multi-criteria decision-making methods during this period, and they are particularly interested in innovatively using sophisticated fuzzy clusters to address uncertainties. The multi-criteria decision-making tools that have gained preference among researchers include OPA & DOBI (Deveci et al., 2022a), OPA & RAFSI (Deveci et al., 2022b), Choquet integral (CI) (Rahnamay Bonab et al., 2023), Entropy-COPRAS, (Gokasar et al., 2023), FUCOM (Gokasar et al., 2023b), DEMATEL & Analytic Network Process (ANP) (Zarbakhshnia and Ma, 2024), COPRAS & VIKOR (Babaei et al., 2024) and ELECTRE IV (Akram et al., 2024). Researchers have mainly used classical fuzzy sets (Deveci et al., 2022a), type-2 neutrosophic numbers (Gokasar et al., 2023), and spherical fuzzy sets (SFS) (Akram et al., 2024; Rahnamay Bonab et al., 2023) to address uncertainties.

At present, user-oriented design is a dynamic and compelling theme in the context of autonomous vehicles. Researchers have explored the psychological, socio-economic, cultural, and demographic factors shaping users' perceptions of autonomous vehicle use and preferences. In this context, Lee et al. (2024) demonstrated that passengers view driverless buses as more risky than traditional buses. They also underscored the dynamic nature of user attitudes and perceptions towards autonomous vehicle technology, which socio-economic and cultural factors can influence.

Another important research focus that has emerged recently is the sustainability of autonomous vehicles (Hasan et al., 2024). suggested that carbon emissions could be reduced by 50 % if autonomous vehicles could be integrated into urban transportation systems. This integration, supported by comprehensive life cycle analyses, has been a key area of research. The environmental sustainability performance of driverless vehicles has been discussed from a broad perspective, and the social acceptance of these vehicles and passengers' experiences with autonomous buses have come to the fore as determining factors in the successful adoption of autonomous vehicle technology.

When considered in a broader context, themes such as social acceptance of autonomous vehicles, sustainability, and adaptation of innovative technologies have consistently emerged from the early literature. Furthermore, the methodological frameworks used in the literature have become more sophisticated over time to address more complex problems. These developments indicate that autonomous vehicle technologies are maturing and should be handled with an interdisciplinary approach. They also suggest that we may be on the brink of significant transformations in driverless vehicles.

2.2. Consensus and disagreements in the literature

It is seen that the literature related to autonomous vehicles generally focuses on the adaptation and integration of driverless public vehicles, especially buses, used in public transportation to urban transportation systems, the social acceptance and adoption of users regarding autonomous public vehicles, and technical issues for improving the safety and other capabilities of these vehicles.

In the relevant literature, researchers share common views on autonomous vehicles. Most researchers agree that safety, as perceived by users, is a critical factor for expanding social acceptance of autonomous vehicles. They also concur on the increasing preference of passengers for driverless public transportation. Further development and advancement of autonomous vehicle technology and more comprehensive regulation by public authorities may serve this purpose. This regulation could influence the social acceptance of autonomous vehicles (Biswas et al., 2022).

Furthermore, the authors in the relevant literature widely agree that autonomous vehicles can contribute to reducing carbon emissions, alleviating vehicle traffic in urban areas, and making energy

consumption more rational. Notably, many authors argue that driverless public transport vehicles will become economically effective and efficient in the medium and long term, bringing long-term benefits that outweigh the high investment costs.

In many previous studies, researchers consistently argue that socio-cultural, economic, and demographic factors play a crucial role in adopting autonomous vehicles in urban transportation. For example, researchers such as Papadima et al. (2020) and Ariza-Álvarez et al. (2023) have shown that users who are more tech-savvy and live in more socio-economically developed regions have higher adoption rates for autonomous vehicles, shedding light on the intricate dynamics of urban transportation.

When the differences of opinion among researchers are examined, some researchers, such as Deveci et al. (2022a), suggest that autonomous vehicles should be integrated into urban transportation systems as quickly as possible. In contrast, some authors, such as Mouratidis and Cobeña Serrano (2021), stress the need for a more cautious and deliberate approach. They argue that the high costs and insufficient social acceptance of autonomous vehicles in urban transportation necessitate thorough planning and consideration.

Another issue that leads to disagreement among researchers is how user experiences affect perceptions of autonomous vehicles. In this regard, researchers reveal different findings. For example, Salonen and Haavisto (2019) suggested an increase in users' perceptions of safety after the autonomous public transportation experience. Wicki (2021) found that travel experiences had a limited impact on users' preferences. Ultimately, researchers have not determined whether autonomous vehicles are more suitable for private or shared mobility. Whether they can effectively solve the last-mile problem in urban transportation is controversial among researchers.

2.3. Research gaps and theoretical deficiencies

Despite the significant and valuable contributions of the studies in the relevant literature, critical research gaps and theoretical evaluation deficiencies still need to be filled. First of all, when the existing research gaps in the literature are considered, it is seen that the most critical research gap is related to the sustainability performance of autonomous vehicles. While the studies in the literature focused only on the environmental sustainability of autonomous vehicles and examined the effects of these vehicles on reducing carbon emissions and energy consumption, they neglected the economic, social, operational, and technical dimensions of sustainability. For example, regarding autonomous vehicles, Hasan et al. (2022) only addressed environmental factors, focusing on the energy efficiency of these vehicles and the reduction of polluting emissions while not considering other dimensions related to the sustainability of autonomous vehicles. The research gaps on this topic could significantly impact urban transportation, making it difficult to achieve sustainability goals and develop comprehensive strategies and policies in this context.

Another crucial research gap in the literature pertains to the methodological frameworks employed in previous studies. Due to their inherent structural issues and limitations, these approaches have not yet successfully addressed complex problems. For instance, Owczarzak and Zak (2015) utilized the classical version of the ELECTREE III method as a decision-making procedure. However, they could not capture the effects of interactions and relationships between criteria in the ranking results, highlighting the method's limitations. These similar limitations in the existing literature have hindered the resolution of decision problems related to selecting autonomous buses and generating appropriate solutions. It underscores the pressing need for further investigation and improvement in our methodological frameworks.

While societal acceptance and passenger experiences of autonomous vehicles are critical to adopting these vehicles, many researchers have examined this problem in the context of surveys, qualitative evaluations, and bilateral interviews. The study proposing a mathematical model or decision support system on this subject is not available in the literature in line with the authors' knowledge. However, although many authors have emphasized the role of the perception of security in adopting these tools, concrete suggestions have yet to be offered for improving these perceptions. Such gaps make it difficult to fully understand the barriers to adopting autonomous buses and greatly slow the development and progress of autonomous vehicle technology. It is crucial to address these barriers to ensure the smooth progression of autonomous vehicle technology.

Moreover, while some studies have touched upon the topic, there remains a significant research gap in the literature regarding the adaptation of autonomous buses to complex urban transportation systems and their integration into existing transportation infrastructures. For instance, Deveci et al. (2022b) explored integrating advanced technologies like the metaverse into autonomous vehicles. Still, the real-time implications of such integration in urban traffic and transportation environments were not adequately tested. This gap is crucial as it hinders a comprehensive understanding of the potential of driverless autonomous vehicles to function effectively in highly variable and dynamic urban transportation systems.

More importantly, previous studies measuring the performance of autonomous vehicles have compared autonomous public vehicles with conventional vehicles used in traditional transportation systems. Still, a study that deals with and comparatively evaluates existing autonomous bus alternatives to create an autonomous bus fleet planned to be integrated into the urban transportation system has yet to be available in the literature to the authors' knowledge. Therefore, no comprehensive model proposal focuses on directly comparing autonomous buses for integration into urban transportation systems.

This research gap may have been because, in previous studies in the literature, the authors tried to prove and convince stakeholders that autonomous vehicles may be used in urban transportation and have critical advantages over traditional transportation systems. This research gap makes it difficult for decision-makers to consider autonomous bus alternatives comparatively and to make rational and optimal decisions when building an autonomous bus fleet. Although the current studies are related to autonomous vehicles in one aspect, the different focal points of these studies cannot generalize the findings, inferences, and insights obtained to consider alternatives and determine the optimal solution for autonomous bus selection. This gap points to the lack of an integrated decision-making tool that addresses how autonomous buses to be used in driverless urban transportation will perform within sustainability's economic, social, environmental, technical, and operational dimensions.

In addition to research gaps, the literature has critical theoretical evaluation deficiencies. While researchers have developed effective models, there is significant potential for improvement. For instance, Biswas et al. (2022) examined the lane-changing behaviour of autonomous vehicles using decision-making tools. However, they could further enhance their model's performance in uncertain situations. Addressing these inadequacies can significantly improve decision-making processes' reliability, consistency, and stability, offering hope for future research.

One of the critical evaluation inadequacies in the literature is the lack of an integrated model that provides a balanced and comprehensive evaluation of all sustainability dimensions. While authors like Zarbakhshnia and Ma (2024) have tried to include various sustainability dimensions in their assessments, they have only minimally addressed the social and economic aspects, such as social acceptance and cost. This imbalance in evaluation makes it challenging to comprehend the performance of autonomous buses across different sustainability dimensions and to achieve a balanced evaluation process.

As a result, the existing research gaps and theoretical evaluation deficiencies in the literature reveal that a more comprehensive and interdisciplinary evaluation and calculation tool should be developed. This tool, when applied to highly complex and sophisticated

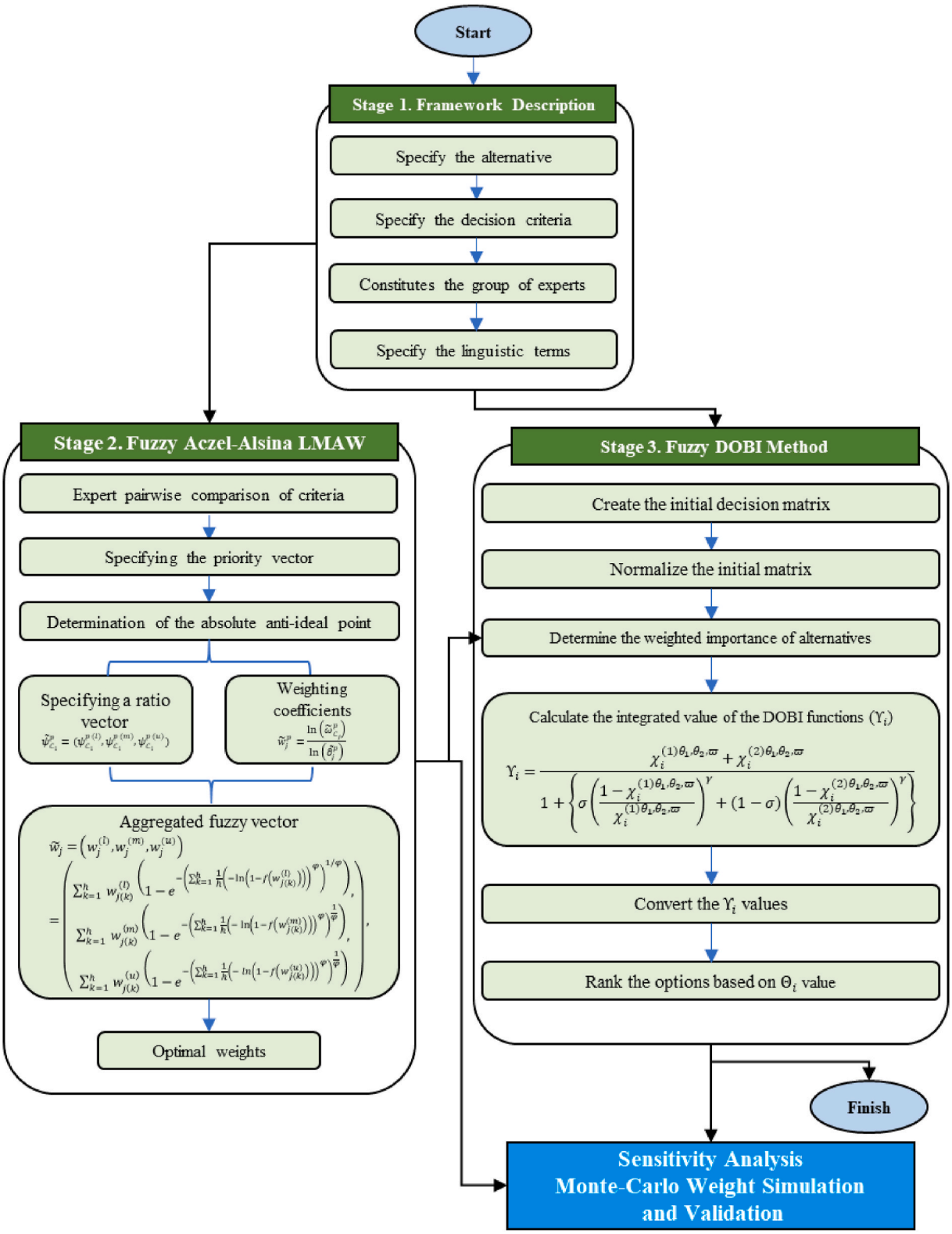


Fig. 1. Flowchart of the proposed study.

technologies, such as autonomous buses, for integration into urban transportation systems, has the potential to provide a reliable and consistent analysis. It will consider this comprehensive decision-making model's economic, social, technical, environmental, and operational sustainability dimensions, significantly impacting urban transportation and technology development.

3. Suggested model

This study aims to address the challenge of selecting sustainable autonomous vehicles. We employ a multi-criteria decision-making approach that integrates fuzzy logic techniques to achieve this. Specifically, we utilize the fuzzy Aczel-Alsina function-based Logarithmic

Table 1
Fuzzy linguistic terms and their corresponding fuzzy numbers for assessing criteria and alternatives.

Linguistic terms	Membership function
Extremely Low (EL)	(1.0, 1.0, 1.0)
Very low (VL)	(1.0, 1.5, 2.5)
Low (LL)	(1.5, 2.5, 3.5)
Medium-low (ML)	(2.5, 3.5, 4.5)
Medium (M)	(3.5, 4.5, 5.5)
Medium-high (MH)	(4.5, 5,5, 6.5)
High (H)	(5.5, 6.5, 7.5)
Very high (VH)	(6.5, 7.5, 8.5)
Very very high (VVH)	(7.5, 8.5, 9.5)
Extremely high (EH)	(8.5, 9.5, 10.0)

Methodology of Additive Weights (LMAW) to determine the criteria weights. This approach allows us to handle the inherent uncertainty and vagueness associated with the importance of various criteria (Fig. 1).

We adopt the Fuzzy Dombi Bonferroni (DOBI) Method for the evaluation phase, which provides a robust framework for aggregating the criteria and ranking the alternatives, reassuring the audience about the research quality. Combining these methodologies ensures a comprehensive and reliable selection process, facilitating the identification of the most sustainable autonomous vehicle option. The sensitivity analysis investigates the changes in criteria weights and alternative scores, while the Monte Carlo simulation ensures the accurate interpretation of the results. The study flowchart is presented in Fig. 1.

Integrating these two methodologies provides a robust analytical framework capable of effectively managing the inherent uncertainties and complexities in MCDM scenarios. The fuzzy LMAW technique initially determines reliable criteria weights, forming a sound basis for accurate alternative evaluation. Subsequently, the fuzzy DOBI method uses these established weights for meticulous alternative ranking, leveraging its advanced aggregation capacities to capture nuanced criteria interactions. This sequential integration ensures a comprehensive decision-making process, improving both the accuracy and reliability of outcomes.

The interactive use of fuzzy LMAW and fuzzy DOBI methods offers distinct advantages. The combination allows for a deeper integration of expert judgment and fuzzy logic, accurately reflecting real-world uncertainties. By first employing fuzzy LMAW, criteria weights are established precisely and transparently, improving the subsequent reliability of the DOBI rankings. The fuzzy DOBI method, capitalizing on fuzzy LMAW-derived weights, applies complex aggregation operators, capturing interaction effects among criteria, which significantly enhances decision accuracy. The structured, stepwise interaction between these methods improves the transparency of decision-making processes,

Table 2
Information about the decision makers.

DMs	Age	Experience (Years)	Expertise	Graduation Degree	Department	Country	Duties	Reputation
DM1	45	20	Transportation Engineering	Bachelor	Civil Engineering	USA	Urban mobility research, policy development	Leading voice in transit innovation
DM2	50	25	Autonomous Systems	Bachelor	Computer Science	Germany	AI implementation in transport systems	Renowned for AI applications in mobility
DM3	38	15	Environmental Science	Master's	Environmental Studies	UK	Sustainability assessment, impact analysis	Influential in green transportation
DM4	42	18	Urban Planning	Bachelor	Urban Studies	Brazil	City infrastructure planning	Highly respected urban planner
DM5	36	12	Electrical Engineering	Master's	Electrical Engineering	Turkey	Electric vehicle technology development	Innovator in electric mobility
DM6	47	22	Social Impact Assessment	Ph.D.	Sociology	Spain	Social implications of transportation projects	Expert in social equity assessments

Table 3
Information about the alternatives.

Brand		Model	Specifications	Advantages	Disadvantages
Navya	**A1**	Navya Arma	Capacity: 15–16 passengers	Fully electric, compact design	Limited range, not suitable for long trips
	A2	Navya Autonom Shuttle	Autonomous driving level: SAE Level 4	Ideal for urban environments	High initial cost, regulatory challenges
EasyMile	**A3**	EasyMile EZ10	Capacity: 12 passengers	Flexible routes, easy to deploy	Limited speed, short operational range
	A4	EasyMile EZ100	Range: Up to 100 km	Scalable solutions	Limited autonomy in complex environments
Volvo	**A5**	Volvo 7900 Electric	Length: 12–18 m	Environmentally friendly, robust design	Requires infrastructure investment
	A6	Volvo 9400	Autonomous pilot projects	High passenger capacity	Limited availability in some regions
Mercedes-Benz	**A7**	Citaro	Capacity: 35–100 passengers	Versatile for various routes	Expensive maintenance
	A8	eCitaro	Fully electric	Low emissions	High upfront costs
Proterra	**A9**	Proterra Catalyst	Range: 350 miles	High efficiency, modular design	Charging infrastructure needed
	A10	Proterra ZX5	Capacity: 40–60 passengers	Proven reliability	Still in testing phases for autonomy
Scania	**A11**	Scania Citywide	Length: 12–18 m	Adaptable for different environments	Complexity in integration
	A12	Scania Interlink	Multiple configurations	High passenger capacity	Regulatory hurdles in urban areas
BYD	**A13**	BYD K9	Range: 250–300 km	All-electric, long range	Limited to urban areas
	A14	BYD C6	Capacity: 30–40 passengers	Cost-effective for transit systems	Not fully autonomous yet
King Long	**A15**	King Long XMQ6127	Length: 12 m	Affordable option	Less brand recognition
	A16	King Long XMQ6119	Capacity: 50 passengers	Flexible configurations	Limited market presence
DAIMLER	**A17**	Daimler Future Bus	Level 4 automation	Cutting-edge technology	Conceptual, not fully deployed
	A18	Mercedes-Benz Future Bus	Various configurations	High-tech features	High cost and complexity
Iveco	**A19**	Iveco Urbanway	Length: 12–18 m	Suitable for urban routes	High initial investment
	A20	Iveco E-WAY	Fully electric	Environmentally friendly	Charging infrastructure required

Table 4
The criteria list of sustainable autonomous vehicles.

Main criteria	Sub-criteria	Description	Evaluation Factors	Types
Social Sustainability (MC_1)				
SO1	Accessibility	Ensures ease of use for all community members	ADA compliance, service coverage, user interfaces	Benefit
SO2	Public Perception	Reflects community attitudes towards autonomous vehicles	Surveys, public forums, media coverage	Benefit
SO3	User Satisfaction	Measures overall satisfaction of passengers using the service	Customer feedback, usage rates, service reliability	Benefit
SO4	Job Impact	Evaluates effects on local employment and job creation	Job creation vs. job displacement, local hiring policies	Benefit
SO5	Community Engagement	Involves the community in decision-making processes	Community meetings, stakeholder participation, feedback mechanisms	Benefit
SO6	Equity	Ensures fairness in service distribution across populations	Service availability in low-income areas, demographic inclusivity	Benefit
SO7	Safety Perception	Assesses public trust in the safety of autonomous systems	Incident rates, safety feature visibility, community safety initiatives	Benefit
Economic Sustainability (MC_2)				
EC1	Initial Investment	Upfront costs for purchasing and deploying vehicles	Purchase price, financing options	Cost
EC2	Operational Costs	Ongoing expenses for running the service	Fuel/energy costs, driverless operation costs	Cost
EC3	Maintenance Costs	Expenses related to vehicle upkeep	Average maintenance costs, frequency of repairs	Cost
EC4	Funding Availability	Access to financial support for investment	Grants, subsidies, partnerships	Benefit
EC5	Cost-Benefit Ratio	Comparison of costs to expected benefits	ROI calculations, net present value	Cost
EC6	Revenue Generation	Potential income from services offered	Projected fare revenue, advertising opportunities	Benefit
EC7	Life Cycle Cost	Total cost of ownership over the vehicle's lifespan	Total cost including purchase, operation, maintenance	Cost
EC8	Economic Impact	Effects on local economies and job creation	Contribution to local GDP, job creation metrics	Benefit
Environmental Sustainability (MC_3)				
EN1	Emissions Reduction	Impact on reducing greenhouse gas emissions	CO2 and NOx emissions levels	Benefit
EN2	Energy Efficiency	Effectiveness of energy use during operation	Energy consumption per mile or passenger	Benefit
EN3	Lifecycle Assessment	Environmental impact throughout the vehicle's lifecycle	Assessment of materials, manufacturing, and disposal	Cost
EN4	Land Use Impact	Effect on urban land use and space efficiency	Land area required for operations	Cost
EN5	Resource Use	Efficiency in the use of raw materials	Percentage of recycled materials used	Cost
EN6	Pollution Reduction	Contribution to reducing air and noise pollution	Noise levels and emissions during operation	Benefit
EN7	Biodiversity Impact	Effects on local ecosystems and biodiversity	Impact assessments on local wildlife	Cost
EN8	Renewable Energy Use	Utilization of renewable energy sources	Percentage of energy from renewable sources	Benefit
EN9	Waste Management	Strategies for end-of-life vehicle disposal	Recycling programs and disposal methods	Benefit
EN10	Water Usage	Amount of water used during operation and manufacturing	Water consumption metrics	Cost
Techhnical & Operational Sustainability (MC_4)				
TO1	Reliability	Consistency of vehicle performance	Uptime percentage, failure rates	Benefit
TO2	Safety Features	Inclusion of safety technologies and systems	Number of safety features, compliance standards	Benefit
TO3	Operational Efficiency	Overall efficiency of service delivery	Average wait time, passenger throughput	Benefit
TO4	Integration Capability	Compatibility with existing transport systems	Ability to integrate with current fleets and infrastructure	Benefit
TO5	Autonomous Level	Degree of automation (e.g., SAE levels)	Level of autonomy, operational scenarios	Benefit
TO6	Maintenance Simplicity	Ease of maintenance and servicing requirements	Maintenance intervals, service complexity	Benefit
TO7	Technology Scalability	Ability to adapt and scale technology	Scalability of software and hardware solutions	Benefit
TO8	Data Security	Measures for protecting operational data and privacy	Data encryption, cybersecurity measures	Benefit

fostering trust among decision-makers and stakeholders.

The synergistic integration of fuzzy LMAW and fuzzy DOBI methods thus provides a powerful and reliable framework, particularly suitable for scenarios characterized by high uncertainty and interdependent criteria. This interactive approach not only improves decision accuracy but also enriches methodological robustness and transparency, representing a valuable advancement in the field of fuzzy multi-criteria decision-making.

In this study, a Monte Carlo simulation was conducted to validate the robustness of the decision-making model. The simulation innovatively generated 1000 random sets of weights within the [0,1] interval, each summing exactly to 1. Subsequently, alternative rankings were recalculated for each set of randomly generated weights. This innovative validation approach emphasizes the reliability and consistency of the results by highlighting that certain alternatives consistently outperform others, even under random weight distributions. Such methodological rigor significantly enhances the credibility and generalizability of the decision-making framework, underscoring the importance of innovation in ensuring robust and dependable analytical outcomes.

3.1. Fuzzy Aczel-Alsina function-based LMAW method

The fuzzy Aczel-Alsina function-based Logarithmic Methodology of Additive Weights (LMAW) is a sophisticated decision-making tool

Table 5
Criteria priority vectors.

	DM1	DM2	DM3	DM4	DM5	DM6
Social Sustainability Criteria (MC1)						
SO1	EH	EH	EH	EH	VVH	EH
SO2	LL	M	MH	M	H	MH
SO3	VVH	EH	EH	EH	VVH	VVH
SO4	LL	M	M	M	MH	MH
SO5	VH	VH	VVH	VVH	VH	VVH
SO6	VH	VH	VH	VVH	VH	VH
SO7	VH	VH	VVH	VVH	VH	VH
Economic Sustainability Criteria (MC2)						
EC1	VVH	VH	VVH	VVH	VVH	VVH
EC2	LL	M	M	M	H	MH
EC3	MH	VH	H	H	MH	H
EC4	LL	M	M	M	MH	M
EC5	VVH	VVH	EH	EH	VVH	VVH
EC6	VH	VH	VH	VVH	VH	H
EC7	H	VH	VH	VH	VH	H
EC8	MH	VH	H	H	M	H
Environmental Sustainability Criteria (MC3)						
EN1	VVH	VVH	EH	VVH	VVH	VVH
EN2	H	VH	H	H	VH	H
EN3	M	MH	MH	H	M	H
EN4	ML	MH	MH	M	ML	MH
EN5	M	H	MH	H	M	H
EN6	H	VH	VH	H	VH	H
EN7	M	MH	MH	H	M	MH
EN8	M	VH	H	H	M	H
EN9	M	H	H	H	M	H
EN10	ML	MH	MH	H	ML	MH
Technical & Operational Sustainability Criteria (MC4)						
TO1	VVH	EH	EH	EH	VVH	EH
TO2	VVH	VVH	VVH	VVH	VVH	VVH
TO3	VH	VH	VVH	VVH	VVH	VVH
TO4	LL	MH	MH	M	ML	MH
TO5	LL	MH	MH	M	H	MH
TO6	VH	VH	VH	VH	VH	H
TO7	H	VH	H	H	H	H
TO8	H	VH	H	H	MH	H

engineered to address uncertainty in complex environments by integrating triangular fuzzy numbers (TFNs). These TFNs, defined by a lower bound, peak value, and upper bound (e.g., a, b, c), enable the method to model ambiguous or incomplete data, such as subjective expert judgments (e.g., "moderate to high priority") (Eti et al., 2025), within a mathematically rigorous framework. Originally designed for crisp (exact) values, the LMAW method was enhanced to handle fuzzy inputs, making it adaptable to real-world scenarios where information is often vague or incomplete (Božanić et al., 2022). Through logarithmic aggregation, it combines these fuzzy inputs to generate precise rankings while minimizing computational complexity. The method's robustness is validated by its superior performance over established techniques like Fuzzy Simple Additive Weighting (FSAW), which struggles with extreme values, and Fuzzy Multi-Attributive Border Approximation Area Comparison (FMABAC) (Mandal and Seikh, 2023), which requires cumbersome reference comparisons. Fuzzy LMAW not only delivers stable, consistent results under fluctuating conditions but also balances theoretical rigor with practical usability, making it invaluable in critical, time-sensitive decisions such as military planning or healthcare logistics. By bridging the gap between uncertainty modeling and actionable outcomes, the fuzzy LMAW method, as demonstrated by Božanić et al. (2022). The fuzzy nature of Aczel-Alsa's LMAW methodology is operationalized through four sequential steps, as delineated in the subsequent section.

Step 1. This step involves the establishment of the priority vector. Assuming a scenario where a total of h experts are engaged in the research, with p falling within the range of 1 to h, a priority vector ($\mathbb{R}$) can be articulated for each expert in the following manner:

$$\mathbb{R}^p = \left(\widetilde{\psi}^p_{C_1}, \widetilde{\psi}^p_{C_2}, .., \widetilde{\psi}^p_{C_n}\right), \tag{1}$$

where $\widetilde{\psi}^p_{C_1} = \left(\psi^{p(l)}_{C_1}, \psi^{p(m)}_{C_1}, \psi^{p(u)}_{C_1}\right)$ denotes the expert p's inclination towards criterion C_1, determined by a fuzzy scale previously utilized.

Step 2. Determining the absolute anti-ideal point (ε) involves establishing the reference value that defines the significance of the criterion, known as the absolute anti-ideal point (AAIP). The value of AAIP is selected arbitrarily to fulfil the specified conditions outlined in Eq. (2).

$$\varepsilon < \min_{\substack{1\le j\le n,\\ 1\le p\le h}} \left(\widetilde{\psi}^p_{C_j}\right), \tag{2}$$

where $\widetilde{\psi}^p_{C_j}$ denotes the component of the priority vector.

Step 3. Defining a ratio vector is crucial in analytical decision-making processes. This vector plays a significant role in establishing the connection between the criteria outlined in the priority vector. The components of the ratio vector, denoted as $\mathbb{Z}^p = \left(\widetilde{\omega}^p_{C_1}, \widetilde{\omega}^p_{C_2}, .., \widetilde{\omega}^p_{C_n}\right)$, are precisely specified through the utilization of Equation (3).

$$\widetilde{\omega}^p_{C_1} = \frac{\widetilde{\psi}^p_{C_j}}{\varepsilon}, \tag{3}$$

where $\widetilde{\psi}^p_{C_j} \in \mathbb{R}^p$, $\widetilde{\psi}^p_{C_1} = \left(\psi^{p(l)}_{C_1}, \psi^{p(m)}_{C_1}, \psi^{p(u)}_{C_1}\right)$, and $1 \le p \le h$.

Step 4. The final values of the fuzzy vector of weighting coefficients are determined using Eqs. (4)–(6). For each expert, the weighting coefficient vectors are established by applying Eqs. (4) and (5):

$$\widetilde{w}^p_j = \frac{\ln\left(\widetilde{\omega}^p_{C_j}\right)}{\ln\left(\widetilde{\partial}^p_j\right)} = \left(\frac{\ln\left(\omega^{p(l)}_{C_j}\right)}{\ln\left(\partial^{p(l)}_j\right)}, \frac{\ln\left(\omega^{p(m)}_{C_j}\right)}{\ln\left(\partial^{p(m)}_j\right)}, \frac{\ln\left(\omega^{p(u)}_{C_j}\right)}{\ln\left(\partial^{p(u)}_j\right)}\right), \tag{4}$$

where the element $\widetilde{\partial}^p_j = \left(\partial^{p(l)}_j, \partial^{p(m)}_j, \partial^{p(u)}_j\right)$ is obtained by applying Eq. (5):

$$\widetilde{\partial}^p_j = \left(\partial^{p(l)}_j, \partial^{p(m)}_j, \partial^{p(u)}_j\right) = \begin{pmatrix} \sum_{j=1}^{n} \omega^{p(l)}_j \left(e^{-\left(\sum_{j=1}^{n} \frac{1}{n}\left(-ln\left(f\left(\omega^{p(l)}_j\right)\right)\right)^{\alpha}\right)^{1/\alpha}}\right), \\ \sum_{j=1}^{n} \omega^{p(m)}_j \left(e^{-\left(\sum_{j=1}^{n} \frac{1}{n}\left(-\ln\left(f\left(\omega^{p(m)}_j\right)\right)\right)^{\alpha}\right)^{\frac{1}{\alpha}}}\right), \\ \sum_{j=1}^{n} \omega^{p(u)}_j \left(e^{-\left(\sum_{j=1}^{n} \frac{1}{n}\left(-ln\left(f\left(\omega^{p(u)}_j\right)\right)\right)^{\alpha}\right)^{1/\alpha}}\right) \end{pmatrix}, \tag{5}$$

where $\alpha > 0$, and $f\left(\widetilde{\omega}^p_j\right) = \widetilde{\omega}^p_j / \sum_{j=1}^{n} \widetilde{\omega}^p_j$.

The aggregated fuzzy vector of weighting coefficients is determined using Eq. (6):

Table 6
Criteria ratio vectors.

	DM1	DM2	DM3	DM4	DM5	DM6
Social Sustainability Criteria (MC1)						
SO1	(14.17,19,25)	(14.17,19,25)	(14.17,19,25)	(14.17,19,25)	(12.5,17,23.75)	(14.17,19,25)
SO2	(2.5,5,8.75)	(5.83,9,13.75)	(7.5,11,16.25)	(5.83,9,13.75)	(9.17,13,18.75)	(7.5,11,16.25)
SO3	(12.5,17,23.75)	(14.17,19,25)	(14.17,19,25)	(14.17,19,25)	(12.5,17,23.75)	(12.5,17,23.75)
SO4	(2.5,5,8.75)	(5.83,9,13.75)	(5.83,9,13.75)	(5.83,9,13.75)	(7.5,11,16.25)	(7.5,11,16.25)
SO5	(10.83,15,21.25)	(10.83,15,21.25)	(12.5,17,23.75)	(12.5,17,23.75)	(10.83,15,21.25)	(12.5,17,23.75)
SO6	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(12.5,17,23.75)	(10.83,15,21.25)	(10.83,15,21.25)
SO7	(10.83,15,21.25)	(10.83,15,21.25)	(12.5,17,23.75)	(12.5,17,23.75)	(10.83,15,21.25)	(10.83,15,21.25)
Economic Sustainability Criteria (MC2)						
EC1	(12.5,17,23.75)	(10.83,15,21.25)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)
EC2	(2.5,5,8.75)	(5.83,9,13.75)	(5.83,9,13.75)	(5.83,9,13.75)	(9.17,13,18.75)	(7.5,11,16.25)
EC3	(7.5,11,16.25)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(7.5,11,16.25)	(9.17,13,18.75)
EC4	(2.5,5,8.75)	(5.83,9,13.75)	(5.83,9,13.75)	(5.83,9,13.75)	(7.5,11,16.25)	(5.83,9,13.75)
EC5	(12.5,17,23.75)	(12.5,17,23.75)	(14.17,19,25)	(14.17,19,25)	(12.5,17,23.75)	(12.5,17,23.75)
EC6	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(12.5,17,23.75)	(10.83,15,21.25)	(9.17,13,18.75)
EC7	(9.17,13,18.75)	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(9.17,13,18.75)
EC8	(7.5,11,16.25)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(5.83,9,13.75)	(9.17,13,18.75)
Environmental Sustainability Criteria (MC3)						
EN1	(12.5,17,23.75)	(12.5,17,23.75)	(14.17,19,25)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)
EN2	(9.17,13,18.75)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(10.83,15,21.25)	(9.17,13,18.75)
EN3	(5.83,9,13.75)	(7.5,11,16.25)	(7.5,11,16.25)	(9.17,13,18.75)	(5.83,9,13.75)	(9.17,13,18.75)
EN4	(4.17,7,11.25)	(7.5,11,16.25)	(7.5,11,16.25)	(5.83,9,13.75)	(4.17,7,11.25)	(7.5,11,16.25)
EN5	(5.83,9,13.75)	(9.17,13,18.75)	(7.5,11,16.25)	(9.17,13,18.75)	(5.83,9,13.75)	(9.17,13,18.75)
EN6	(9.17,13,18.75)	(10.83,15,21.25)	(10.83,15,21.25)	(9.17,13,18.75)	(10.83,15,21.25)	(9.17,13,18.75)
EN7	(5.83,9,13.75)	(7.5,11,16.25)	(7.5,11,16.25)	(9.17,13,18.75)	(5.83,9,13.75)	(7.5,11,16.25)
EN8	(5.83,9,13.75)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(5.83,9,13.75)	(9.17,13,18.75)
EN9	(5.83,9,13.75)	(9.17,13,18.75)	(9.17,13,18.75)	(9.17,13,18.75)	(5.83,9,13.75)	(9.17,13,18.75)
EN10	(4.17,7,11.25)	(7.5,11,16.25)	(7.5,11,16.25)	(9.17,13,18.75)	(4.17,7,11.25)	(7.5,11,16.25)
Technical & Operational Sustainability Criteria (MC4)						
TO1	(12.5,17,23.75)	(14.17,19,25)	(14.17,19,25)	(14.17,19,25)	(12.5,17,23.75)	(14.17,19,25)
TO2	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)
TO3	(10.83,15,21.25)	(10.83,15,21.25)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)	(12.5,17,23.75)
TO4	(2.5,5,8.75)	(7.5,11,16.25)	(7.5,11,16.25)	(5.83,9,13.75)	(4.17,7,11.25)	(7.5,11,16.25)
TO5	(2.5,5,8.75)	(7.5,11,16.25)	(7.5,11,16.25)	(5.83,9,13.75)	(9.17,13,18.75)	(7.5,11,16.25)
TO6	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(10.83,15,21.25)	(9.17,13,18.75)
TO7	(9.17,13,18.75)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(9.17,13,18.75)	(9.17,13,18.75)
TO8	(9.17,13,18.75)	(10.83,15,21.25)	(9.17,13,18.75)	(9.17,13,18.75)	(7.5,11,16.25)	(9.17,13,18.75)

$$\widetilde{w}_j=\left(w_j^{(l)},w_j^{(m)},w_j^{(u)}\right)=\left(\begin{array}{c}\sum_{k=1}^{h} w_{j(k)}^{(l)}\left(1-e^{-\left(\sum_{k=1}^{h}\frac{1}{h}\left(-\ln\left(1-f\left(w_{j(k)}^{(l)}\right)\right)\right)^{\varphi}\right)^{1/\varphi}}\right),\\ \sum_{k=1}^{h} w_{j(k)}^{(m)}\left(1-e^{-\left(\sum_{k=1}^{h}\frac{1}{h}\left(-\ln\left(1-f\left(w_{j(k)}^{(m)}\right)\right)\right)^{\varphi}\right)^{\frac{1}{\varphi}}}\right),\\ \sum_{k=1}^{h} w_{j(k)}^{(u)}\left(1-e^{-\left(\sum_{k=1}^{h}\frac{1}{h}\left(-\ln\left(1-f\left(w_{j(k)}^{(u)}\right)\right)\right)^{\varphi}\right)^{\frac{1}{\varphi}}}\right)\end{array}\right), \tag{6}$$

where $\varphi>0, f\left(\widetilde{w}_{j(k)}\right)=\widetilde{w}_{j(k)}/\sum_{k=1}^{h}\widetilde{w}_{j(k)}$, and h denotes the number of experts.

3.2. Fuzzy Dombi Bonferroni method

The Fuzzy Dombi Bonferroni methodology is a versatile and adaptable framework designed to address multi-attribute decision-making (MADM) challenges, particularly under uncertainty. By extending the Dombi operator (Seikh and Mandal, 2023)—which allows parameterized flexibility to model decision-maker preferences—and the Bonferroni mean—which accounts for attribute interdependencies through pairwise aggregation—into advanced fuzzy environments like T-spherical fuzzy sets (T-SFS) and q-rung orthopair fuzzy sets (q-ROFS), the approach effectively manages imprecise or ambiguous data. These fuzzy frameworks enable nuanced representation of membership, non-membership, and neutrality degrees (in T-SFS) or independent adjustment of membership and non-membership within a q-ROFS (Yang and Pang, 2020, 2022). The methodology employs specialized operators such as the interaction Bonferroni mean (to handle attribute correlations), geometric Bonferroni mean (for multiplicative aggregation), weighted operators (to assign priority to criteria), and partitioned operators (to cluster attributes into subgroups), ensuring tailored solutions for diverse problem complexities. The steps to implement this method can be summarized as follows (Deveci et al., 2023):

The decision-making model is designed to evaluate m alternatives, provided by a committee of experts $A_i(i=1,2,\ldots,m)$, based on n criteria $C_j(j=1,2,\ldots,n)$, with the assistance of a set of experts $H_e(e=1,2,\ldots,b)$.

Step 1. Create the initial decision matrix $Z=\left[\Omega_{ij}\right]_{mxn}$ by using the fuzzy Dombi weighted geometric averaging (FDWGA) operator, as specified in Eq. (7).

Table 7
Vectors of weighting coefficients.

	DM1	DM2	DM3	DM4	DM5	DM6
SO1	(0.177,0.18,0.187)	(0.165,0.164,0.161)	(0.166,0.164,0.161)	(0.166,0.164,0.162)	(0.161,0.163,0.167)	(0.166,0.164,0.162)
SO2	(0.091,0.13,0.153)	(0.163,0.161,0.159)	(0.187,0.176,0.17)	(0.163,0.162,0.16)	(0.21,0.195,0.188)	(0.187,0.177,0.17)
SO3	(0.171,0.176,0.185)	(0.168,0.166,0.162)	(0.168,0.166,0.162)	(0.168,0.166,0.163)	(0.164,0.166,0.168)	(0.161,0.16,0.16)
SO4	(0.094,0.134,0.156)	(0.17,0.166,0.162)	(0.17,0.166,0.163)	(0.171,0.166,0.163)	(0.199,0.187,0.182)	(0.195,0.182,0.174)
SO5	(0.17,0.175,0.183)	(0.159,0.159,0.158)	(0.169,0.167,0.164)	(0.169,0.167,0.164)	(0.163,0.165,0.166)	(0.169,0.167,0.164)
SO6	(0.173,0.178,0.186)	(0.162,0.161,0.16)	(0.163,0.162,0.16)	(0.172,0.169,0.166)	(0.166,0.167,0.168)	(0.163,0.162,0.161)
SO7	(0.172,0.176,0.185)	(0.161,0.16,0.159)	(0.171,0.168,0.165)	(0.171,0.168,0.165)	(0.165,0.166,0.167)	(0.161,0.161,0.16)
EC1	(0.177,0.18,0.188)	(0.156,0.157,0.156)	(0.166,0.164,0.162)	(0.166,0.164,0.162)	(0.17,0.17,0.17)	(0.166,0.165,0.162)
EC2	(0.093,0.132,0.155)	(0.167,0.163,0.161)	(0.167,0.164,0.161)	(0.167,0.164,0.162)	(0.215,0.198,0.19)	(0.191,0.179,0.172)
EC3	(0.162,0.17,0.179)	(0.18,0.174,0.169)	(0.167,0.165,0.163)	(0.167,0.165,0.163)	(0.156,0.16,0.163)	(0.168,0.166,0.163)
EC4	(0.097,0.136,0.158)	(0.174,0.168,0.164)	(0.175,0.168,0.164)	(0.175,0.169,0.165)	(0.204,0.19,0.184)	(0.175,0.169,0.165)
EC5	(0.172,0.177,0.186)	(0.161,0.161,0.16)	(0.17,0.167,0.163)	(0.17,0.167,0.163)	(0.165,0.167,0.168)	(0.162,0.161,0.161)
EC6	(0.175,0.179,0.187)	(0.164,0.163,0.161)	(0.164,0.163,0.161)	(0.174,0.171,0.167)	(0.168,0.169,0.169)	(0.153,0.155,0.155)
EC7	(0.167,0.173,0.182)	(0.168,0.166,0.163)	(0.168,0.166,0.163)	(0.168,0.166,0.164)	(0.172,0.172,0.172)	(0.157,0.158,0.157)
EC8	(0.166,0.172,0.181)	(0.183,0.177,0.171)	(0.171,0.167,0.164)	(0.171,0.168,0.165)	(0.139,0.148,0.154)	(0.171,0.168,0.165)
EN1	(0.174,0.178,0.186)	(0.163,0.162,0.16)	(0.171,0.168,0.163)	(0.163,0.162,0.161)	(0.167,0.168,0.169)	(0.163,0.162,0.161)
EN2	(0.171,0.176,0.184)	(0.172,0.169,0.165)	(0.16,0.16,0.159)	(0.16,0.16,0.159)	(0.176,0.175,0.174)	(0.16,0.16,0.159)
EN3	(0.155,0.165,0.176)	(0.166,0.164,0.161)	(0.166,0.164,0.162)	(0.182,0.176,0.17)	(0.149,0.156,0.16)	(0.183,0.176,0.171)
EN4	(0.141,0.158,0.172)	(0.186,0.177,0.17)	(0.187,0.177,0.17)	(0.164,0.162,0.161)	(0.135,0.149,0.156)	(0.187,0.178,0.171)
EN5	(0.152,0.163,0.175)	(0.179,0.173,0.168)	(0.163,0.162,0.16)	(0.18,0.174,0.169)	(0.146,0.154,0.158)	(0.18,0.174,0.169)
EN6	(0.169,0.174,0.183)	(0.17,0.167,0.164)	(0.17,0.167,0.164)	(0.158,0.159,0.158)	(0.174,0.173,0.173)	(0.159,0.159,0.158)
EN7	(0.157,0.167,0.178)	(0.168,0.166,0.163)	(0.169,0.166,0.163)	(0.186,0.178,0.172)	(0.151,0.157,0.161)	(0.169,0.166,0.164)
EN8	(0.148,0.16,0.172)	(0.187,0.179,0.173)	(0.174,0.17,0.166)	(0.174,0.17,0.166)	(0.142,0.151,0.156)	(0.174,0.17,0.167)
EN9	(0.15,0.162,0.173)	(0.176,0.171,0.167)	(0.177,0.171,0.167)	(0.177,0.172,0.168)	(0.144,0.152,0.157)	(0.177,0.172,0.168)
EN10	(0.135,0.154,0.169)	(0.179,0.172,0.167)	(0.179,0.172,0.167)	(0.197,0.185,0.176)	(0.13,0.145,0.153)	(0.179,0.173,0.168)
TO1	(0.17,0.175,0.185)	(0.167,0.165,0.161)	(0.167,0.165,0.162)	(0.167,0.165,0.162)	(0.163,0.164,0.167)	(0.167,0.166,0.162)
TO2	(0.175,0.179,0.187)	(0.164,0.163,0.161)	(0.164,0.163,0.161)	(0.164,0.163,0.161)	(0.168,0.169,0.169)	(0.164,0.163,0.161)
TO3	(0.168,0.174,0.182)	(0.158,0.158,0.157)	(0.167,0.165,0.163)	(0.167,0.166,0.163)	(0.171,0.171,0.171)	(0.168,0.166,0.163)
TO4	(0.095,0.134,0.157)	(0.196,0.182,0.173)	(0.197,0.182,0.174)	(0.172,0.167,0.164)	(0.143,0.153,0.158)	(0.197,0.182,0.174)
TO5	(0.088,0.128,0.152)	(0.182,0.173,0.168)	(0.182,0.173,0.168)	(0.16,0.159,0.158)	(0.205,0.192,0.186)	(0.183,0.174,0.169)
TO6	(0.177,0.181,0.188)	(0.166,0.164,0.162)	(0.166,0.164,0.162)	(0.166,0.165,0.162)	(0.17,0.17,0.17)	(0.155,0.156,0.156)
TO7	(0.173,0.178,0.185)	(0.174,0.17,0.166)	(0.162,0.161,0.16)	(0.162,0.162,0.16)	(0.166,0.167,0.168)	(0.162,0.162,0.16)
TO8	(0.176,0.179,0.187)	(0.177,0.172,0.168)	(0.165,0.163,0.161)	(0.165,0.163,0.162)	(0.153,0.158,0.161)	(0.165,0.164,0.162)

Table 8
Final fuzzy vector of weight coefficients.

	Fuzzy Value	Defuzzification
SO1	(0.0273,0.0347,0.0445)	0.0351
SO2	(0.0185,0.0264,0.0367)	0.0268
SO3	(0.0268,0.0343,0.0443)	0.0347
SO4	(0.0177,0.0257,0.0360)	0.0260
SO5	(0.0254,0.0329,0.0431)	0.0334
SO6	(0.0250,0.0324,0.0426)	0.0329
SO7	(0.0252,0.0326,0.0429)	0.0331
EC1	(0.0259,0.0334,0.0437)	0.0339
EC2	(0.0181,0.0260,0.0363)	0.0264
EC3	(0.0225,0.0300,0.0402)	0.0305
EC4	(0.0173,0.0253,0.0356)	0.0257
EC5	(0.0266,0.0341,0.0442)	0.0345
EC6	(0.0247,0.0321,0.0424)	0.0326
EC7	(0.0241,0.0316,0.0418)	0.0320
EC8	(0.0221,0.0297,0.0398)	0.0301
EN1	(0.0264,0.0339,0.0441)	0.0343
EN2	(0.0236,0.0310,0.0412)	0.0315
EN3	(0.0207,0.0283,0.0385)	0.0287
EN4	(0.0184,0.0262,0.0365)	0.0266
EN5	(0.0210,0.0286,0.0388)	0.0291
EN6	(0.0238,0.0313,0.0415)	0.0318
EN7	(0.0203,0.0280,0.0382)	0.0284
EN8	(0.0217,0.0292,0.0394)	0.0297
EN9	(0.0214,0.0289,0.0391)	0.0294
EN10	(0.0192,0.0270,0.0372)	0.0274
TO1	(0.0270,0.0345,0.0444)	0.0349
TO2	(0.0262,0.0337,0.0439)	0.0341
TO3	(0.0257,0.0331,0.0434)	0.0336
TO4	(0.0175,0.0255,0.0358)	0.0259
TO5	(0.0189,0.0268,0.0370)	0.0272
TO6	(0.0244,0.0319,0.0421)	0.0324
TO7	(0.0233,0.0307,0.0409)	0.0312
TO8	(0.0229,0.0304,0.0406)	0.0309

$$FDGWP_i^{\varpi} = \left(FDGWP_i^{\varpi(l)}, FDGWP_i^{\varpi(m)}, FDGWP_i^{\varpi(u)} \right) = \left(\frac{\sum_{j=1}^{n} \left(\Omega_{ij}^{(l)} \right)}{1 + \left\{ \sum_{j=1}^{n} \psi_j \left(\frac{1 - f\left(\Omega_{ij}^{(l)} \right)}{f\left(\Omega_{ij}^{(l)} \right)} \right)^{\varpi} \right\}^{1/\varpi}}, \frac{1}{1 + \left\{ \sum_{j=1}^{n} \psi_j \left(\frac{1 - f\left(\Omega_{ij}^{(m)} \right)}{f\left(\Omega_{ij}^{(m)} \right)} \right)^{\varpi} \right\}^{1/\varpi}}, \frac{1}{1 + \left\{ \sum_{j=1}^{n} \psi_j \left(\frac{1 - f\left(\Omega_{ij}^{(u)} \right)}{f\left(\Omega_{ij}^{(u)} \right)} \right)^{\varpi} \right\}^{1/\varpi}} \right), \tag{7}$$

where $\psi_j = (\psi_1, \psi_2, \ldots, \psi_\eta)^T$ denotes the vector of the weight coefficients of the criteria and

$$f(\Phi_j) = \left(\Phi_j^l \Big/ \sum_{j=1}^{n} \Phi_j^{(l)}, \Phi_j^{(m)} \Big/ \sum_{j=1}^{n} \Phi_j^{(m)}, \Phi_j^{(u)} \Big/ \sum_{j=1}^{n} \Phi_j^{(u)} \right).$$

Then, $FDGWP_i^{\varpi}$ represents the fuzzy Dombi weighted averaging function.

Step 2. Normalize the initial matrix $Z = \left[\Omega_{ij} \right]_{mxn}$. The normalized matrix $N = \left[\Phi_{ij} \right]_{mxn}$ is derived using equations (8) and (9). The elements $\Phi_{ij} = \left(\Phi_{ij}^{(l)}, \Phi_{ij}^{(m)}, \Phi_{ij}^{(u)} \right)$ are given by:

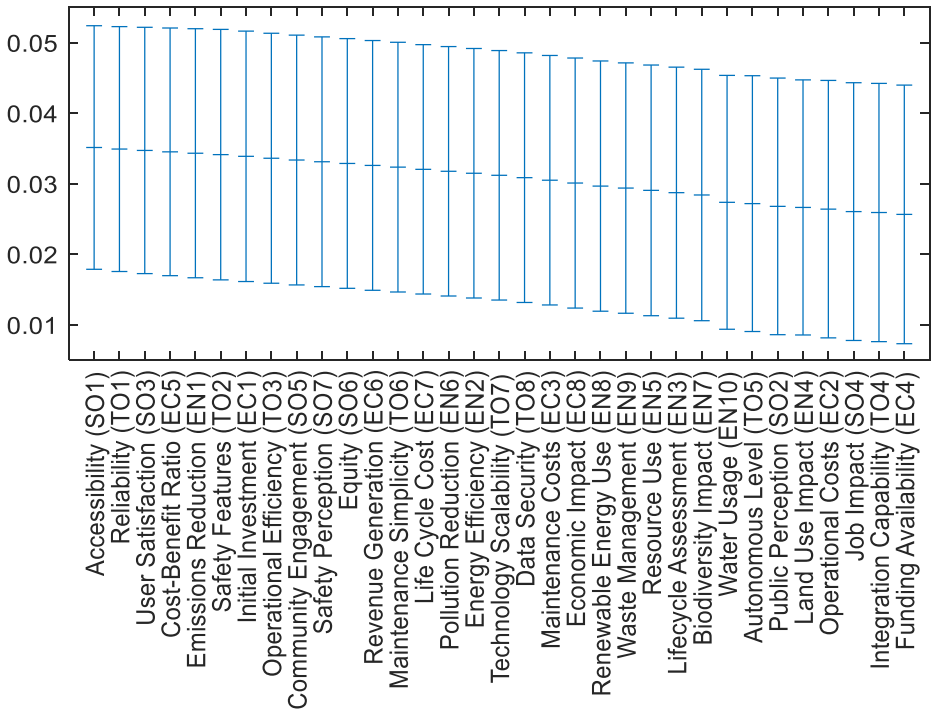


Fig. 2. Fuzzy weight coefficients of criteria.

$$\Phi_{ij}=\left(\Phi_{ij}^{(l)},\Phi_{ij}^{(m)},\Phi_{ij}^{(u)}\right)=\begin{cases}\Phi_{ij}^{(l)}=\dfrac{\Phi_{ij}^{(l)}}{\Phi_{ij}^{(u)+}}\\ \Phi_{ij}^{(m)}=\dfrac{\Phi_{ij}^{(m)}}{\Phi_{ij}^{(u)+}}\ \text{if } j\in B\\ \Phi_{ij}^{(u)}=\dfrac{\Phi_{ij}^{(u)}}{\Phi_{ij}^{(u)+}}\end{cases} \tag{8}$$

$$\Phi_{ij}=\left(\Phi_{ij}^{(l)},\Phi_{ij}^{(m)},\Phi_{ij}^{(u)}\right)=\begin{cases}\Phi_{ij}^{(l)}=\dfrac{\Phi_{ij}^{(l)-}}{\Phi_{ij}^{(u)}}\\ \Phi_{ij}^{(m)}=\dfrac{\Phi_{ij}^{(l)-}}{\Phi_{ij}^{(m)}};\ \text{if } j\in C\\ \Phi_{ij}^{(u)}=\dfrac{\Phi_{ij}^{(l)}}{\Phi_{ij}^{(l)-}}\end{cases} \tag{9}$$

where $\Phi_{ij}{}^{+}=max(\Phi_{ij})$ and $\Phi_{ij}{}^{-}=min(\Phi_{ij})$ representing the benefit (B) and cost (C) criteria, respectively.

Step 3. Determine the weighted importance of alternatives. The DOBI weighted averaging function $\left(\chi_i^{(1)\theta_1,\theta_2,\varpi}\right)$ and the DOBI weighted geometric function $\left(\chi_i^{(2)\theta_1,\theta_2,\varpi}\right)$ are designed to compute this weighted importance. These functions are represented by equations (10) and (11), where $f(\Phi_j)=\Phi_j/\sum_{j=1}^{n}\Phi_i^n$. In this context, $\left(\chi_i^{(1)\theta_1,\theta_2,\varpi}\right)$ represents the DOBI weighted averaging function, whereas $\left(\chi_i^{(2)\theta_1,\theta_2,\varpi}\right)$ signifies the DOBI weighted geometric averaging function.

$$\chi_i^{(1)\theta_1,\theta_2,\varpi}=\frac{\sum_{j=1}^{n}\Phi_{ij}}{1+\left\{\frac{1}{\psi_i\psi_j(\theta_1+\theta_2)}\sum_{\substack{i,j=1\\ i\neq j}}^{n}\frac{1-\psi_i}{\frac{1}{(\theta_1((1-f(\Phi_i)))/f(\Phi_i))^{\varpi}+\theta_2\left(\left(1-f\left(\Phi_j\right)\right)/f\left(\Phi_j\right)\right)^{\varpi}}}\right\}^{1/\varpi}} \tag{10}$$

$$\chi_i^{(2)\theta_1,\theta_2,\varpi}=\sum_{j=1}^{n}\Phi_{ij}-\frac{\sum_{j=1}^{n}\Phi_{ij}}{1+\left\{\frac{1}{(\theta_1+\theta_2)\psi_i\psi_j}\sum_{\substack{i,j=1\\ i\neq j}}^{n}\frac{1-\psi_i}{\frac{1}{\left(\theta_1\left(f(\Phi_i)/(1-f(\Phi_i))\right)^{\varpi}+\left(\theta_2\left(f\left(\Phi_j\right)/\left(1-f\left(\Phi_j\right)\right)\right)^{\varpi}\right.\right.}}\right\}^{1/\varpi}} \tag{11}$$

Step 4. Calculate the integrated value of the DOBI functions Υ_i. Subsequently, the DOBI functions can be expressed as:

$$\Upsilon_i=\frac{\chi_i^{(1)\theta_1,\theta_2,\varpi}+\chi_i^{(2)\theta_1,\theta_2,\varpi}}{1+\left\{\sigma\left(\frac{1-\chi_i^{(1)\theta_1,\theta_2,\varpi}}{\chi_i^{(1)\theta_1,\theta_2,\varpi}}\right)^{\gamma}+(1-\sigma)\left(\frac{1-\chi_i^{(2)\theta_1,\theta_2,\varpi}}{\chi_i^{(2)\theta_1,\theta_2,\varpi}}\right)^{\gamma}\right\}} \tag{12}$$

where $\gamma\geq 0$.

Step 5. Convert the Υ_i values to obtain numerical values for the alternatives.

Table 9
Aggregated initial matrix.
Step 2: The normalized initial matrix for each alternatives in term sof criteria is obtained by the values in Table above with the help of Eqs. (8) and (9) and is given in Table 10

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
SO1	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
SO2	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)
SO3	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)
SO4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)
SO5	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
SO6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
SO7	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EC1	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)
EC2	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EC3	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EC4	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EC5	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
EC6	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EC7	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)
EC8	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN1	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(6.87,7.88,8.83)
EN2	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN3	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN5	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN7	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN8	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)
EN9	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
EN10	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
TO1	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(6.87,7.88,8.83)
TO2	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)
TO3	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)
TO4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
TO5	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
TO6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
TO7	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
TO8	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)
	A11	**A12**	**A13**	**A14**	**A15**	**A16**	**A17**	**A18**	**A19**	**A20**
SO1	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
SO2	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(6.87,7.88,8.83)
SO3	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(6.87,7.88,8.83)
SO4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)
SO5	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
SO6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
SO7	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(5.84,6.87,7.88)	(6.87,7.88,8.83)
EC1	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EC2	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EC3	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EC4	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EC5	(2.65,3.76,4.81)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(3.76,4.81,5.84)
EC6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EC7	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)
EC8	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EN1	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(6.87,7.88,8.83)
EN2	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EN3	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)

(*continued on next page*)

Table 9 (*continued*)

	A11	A12	A13	A14	A15	A16	A17	A18	A19	A20
EN4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EN5	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EN6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EN7	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
EN8	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
EN9	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
EN10	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
TO1	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(6.87,7.88,8.83)
TO2	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(4.81,5.84,6.87)	(6.87,7.88,8.83)
TO3	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(6.87,7.88,8.83)	(6.87,7.88,8.83)	(3.76,4.81,5.84)	(6.87,7.88,8.83)
TO4	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)
TO5	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
TO6	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(2.65,3.76,4.81)	(2.65,3.76,4.81)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(3.76,4.81,5.84)	(5.84,6.87,7.88)
TO7	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)
TO8	(4.81,5.84,6.87)	(4.81,5.84,6.87)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(3.76,4.81,5.84)	(3.76,4.81,5.84)	(5.84,6.87,7.88)	(5.84,6.87,7.88)	(4.81,5.84,6.87)	(5.84,6.87,7.88)

$$\Theta_i = \frac{\Upsilon_i^l + 4\Upsilon_i^m + \Upsilon_i^u}{6} \tag{13}$$

where Θ_i denotes the score values of options.

Step 6. Rank the options based on their defuzzified values of Θ_i value. The best option is chosen as the one with the highest Θ_i value.

4. Case study

The decision-making model proposed in this study was applied to address the pressing decision problem regarding evaluating the economic, social, environmental, technical, and operational sustainability performances of autonomous (driverless) buses and to produce appropriate solutions. The transportation authorities expressed this decision problem in a recent workshop on urbanization and urban transportation, and it was decided that we would work on it with our research team. After an extensive preliminary research and literature review, the research team and the professionals of the competent authority unanimously identified this as a decision-making problem with complex uncertainties. We all recognized the need for a consistent and reliable decision-making model that provides a sophisticated and comprehensive analysis. In response, the researchers meticulously designed a process that we believe will reasonably address the decision-making problem and yield logical and applicable results.

First, the researchers defined the research problem in detail to structure the decision-making problem correctly. Accordingly, analyzing the sustainability performance of autonomous buses in a multidimensional way and adapting them to the cities' transportation systems by considering their sustainability performance is a critical research problem. Autonomous buses have the potential to minimize the risks of accidents, control carbon emissions, and improve operational and economic efficiency by reducing driver-related errors compared to conventional ones. Integrating these vehicles into urban transportation systems is not just a problem but a complex decision-making challenge involving extremely complex uncertainties, underscoring the importance of our research.

In this context, instead of addressing this decision problem only in Istanbul, the researchers handled the problem more comprehensively to generalize the results. In this direction, to minimize the biased attitudes and subjective evaluations of decision-makers, the board of experts preferred to create a broader perspective and to include responsible and authorized people in the field of urban transportation in the big cities of different countries. Next, professional businesses' social platforms (i.e., LinkedIn, etc.) were examined using combinations of keywords such as autonomous vehicles, driverless vehicles, and buses. Among 800 professionals related to the approximate field by the scope and requirements of the study, 34 candidates with vast knowledge and experience in autonomous vehicle technology and urban transportation were determined. An invitation was sent to these candidates to become members of the expert board, and in this invitation, the scope of the work and their expectations were specified. An online meeting was held with nine candidates who responded positively to the invitation, and it was decided to continue with six professionals.

Afterwards, they conducted a comprehensive review process involving experts and determined sustainability criteria and autonomous bus alternatives. The researchers did not work in isolation; they collaborated with experts to ensure the credibility of the criteria. First, a detailed literature review was conducted to determine the criteria, and the researchers noted the criteria used in the relevant literature. The researchers then updated the list of criteria by eliminating the repeated criteria. In this process, a blank evaluation form was sent to the experts, who were asked to note the criteria affecting autonomous buses' sustainability performance. After all forms were collected, the criteria in the list obtained by the literature review were excluded, and the criteria that were not in the list and reflected the opinions of experts were added to the list. Likewise, care was taken to ensure that each newly added

Table 10
Normalized matrix.

	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
SO1	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)
SO2	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)
SO3	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)
SO4	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.39,0.55,0.7)	(0.55,0.7,0.85)	(0.7,0.85,1)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.7,0.85,1)	(0.7,0.85,1)	(0.55,0.7,0.85)
SO5	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
SO6	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)
SO7	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)
EC1	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.55,0.64,0.78)
EC2	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EC3	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EC4	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EC5	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.55,0.71,1)	(0.45,0.55,0.71)	(0.34,0.39,0.45)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.45,0.55,0.71)
EC6	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EC7	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.39,0.45,0.55)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.39,0.45,0.55)	(0.39,0.45,0.55)	(0.39,0.45,0.55)
EC8	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EN1	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.78,0.89,1)
EN2	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EN3	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EN4	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EN5	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EN6	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EN7	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
EN8	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.7,0.85,1)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.7,0.85,1)	(0.55,0.7,0.85)
EN9	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
EN10	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.48,0.55,0.64)
TO1	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.78,0.89,1)
TO2	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)
TO3	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.43,0.54,0.66)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)
TO4	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)
TO5	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
TO6	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
TO7	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
TO8	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.74,0.87,1)
	A11	**A12**	**A13**	**A14**	**A15**	**A16**	**A17**	**A18**	**A19**	**A20**
SO1	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.66,0.78,0.89)
SO2	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.78,0.89,1)
SO3	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.78,0.89,1)
SO4	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.39,0.55,0.7)	(0.39,0.55,0.7)	(0.7,0.85,1)	(0.7,0.85,1)	(0.55,0.7,0.85)	(0.7,0.85,1)
SO5	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)
SO6	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.45,0.64,0.82)	(0.45,0.64,0.82)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)	(0.64,0.82,1)
SO7	(0.66,0.78,0.89)	(0.66,0.78,0.89)	(0.78,0.89,1)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.78,0.89,1)	(0.78,0.89,1)	(0.66,0.78,0.89)	(0.78,0.89,1)
EC1	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.48,0.55,0.64)
EC2	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.48,0.55,0.64)
EC3	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.48,0.55,0.64)
EC4	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)
EC5	(0.55,0.71,1)	(0.45,0.55,0.71)	(0.39,0.45,0.55)	(0.45,0.55,0.71)	(0.55,0.71,1)	(0.55,0.71,1)	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.55,0.71,1)	(0.45,0.55,0.71)
EC6	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
EC7	(0.45,0.55,0.71)	(0.45,0.55,0.71)	(0.39,0.45,0.55)	(0.39,0.45,0.55)	(0.55,0.71,1)	(0.55,0.71,1)	(0.39,0.45,0.55)	(0.39,0.45,0.55)	(0.45,0.55,0.71)	(0.39,0.45,0.55)
EC8	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
EN1	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.43,0.54,0.66)	(0.43,0.54,0.66)	(0.78,0.89,1)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.78,0.89,1)
EN2	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
EN3	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.48,0.55,0.64)
EN4	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.48,0.55,0.64)

(*continued on next page*)

Table 10 (*continued*)

	A11	A12	A13	A14	A15	A16	A17	A18	A19	A20
EN5	(0.55,0.64,0.78)	(0.55,0.64,0.78)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.48,0.55,0.64)
EN6	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.34,0.48,0.61)	(0.34,0.48,0.61)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
EN7	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.55,0.64,0.78)	(0.48,0.55,0.64)
EN8	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)	(0.55,0.7,0.85)
EN9	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
EN10	(0.64,0.78,1)	(0.64,0.78,1)	(0.55,0.64,0.78)	(0.64,0.78,1)	(0.64,0.78,1)	(0.64,0.78,1)	(0.48,0.55,0.64)	(0.48,0.55,0.64)	(0.64,0.78,1)	(0.48,0.55,0.64)
TO1	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.43,0.54,0.66)	(0.43,0.54,0.66)	(0.78,0.89,1)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.78,0.89,1)
TO2	(0.54,0.66,0.78)	(0.54,0.66,0.78)	(0.66,0.78,0.89)	(0.54,0.66,0.78)	(0.43,0.54,0.66)	(0.43,0.54,0.66)	(0.78,0.89,1)	(0.78,0.89,1)	(0.54,0.66,0.78)	(0.78,0.89,1)
TO3	(0.43,0.54,0.66)	(0.43,0.54,0.66)	(0.66,0.78,0.89)	(0.43,0.54,0.66)	(0.3,0.43,0.54)	(0.3,0.43,0.54)	(0.78,0.89,1)	(0.78,0.89,1)	(0.43,0.54,0.66)	(0.78,0.89,1)
TO4	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)
TO5	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
TO6	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.34,0.48,0.61)	(0.34,0.48,0.61)	(0.74,0.87,1)	(0.74,0.87,1)	(0.48,0.61,0.74)	(0.74,0.87,1)
TO7	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)
TO8	(0.61,0.74,0.87)	(0.61,0.74,0.87)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.48,0.61,0.74)	(0.48,0.61,0.74)	(0.74,0.87,1)	(0.74,0.87,1)	(0.61,0.74,0.87)	(0.74,0.87,1)

Table 11
The weighted significance of alternatives.

Alternatives	$\chi_i^{(1)\theta_1,\theta_2,\varpi}$	$\chi_i^{(2)\theta_1,\theta_2,\varpi}$
A1	(0.216,0.465,0.985)	(2.105,1.459,0.984)
A2	(0.216,0.467,0.994)	(2.109,1.467,0.994)
A3	(0.202,0.441,0.946)	(1.963,1.384,0.946)
A4	(0.219,0.452,0.957)	(2.042,1.419,0.957)
A5	(0.235,0.488,0.995)	(2.294,1.531,0.995)
A6	(0.208,0.449,0.952)	(2.028,1.410,0.952)
A7	(0.219,0.467,0.979)	(2.138,1.466,0.979)
A8	(0.236,0.488,0.996)	(2.298,1.533,0.996)
A9	(0.236,0.489,1.000)	(2.299,1.536,1.000)
A10	(0.228,0.475,0.977)	(2.222,1.492,0.977)
A11	(0.204,0.445,0.954)	(1.985,1.397,0.954)
A12	(0.203,0.444,0.950)	(1.980,1.393,0.950)
A13	(0.222,0.469,0.975)	(2.159,1.471,0.975)
A14	(0.207,0.446,0.946)	(2.013,1.401,0.946)
A15	(0.182,0.412,0.905)	(1.776,1.293,0.905)
A16	(0.182,0.412,0.905)	(1.776,1.293,0.905)
A17	(0.234,0.486,0.993)	(2.285,1.526,0.993)
A18	(0.236,0.488,0.996)	(2.298,1.533,0.996)
A19	(0.202,0.440,0.941)	(1.968,1.381,0.940)
A20	(0.234,0.486,0.993)	(2.285,1.526,0.993)

Table 12
The DOBI values of alternatives.

Alternatives	γ_i			Score Values (Θ_i)	Rank
A1	0.910	1.356	1.938	1.379	11
A2	0.913	1.371	1.976	1.396	8
A3	0.791	1.220	1.790	1.244	17
A4	0.856	1.283	1.831	1.303	12
A5	1.079	1.493	1.981	1.505	4
A6	0.844	1.268	1.814	1.288	13
A7	0.938	1.369	1.915	1.389	10
A8	1.083	1.498	1.986	1.510	2
A9	1.085	1.503	2.000	1.516	1
A10	1.013	1.419	1.907	1.433	7
A11	0.809	1.243	1.820	1.267	15
A12	0.805	1.236	1.804	1.259	16
A13	0.957	1.379	1.902	1.395	9
A14	0.831	1.251	1.788	1.271	14
A15	0.647	1.066	1.638	1.091	19
A16	0.647	1.066	1.638	1.091	20
A17	1.071	1.483	1.971	1.496	5
A18	1.083	1.498	1.986	1.510	3
A19	0.795	1.216	1.769	1.238	18
A20	1.071	1.483	1.971	1.496	6

criterion was not on the list before. Finally, the updated list of criteria was evaluated by the experts, and the list of criteria was finalized.

Finally, the researchers, taking experts' opinions, identified autonomous buses produced by bus companies to be included in the bus fleets of urban transportation authorities, apart from autonomous buses still in prototype form, and included them as options in the evaluation process. Then, online meetings were held with the experts, and A group of experts denoted as $\mathbb{Z}_l(l=1,2,...,6)$ is tasked with assessing twenty alternatives $\mathbb{R}_i(i=1,...,20)$ across thirty three criteria $\mathbb{C}_j(j=1,2,...,33)$. The linguistic terms scale and their corresponding values are outlined in Table 1 to solicit the viewpoints of the experts.

Table 2 provides a comprehensive profile of the experts involved in the fuzzy multi-criteria decision-making study. It includes key demographic and professional details such as age, country, educational background, current position, and reputation. This data is crucial for understanding the diverse perspectives and expertise that have informed the study's outcomes.

Table 3 introduces the alternatives considered in the fuzzy multi-criteria decision-making study. Each alternative is described in detail, highlighting its key attributes and relevant characteristics. This information is essential for evaluating the alternatives' performance and understanding the rationale behind their selection.

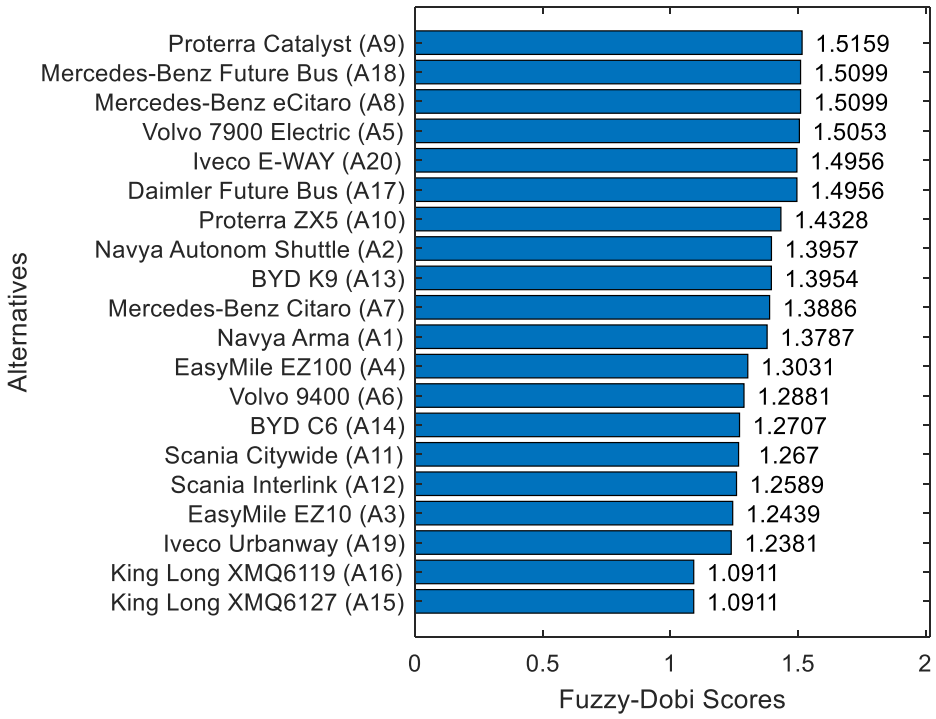


Fig. 3. Alternative evaluation DOBI scores in descending order.

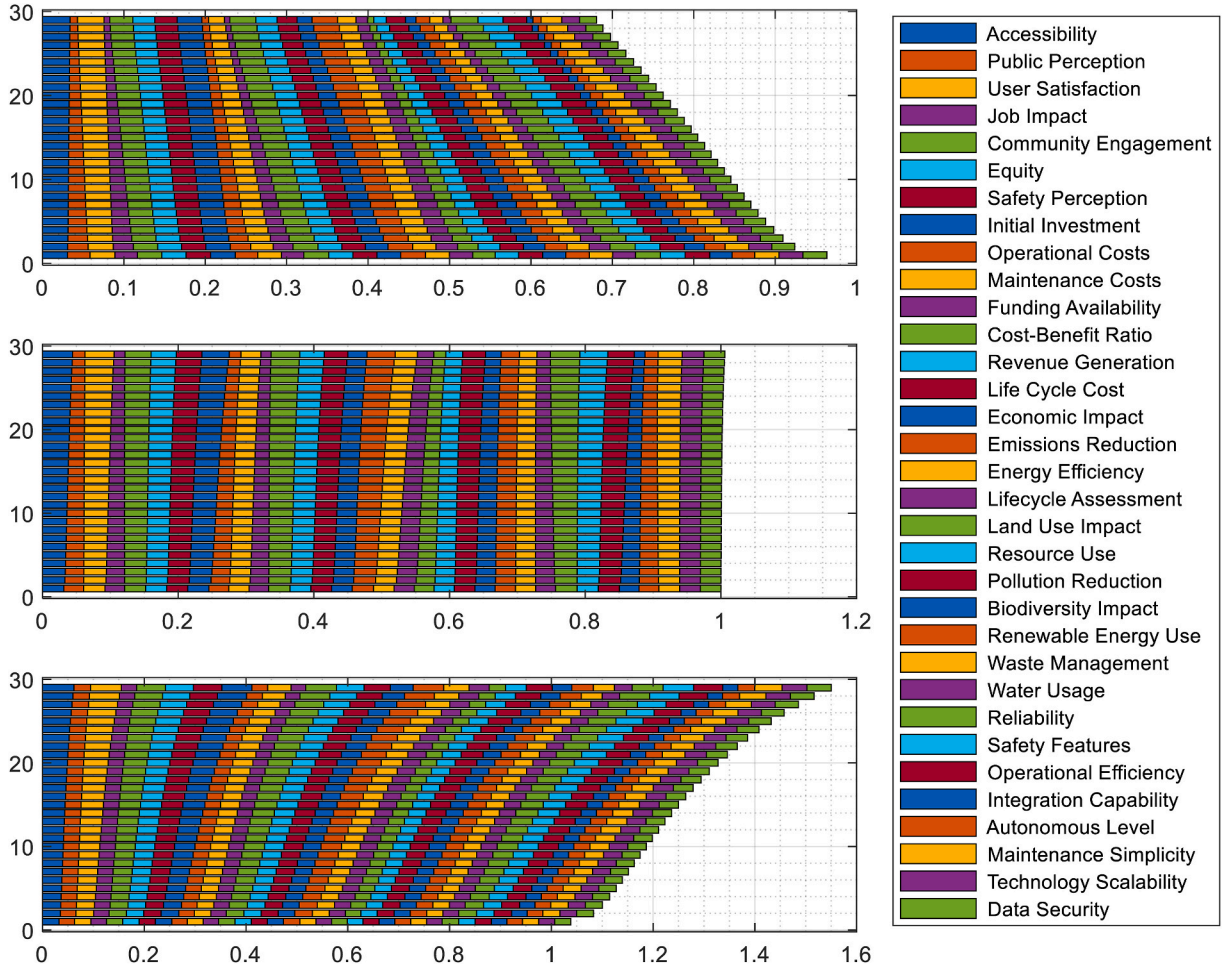


Fig. 4. The influence of AAIP on the change of weight coefficients of criteria.

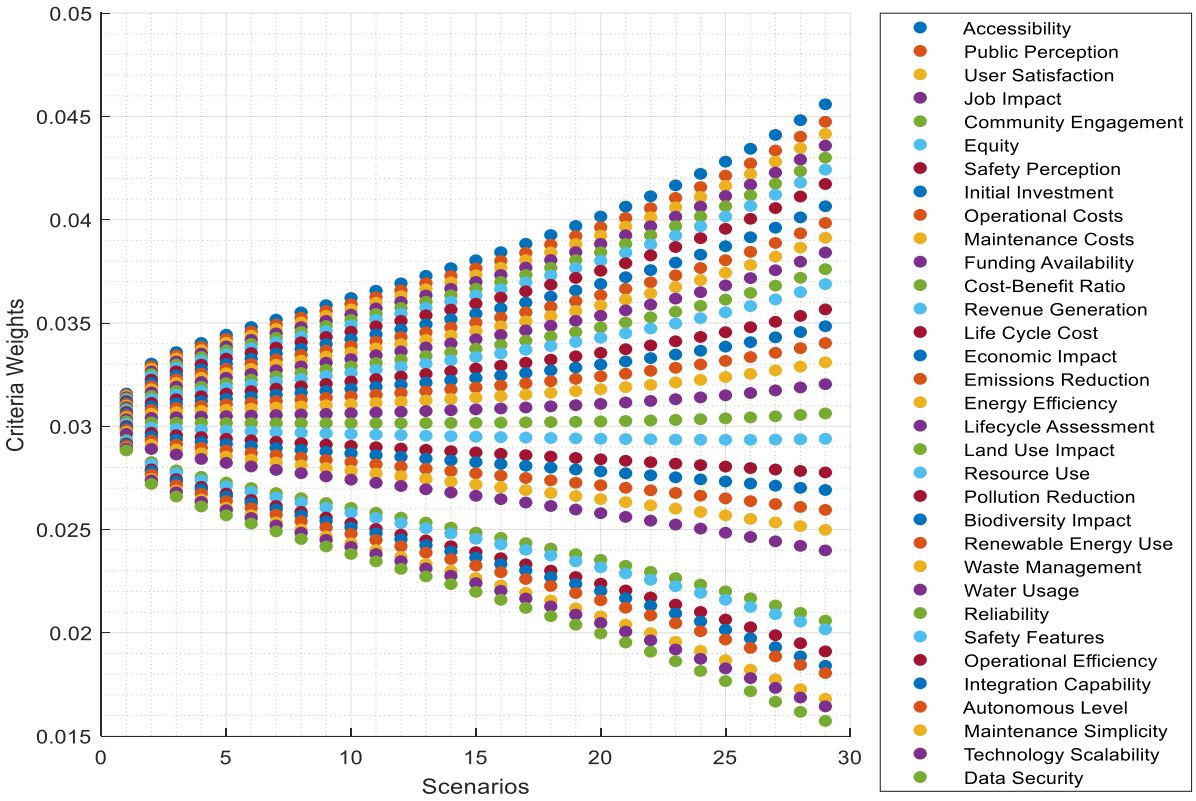


Fig. 5. Scatter diagram of criteria weights vs AAIP.

4.1. Determination of weight coefficients using the Fuzzy Aczel-Alsina LMAW methodology

The study involved six experts who expressed their preferences on the importance of the criteria via a questionnaire. Consequently, thirty-three criteria were identified and categorized into four clusters, as detailed in Table 4.

Subsequently, the methodology of fuzzy Aczel-Alsina LMAW is employed to define the weighting coefficients of the criteria.

Step 1. The experts articulated their preferences through the fuzzy priority vector utilizing the fuzzy scale provided in Table 1. The significance of the criteria is captured through the priority vector of the criteria, as shown in Table 5

Steps 2 and 3: Applying condition (2) and defining the relationship vector, the value of AAIP $\varepsilon = (0.4, 0.5, 0.6)$ was adopted. Finally, AAIP was used to determine the ratio vector using Equation (3). For example, decision maker 1 evaluated the criteria SO1 by using Extremely High (8.5, 9.5, 10) degree. Since AAIP is determined as ε=(0.4,0.5,0.6), by using Equation (3) the following calculations are performed:

$$\widetilde{\omega}_{C_1}^{p} = \left(\frac{8.5}{0.6}, \frac{9.5}{0.5}, \frac{10}{0.4}\right) = (14.17, 19.0, 25.0)$$

Similar calculations are performed for other evaluations and the results are presented in Table 6.

Using Equations (4) and (5), the vectors of weighting coefficients are defined within expert groups are presented in Table 7.

Aggregated elements of the fuzzy vector from Table 7 are defined by applying Equation (14)where it is adopted that all experts have the same significiance; i.e. $w_p = \frac{1}{6} = 0.167, 1 \le p \le h$. Furthermore, since the condition is that $\varphi > 0$, the value of the stabilization parameter of the Aczel-Alsina function $\varphi = 1$ is adopted. In the following section, the aggregation of the weighting coefficient criterion C1 is presented:

$$w_{C1} = \left(w_{C1}^{l}, w_{C1}^{m}, w_{C1}^{h}\right)$$

$$= \begin{cases} w_{C1}^{l} = 0.1636\left(1 - e^{-\left(\frac{1}{6}\left((-\ln(1-0.177))^{1}+(-\ln(1-0.165))^{1}+\cdots+(-\ln(1-0.166))^{1}\right)\right)^{\frac{1}{1}}}\right) = 0.0273 \\ w_{C1}^{m} = 0.2085\left(1 - e^{-\left(\frac{1}{6}\left((-ln(1-0.18))^{1}+(-ln(1-0.164))^{1}+\cdots+(-ln(1-0.164))^{1}\right)\right)^{\frac{1}{1}}}\right) = 0.0398 \\ w_{C1}^{u} = 0.2671\left(1 - e^{-\left(\frac{1}{6}\left((-\ln(1-0.187))^{1}+(-\ln(1-0.161))^{1}+\cdots+(-\ln(1-0.162))^{1}\right)\right)^{\frac{1}{1}}}\right) = 0.0445 \end{cases}$$

Table 13
Correlation coefficients of the criteria with AAIP.

Criteria	Correlation with AAIP	Criteria	Correlation with AAIP
Reliability (TO1)	0.9940	Integration Capability (TO4)	−0.9968
Accessibility (SO1)	0.9938	Funding Availability (EC4)	−0.9967
User Satisfaction (SO3)	0.9938	Operational Costs (EC2)	−0.9967
Cost-Benefit Ratio (EC5)	0.9936	Job Impact (SO4)	−0.9966
Emissions Reduction (EN1)	0.9934	Public Perception (SO2)	−0.9966
Operational Efficiency (TO3)	0.9932	Autonomous Level (TO5)	−0.9965
Safety Features (TO2)	0.9932	Water Usage (EN10)	−0.9963
Community Engagement (SO5)	0.9929	Land Use Impact (EN4)	−0.9963
Initial Investment (EC1)	0.9928	Biodiversity Impact (EN7)	−0.9963
Safety Perception (SO7)	0.9925	Lifecycle Assessment (EN3)	−0.9962
Equity (SO6)	0.9919	Resource Use (EN5)	−0.9961
Revenue Generation (EC6)	0.9910	Waste Management (EN9)	−0.9958
Life Cycle Cost (EC7)	0.9903	Renewable Energy Use (EN8)	−0.9951
Maintenance Simplicity (TO6)	0.9901	Economic Impact (EC8)	−0.9353
Pollution Reduction (EN6)	0.9883		
Energy Efficiency (EN2)	0.9851		
Technology Scalability (TO7)	0.9796		
Data Security (TO8)	0.9621		
Maintenance Costs (EC3)	0.7818		

The fuzzy DOBI procedure was applied for different AAIP values, and the resulting scores of the alternatives were examined. Fig. 6 presents a graphical representation of the scores obtained by each alternative as the AAIP values change. The findings indicate that for alternatives A1, A2, A6, A11, A12, A14, A15, A16, and A19, an increase in AAIP values corresponds to a decrease in their fuzzy DOBI scores. In contrast, a positive correlation was identified between AAIP values and fuzzy DOBI scores for alternatives A8, A9, A10, A17, A18, and A20. For the remaining alternatives, A3, A4, A5, A7, and A13, a non-linear relationship between AAIP values and their scores was observed.

$= (0.0273, 0.0398, 0.0445)$

The remaining elements are calculated similarly and all of the results are presented in Table 8.

The fuzzy vector of weight coefficients is depicted in Fig. 2. The values in the figure are presented in descending order. Accordingly, the criteria of SO1-Accessibility (0.0351), TO1-Reliability (0.0349), and SO3-User Satisfaction (0.0347) emerge as the most significant criteria. Conversely, SO4-Job Impact (0.026), TO4-Integration Capability (0.0259), and EC4-Funding Capability (0.0257) are identified as possessing the lowest weights.

4.2. Results of the fuzzy DOBI method

This section outlines the methodology employed to rank the alternatives and presents the corresponding results. Initially, twenty alternatives were evaluated across thirty-three criteria using the linguistic terms specified in Table 1. The linguistic evaluations for each alternative are detailed in Tables A1–A6 (Appendix) for each decision maker.

Step 1: The fuzzy initial decision matrix is first created using the linguistic evaluations in Table above using Eq. (7). The aggregated initial matrix for each alternative is reported in Table 9.

Step 3: The weighted significance $\chi_i^{(1)\theta_1,\theta_2,\varpi}$, and $\chi_i^{(2)\theta_1,\theta_2,\varpi}$, of alternatives are calculated using the values in Table above and Table (fuzzy weight vector with destrandardization) with the help of Eqs. (10) and (11). The values are presented in Table 11.

Step 4: The integrated value (γ_i) of the DOBI functions is obtained using Eq. (12) and Table 11. The DOBI values of alternatives are presented in Table 12.
Step 5: The DOBI values are defuzzified by Eq (13) and ϒi values in Table 11 to get score values (Θ_i) of alternatives. The score values is reported in Table 12.
Step 6: The alternatives are ranked the basis of decreasing value of score values (Θ_i). This ranking can be seen in Table 12 and Fig. 3.

The defuzzified DOBI scores are presented as a bar chart in Fig. 3 below. It is clear from the figure that the best alternative is Proterra Catalyst (A9), Mercedes-Benz Future Bus (A18), Mercedes-Benz eCitaro (A8). The lowest scores belong to the King Long Vehicles (A15 and A16).

5. Sensitivity analysis

In this study, sensitivity analysis has been examined under three subsections: (1) sensitivity analysis related to the parameters of the fuzzy LMAW technique used for determining criterion weights, and (2) sensitivity analysis concerning the parameters of the fuzzy DOBI technique employed in the calculation of alternative scores. Additionally, a Monte Carlo simulation has been conducted as the third subsection.

5.1. Parameters in fuzzy LMAW technique

Parameters of the fuzzy LMAW technique is investigated in two subsections (1) anti-ideal point (AAIP) and (2) stabilization parameter of Aczel-Alsina function (φ).

5.1.1. Simulation of change of absolute anti-ideal point (ε)

In this study, the value of absolute anti-ideal point $\varepsilon = (0.4, 0.5, 0.6)$ was arbitrarily adopted. The specified value is adopted based on condition (2). Since $\min_{1\le j\le 33, 1\le p\le 6}\left(\psi_{c_j}^{p}\right) = 3$ and condition (2) defines that $0 < \varepsilon < 3$, twenty-nine scenarios were formed in which that AAIP change was simulated. In the initial scenario, a value of $\varepsilon = 0.001$ was applied. For each subsequent scenario, the AAIP value was incrementally increased by 0.1. Corresponding to each new AAIP value, a distinct vector of criteria weighting coefficients was generated, as illustrated in Fig. 4.

The change in AAIP and the corresponding criterion values are illustrated as a scatter plot in Fig. 5. The horizontal axis displays the AAIP values in ascending order, while the vertical axis represents the criterion weights. The most striking observation from the graph is that the weights of some criteria increase as AAIP rises, whereas the weights of others decrease as AAIP increases.

The correlation coefficients for each criterion weight with AAIP values have been calculated and are presented in Table 13. In the table, criteria that have a positive or negative correlation with AAIP values are reported separately. The results indicate that the criteria of reliability and accessibility exhibit a high positive correlation with AAIP. In contrast, the criteria of integration capability and funding availability show a strong negative correlation with AAIP.

The impact of AAIP on the ranking changes of alternatives is depicted as a heatmap in Fig. 7. According to the figure, the rankings of alternatives A2, A3, A5, A7, A8, A12, A13, A16, A18, and A20 are notably affected. In contrast, the rankings of the remaining 10 alternatives

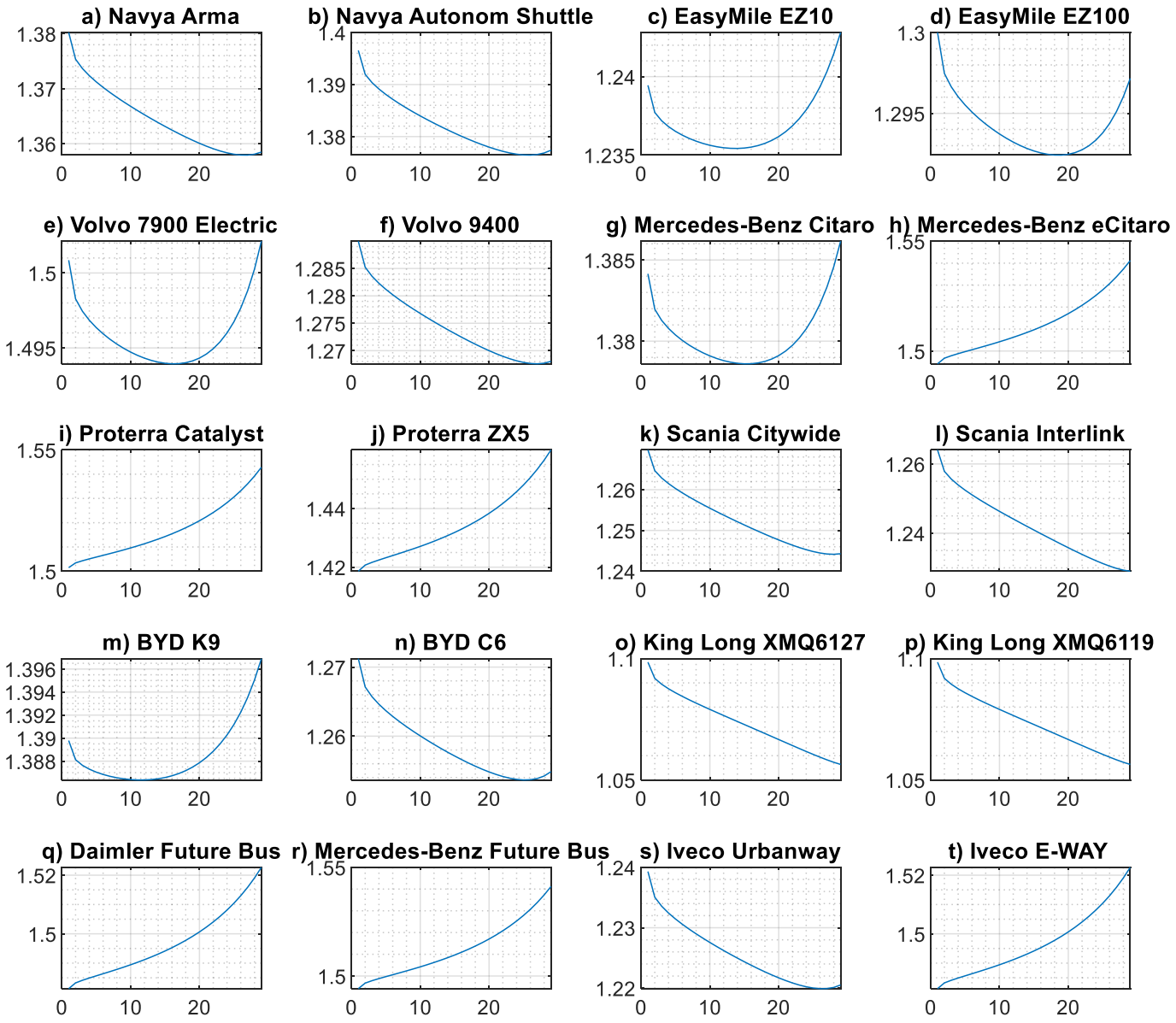


Fig. 6. The influence of AAIP on the change of scores of alternatives.

remain insensitive to changes in AAIP. One of the significant findings is that variations in AAIP do not cause substantial fluctuations in the rankings of the top three alternatives.

5.1.2. Simulation of the change in the stabilization Parameter of the Aczel-Alsina function

To define the initial solution, a stabilization parameter value of $\varphi = 1$ was chosen for the Aczel-Alsina function. Given that the condition requires $\varphi > 0$, the subsequent section explores the impact of different φ values on the initial solution. In the experiment detailed in this section, φ was varied within the interval $1 \leq \varphi \leq 100$. The first scenario utilized $\varphi = 1$, with each following scenario increasing φ by one unit.

Fig. 8 presents the distribution graph of criterion weights obtained under different scenarios. According to the values depicted in the figure, it is evident that, without exception, the weights of the criteria increase as the stabilization parameter of the Aczel-Alsina function rises. Additionally, it is observed that several criteria tend to cluster at higher stabilization values.

Fig. 9 illustrates the changes in the criterion functions of the alternatives across 100 scenarios. When the stabilization parameter takes values between 1 and 20, it exerts a diminishing effect on the fuzzy DOBI values of the alternatives. However, when the parameter ranges between 21 and 100, an increase in the scores of the alternatives is observed.

The impact of the stabilization parameter on the ranking of alternatives is presented in Fig. 10. According to the figure, a single change occurs between the rankings of A2 and A13. In the first 69 scenarios, the A2 alternative is ranked 8th, while the A13 alternative is ranked 9th. However, when the stabilization value reaches 70, the A2 alternative drops to 9th place, and the A13 alternative rises to 8th. No effect of the stabilization parameter is observed on the rankings of other alternatives.

5.2. Parameters of fuzzy DOBI technique

The upcoming section showcases a rigorous sensitivity analysis designed to validate the initial solution. This analysis simulates variations in the parameters θ_1, θ_2, ϖ, γ, and σ. The parameters θ_1,θ_2, ϖ and represent the stabilization factors within the fuzzy Dombi-Bonferroni function, with possible values ranging from [1,∞). Meanwhile, γ and σ are integral to the aggregation function of the fuzzy DOBI model, where

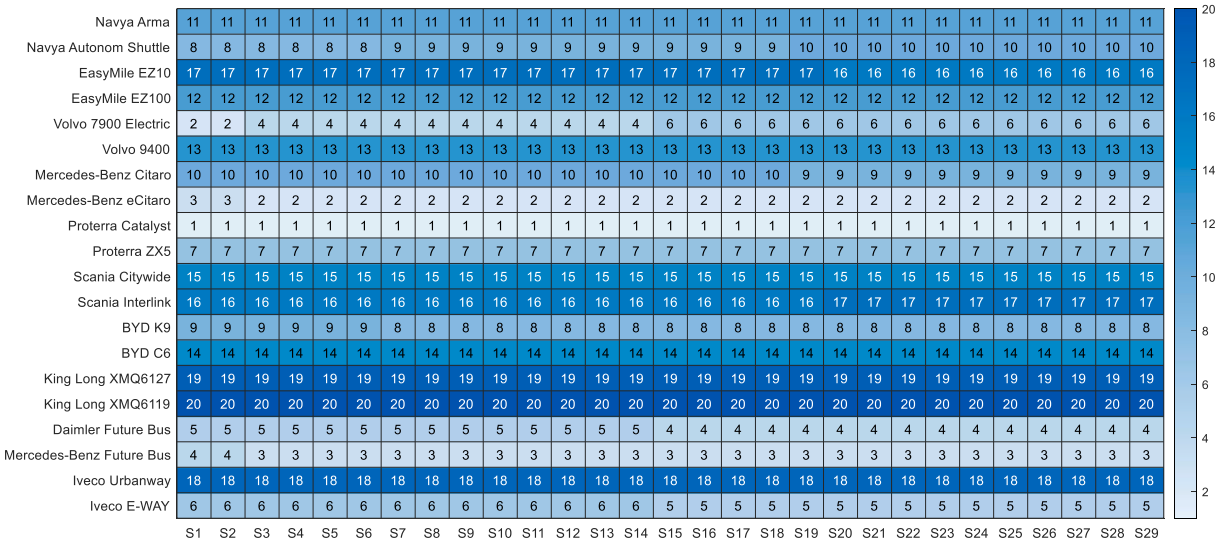


Fig. 7. The influence of AAIP on the change of ranking of alternatives.

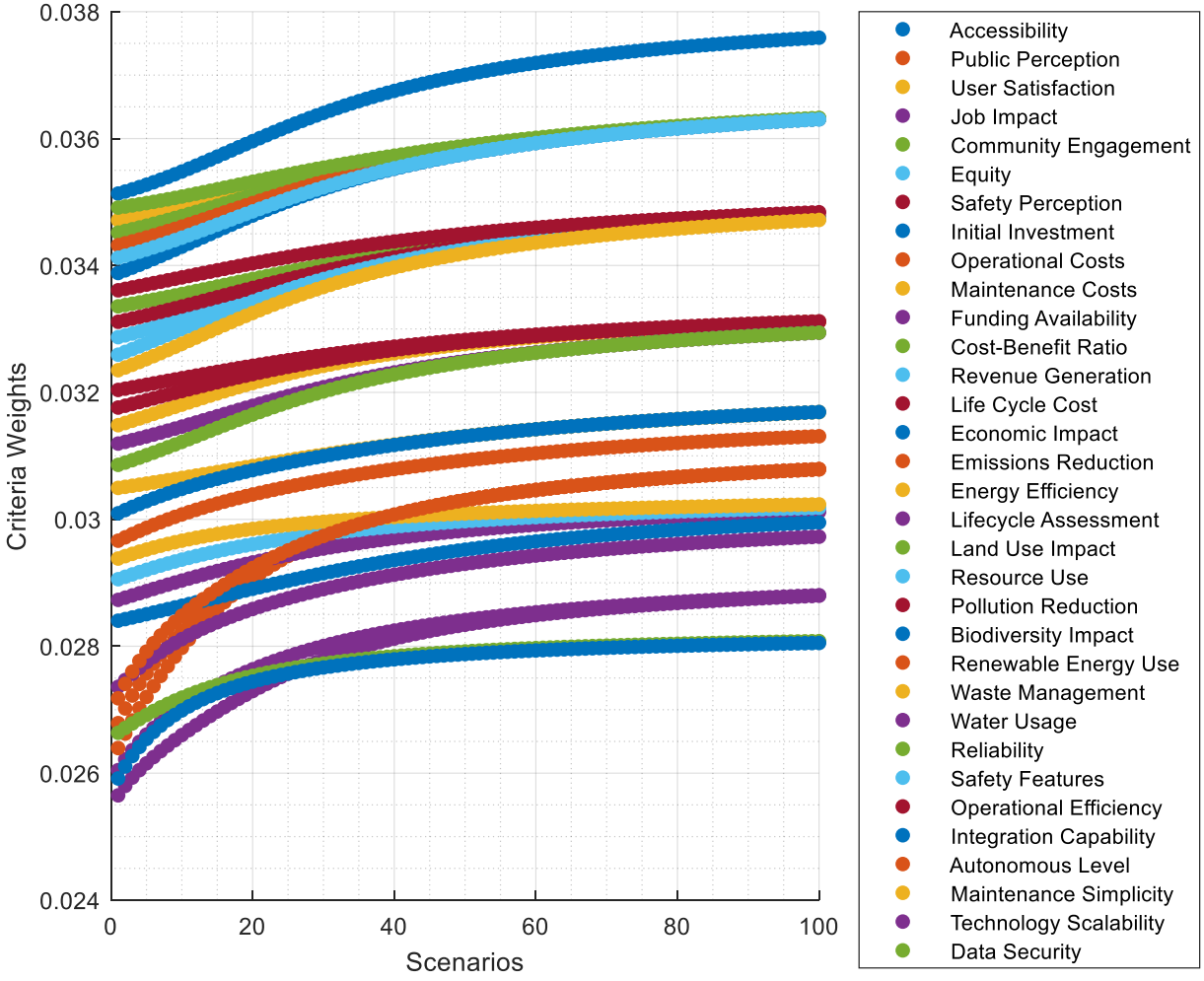


Fig. 8. Scatter diagram of criteria weights under different scenarios.

γ spans the interval [1,∞) and σ ranges between [0,1].

5.2.1. Variation of stabilization parameters $\theta_1, \theta_2,$ and ϖ

In defining the initial solution, the parameters ($\theta_1 = \theta_2 = \varpi = 1$) were set arbitrarily. The subsequent analysis delves into how variations in these parameters influence the initial solution, conducted over two distinct phases. The first phase explores the impact of adjusting θ_1 and θ_2 within the range [1100], with ϖ fixed at one. The second phase simulates changes in ϖ within the same interval, while keeping θ_1 and θ_2 constant at one. Fig. 11 illustrates the relationship between the

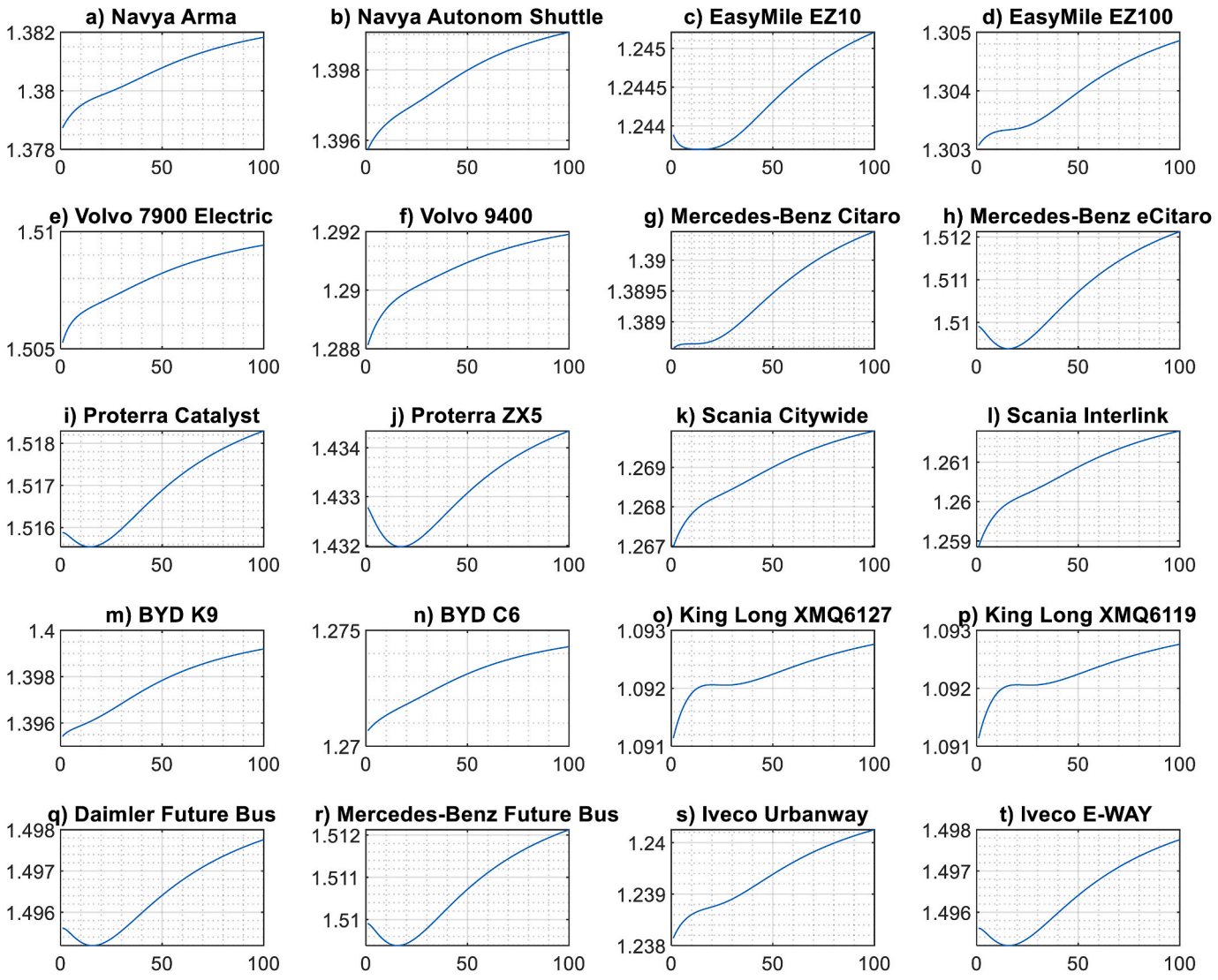


Fig. 9. The influence of the parameter φ on the change of criterion functions of alternatives.

integrated fuzzy DOBI alternative functions and the variations in θ_1 and θ_2.

Fig. 11 presents the values of the integrated fuzzy DOBI alternative functions across one hundred scenarios, where changes in parameters θ1 and θ2 were simulated. In the initial scenario, the values were set to θ1 = θ2 = 1. With each subsequent scenario, the values of θ1 and θ2 were incrementally increased by one, while the parameter ϖ remained constant at a value of one. The results in Fig. 2 demonstrate that variations in the importance of parameters θ1 and θ2 influence the changes in the integrated fuzzy DOBI functions of the alternatives.

The study investigates whether variations in the variables θ_1 and θ_2 influence the ranking of alternatives. In Fig. 12, while θ_1 is held constant at 1, the impact of changes in θ_2 on the ranking of alternatives is observed. According to the figure, no alterations occur in the ranking. A similar pattern is evident for other components as well. Regardless of the specific values that θ assumes within the range [1100], no changes occur in the ranking of alternatives, even though their scores fluctuate.

In the following part, a similar experiment was conducted in which fixed values of the parameters $\theta_1 = \theta_2 = 1$ were considered. At the same time, the change of parameter ϖ in the interval $1 \leq \leq 100$ was simulated. Fig. 13 includes the changes in the DOBI scores of the alternatives when the parameter ϖ increases one-by-one from 1 to 100. Fig. 13 indicates the changes in the ranking of alternatives under different scenarios.

The results from Figs. 13 and 14 show that changes in the parameter ϖ affect the changes in the integrated fuzzy DOBI alternative functions. However, these changes do not affect the change in the rankings of the alternatives.

5.2.2. *Variation of stabilization parameters γ and σ*

The initial solution is defined for the parameter values γ = 1 and σ = 0.5. In the following section, we analyze how varying these parameters affects the integrated DOBI functions of alternatives. The analysis was repeated with 100 different values, with γ ranging from [1100] and σ from [0,1]. The results are presented in Fig. 15. As illustrated, an increase in the γ value consistently reduces the fuzzy DOBI scores for all alternatives, while the σ value does not significantly impact the alternative scores.

5.3. *Monte Carlo simulation*

To determine whether the weight sets and alternative scores exhibit extreme values, a Monte Carlo simulation was employed. In this process, random weight sets were generated, and the MCDM technique was reapplied using these sets. The resulting scores were then recorded and analyzed.

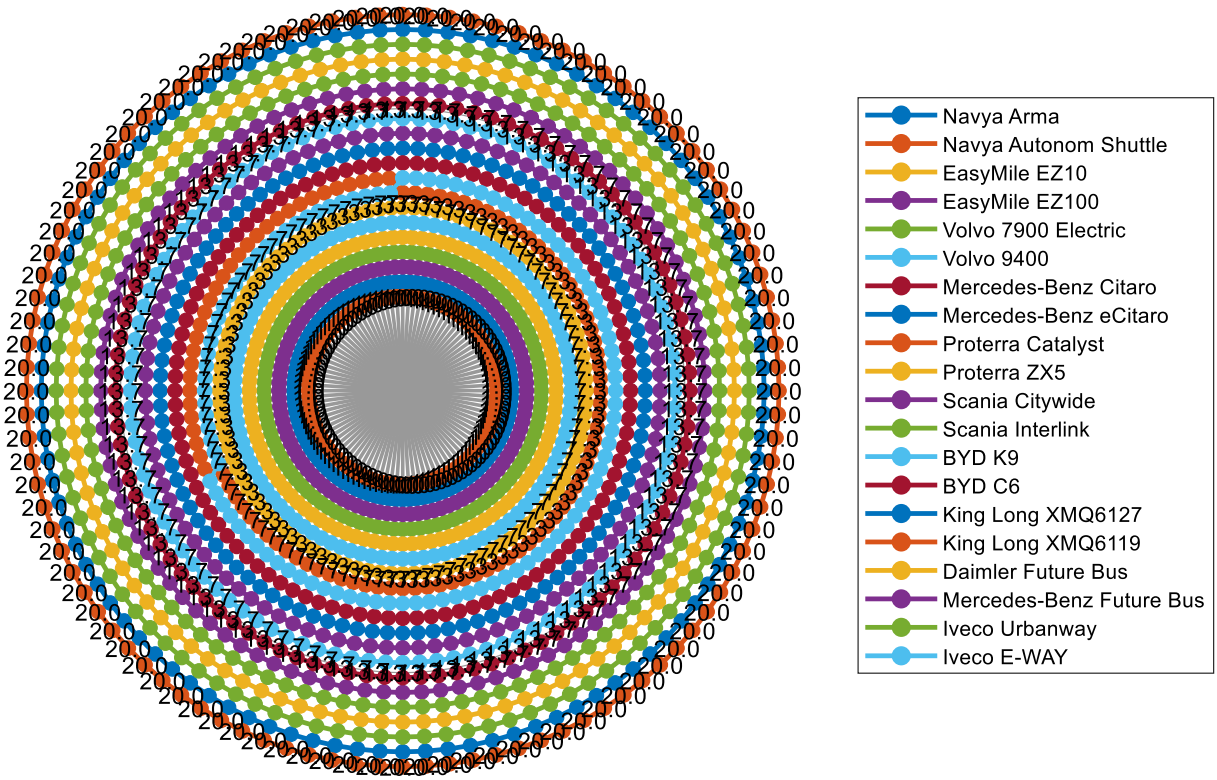


Fig. 10. Comparative presentation of the change in φ in the ranking of alternatives.

Uniformly distributed random numbers within the [0,1] interval were generated. Given that the study involves 33 criteria, 33 random numbers were produced. Each of these random numbers was then divided by the sum of all 33 numbers, resulting in a weight set composed of random values within the [0,1] range, with the sum of the weights equal to 1. This procedure was repeated 1000 times, generating 1000 distinct weight sets.

The implications of this extensive simulation are significant. By applying the MCDM technique across a broad spectrum of weight sets, the robustness of the alternative scores can be assessed. Specifically, the analysis can reveal whether certain alternatives consistently perform well or poorly across various weight configurations, or if the scores are highly sensitive to changes in the weight sets. This approach ensures a thorough evaluation of the stability and reliability of the decision-making process, providing deeper insights into the potential variability and extremities in the alternative rankings.

The descriptive profile for the randomly generated weight sets, created using uniformly distributed random weights, is presented in Table 14. The minimum weight is approximately 0, while the maximum weight is around 0.08. It is expected that the average weight is 0.03, which corresponds to 1/33.

Fig. 16 presents a summary of the results for the criteria weights obtained through the Monte Carlo simulation. The figure includes the following values for each criterion.

- Sim: This refers to the histogram distribution of the weights generated using random values.
- F-LMAW: The F-LMAW scores, as detailed in the study and provided in Table 3, are marked here.
- KDE: Kernel Density Estimation, which represents the curve calculated by using the histogram distribution of weights generated with random values.
- LEFT: The area under the KDE curve to the left of the F-LMAW score.
- RIGHT: The area under the KDE curve to the right of the F-LMAW score.

It is generally expected that the area under the KDE curve to the right or left of the F-LMAW score will be relatively large. A smaller area suggests that the F-LMAW score is an outlier, which could complicate the interpretation of the results.

The implementation described here is crucial for evaluating the robustness and reliability of the F-LMAW scores. By comparing the F-LMAW scores to the KDE curve, researchers can assess whether the scores fall within a typical range or if they are potential outliers. If the area under the KDE curve near the F-LMAW score is small, it indicates that the score is unusual compared to what would be expected from random distributions. This could suggest that the score is not representative of the underlying data or that it is sensitive to particular conditions or assumptions in the model.

Understanding this implementation helps ensure that the conclusions drawn from the simulation are valid and that the F-LMAW scores are reliable indicators. If the scores are outliers, additional scrutiny or alternative approaches may be necessary to ensure an accurate interpretation of the results. This step is fundamental in ensuring that the study's findings are both robust and scientifically sound.

The weight assigned to the Operational Costs criterion (EC2) is 0.40, making it the lowest among the criteria under consideration. From this observation, it can be concluded that none of the weights derived using the F-LMAW technique can be regarded as outliers.

Using a set of random weights (1000 different sets), the Fuzzy DOBI scores were recalculated for each alternative. The results of the Monte Carlo simulation are summarized in Fig. 17. The figure presents the following information for each alternative.

- Sim: Histogram distribution of the alternative's scores (1000 scores)
- F-DOBI: Fuzzy DOBI scores, as detailed in the study (Table 3)
- KDE: Kernel density estimation curve calculated from the histogram distribution of the alternative's scores
- LEFT: The area under the KDE curve to the left of the F-DOBI score
- RIGHT: The area under the KDE curve to the right of the F-DOBI score

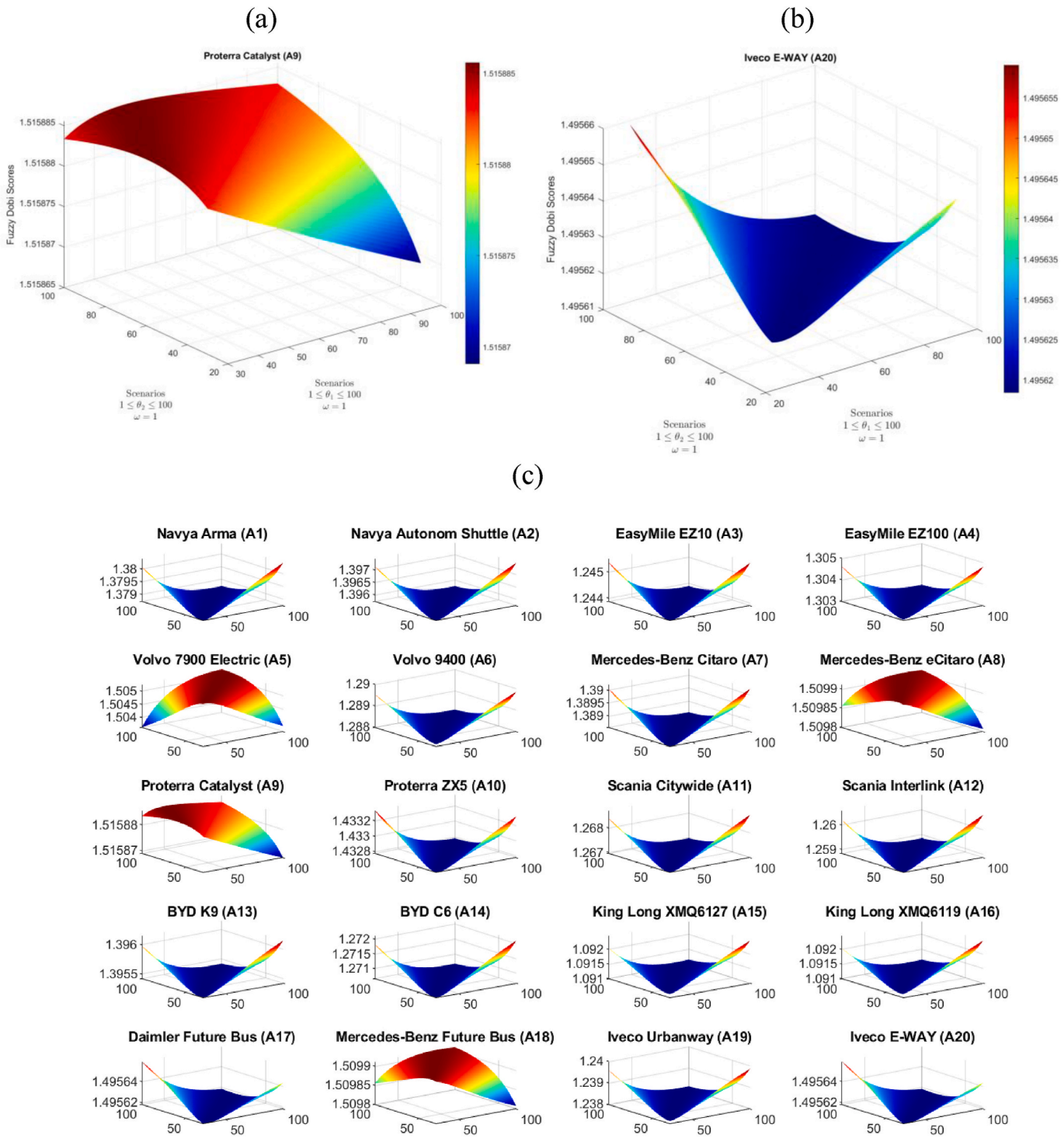


Fig. 11. Parameters θ_1 and θ_2: Changing the integrated DOBI functions of alternatives (a) Alternative 9, (b) Alternative 20, and (c) All alternatives.

The interpretations provided for the criteria are also applicable to the scores of the alternatives. The lowest area value, 0.03, corresponds to the score of the Proterra Catalyst alternative. This suggests that the calculated Fuzzy DOBI scores for the alternatives are not outliers at a 1 % confidence level.

After calculating the scores using the weight sets generated by the Monte Carlo simulation, ranking values for each weight set were also determined. Fig. 18 presents a heat map illustrating the percentage of simulations in which the alternative in the row ranks higher than the alternative in the column. The ranking in the figure is arranged in descending order based on the row totals.

According to Fig. 18, for instance, the Mercedes-Benz eCitaro ranks higher than the Proterra Catalyst in 33.3 % of the weight sets. In the remaining weight sets, the Proterra Catalyst ranks higher than the Mercedes-Benz eCitaro. Similarly, the Mercedes-Benz eCitaro ranks above the Volvo 7900 Electric in 40.5 % of the weight sets. Additionally, the Mercedes-Benz eCitaro consistently ranks higher than the Mercedes-Benz Future Bus, Daimler Future Bus, IVECO E-WAY, and Proterra ZX5 alternatives across all weight sets. In other words, regardless of the weight set used, the Mercedes-Benz eCitaro consistently outperforms

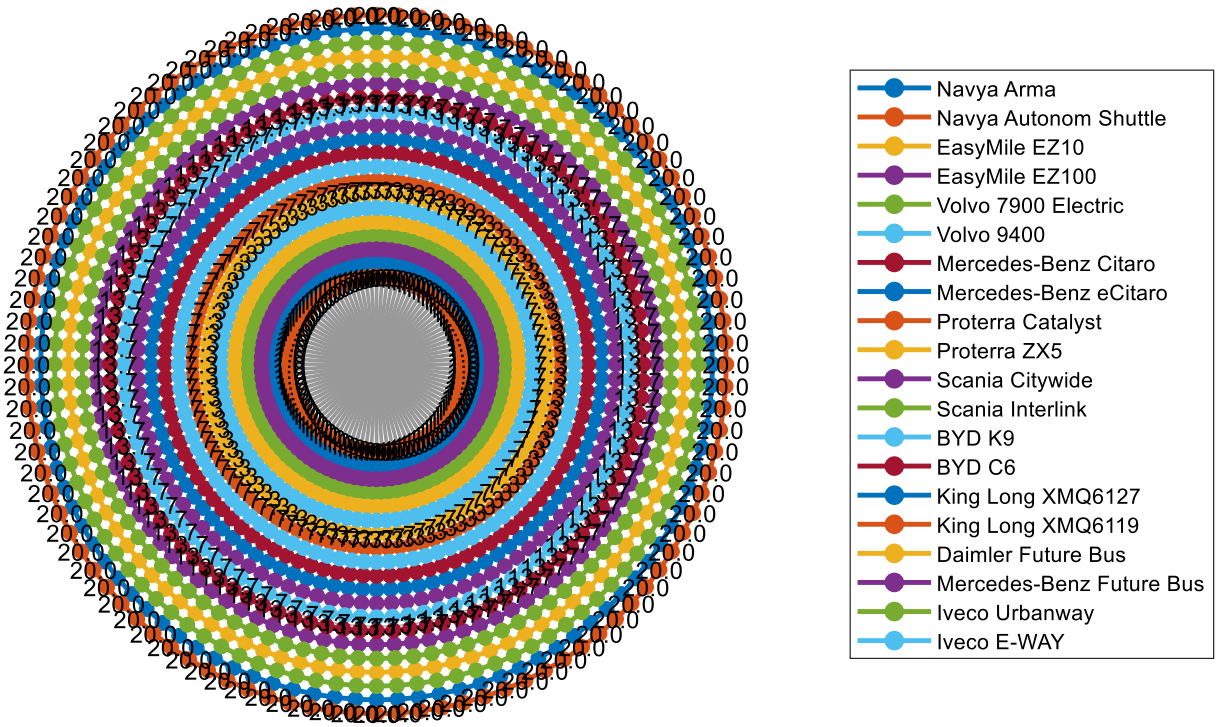


Fig. 12. Ranks for alternatives over different θ_2 values ($\theta_1 = 1$).

these alternatives, indicating an absolute advantage.

To better understand the relative superiority of the alternatives, three-dimensional distribution graphs have been created (Fig. 19). In these graphs, the horizontal axes represent the two criteria with the highest importance: Accessibility (SO1) and Reliability (TO1). The vertical axis displays the Fuzzy DOBI scores of the alternatives. In section (a) of the figure, the scores of the alternatives with the highest and lowest values in the main ranking, specifically alternative 9 (Proterra Catalyst) and alternative A15 (King Long XMQ6127), are compared. As shown in the figure, there is a discernible difference between the scores of the alternatives, and this difference is independent of the weighting set.

Section (b) of Fig. 19 presents the values of alternatives A1 (Navya Arma) and A13 (BYD K9), which are unable to establish superiority over each other in the Monte Carlo simulation. As illustrated in the figure, these alternatives do not exhibit a clear superiority over one another.

5.4. Comparison with other MCDM techniques

In this section, the outcomes obtained through the application of the Fuzzy DOBI technique within the scope of the present study are comparatively analyzed against the results derived from other established methodologies, including Fuzzy MACONT (Seikh and Chatterjee, 2025; Mishra et al., 2024), Fuzzy TOPSIS (Biswas et al., 2025), and Fuzzy MARCOS (Tas et al., 2021). Table 15 shows the ranking outcomes for 20 autonomous and electric bus alternatives, each evaluated using four fuzzy MCDM methods—Fuzzy DOBI, Fuzzy MACONT, Fuzzy TOPSIS, and Fuzzy MARCOS—under the same fuzzy LMBAW weighting scheme. Although the identical weights constrain one source of variation, the different computational approaches to aggregation and distance measurement can still yield divergent rankings, which is precisely what we observe.

Consistently High-Performing Alternatives: Proterra Catalyst (A9) emerges as a top choice under Fuzzy DOBI and TOPSIS (rank 1), but is ranked 2nd by MACONT and 6th by MARCOS. Volvo 7900 Electric (A5) also stands out. MACONT and MARCOS both assign it rank 1, whereas DOBI places it 4th, and TOPSIS ranks it 6th. These variations indicate that while A9 and A5 are widely considered strong alternatives, the precise ranking depends on how each method calculates the distance to the ideal and anti-ideal references.

Lowest-Ranked Alternatives: King Long XMQ6127 (A15) and King Long XMQ6119 (A16) consistently occupy ranks 19 and 20, respectively, across all methods, suggesting near-unanimous agreement that they are less competitive under the criteria and weights employed.

Mid-Range and Moderate Divergence: Alternatives such as Navya Arma (A1) and Volvo 9400 (A6) display fairly consistent mid-range positions across DOBI, MACONT, and TOPSIS, yet Fuzzy MARCOS sometimes shift them slightly. Navya Autonom Shuttle (A2) is another example: it ranks 8th or 10th for three methods but drops to 18th under MARCOS, highlighting that MARCOS' formulation for compromise solutions may give substantially different outcomes in certain scenarios.

The correlation matrix (Fig. 20) presented compares four multi-criteria decision-making (MCDM) techniques—DOBI, MACONT, TOPSIS, and MARCOS—under a fuzzy setting. The table shows high correlation coefficients among DOBI, MACONT, and TOPSIS (all exceeding 0.96). In contrast, MARCOS exhibits moderate-to-high correlations with the other three methods (ranging from approximately 0.85 to 0.87). While these values still indicate a relatively strong association, they are notably lower than the nearly perfect correlations observed among DOBI, MACONT, and TOPSIS.

High Consistency Among DOBI, MACONT, and TOPSIS: The correlation coefficients between these three methods are extremely close to 1 ($\geq$0.96). This suggests that despite employing different computational procedures, DOBI, MACONT, and TOPSIS likely prioritize alternatives in a very similar manner within the given fuzzy data framework. In many MCDM contexts, high correlation among methods may reflect strong consistency in the underlying preference structure or similar handling of criteria weights and normalization processes.

Moderately Lower Correlation with MARCOS: In contrast, MARCOS exhibits correlation values in the 0.85–0.87 range with the other methods. These coefficients are still indicative of a positive and relatively high association, but they reveal that MARCOS' final rankings or preference scores deviate more noticeably compared to the other three. Conceptually, MARCOS' approach to constructing a "compromise

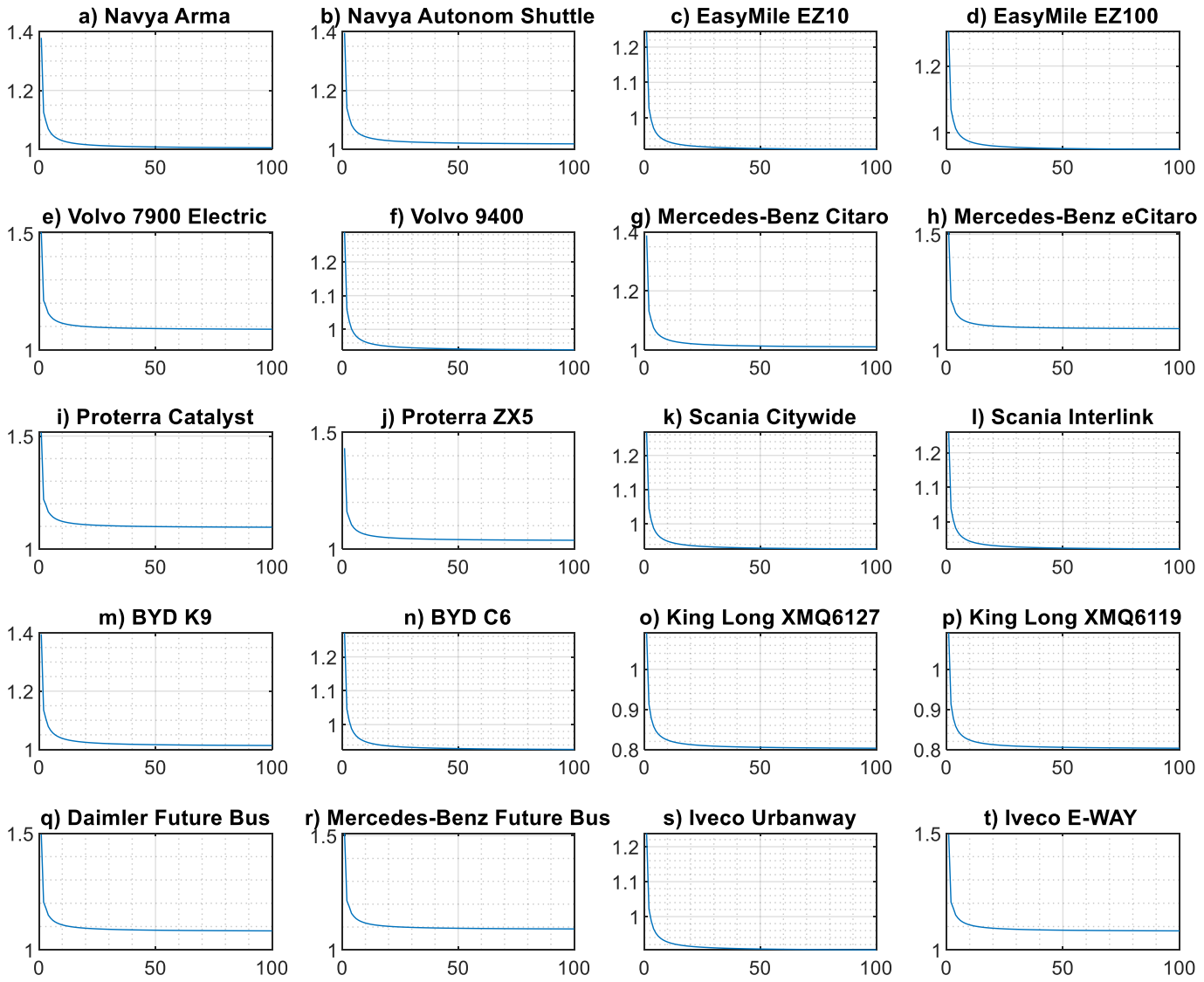


Fig. 13. The influence of the parameter ϖ on the change of criterion functions of alternatives.

solution" and handling the distance to ideal/anti-ideal points might introduce slight variations in how it aggregates fuzzy criteria, hence leading to a lower degree of similarity.

6. Results and discussion

The rapid development of autonomous (driverless) vehicle technology offers unprecedented opportunities to address the problems faced in urban transportation and to produce rational solutions. In particular, driverless buses have critical advantages. They can potentially reduce externalities such as accidents caused by driver errors, optimizing energy consumption and resource use, and depending on their environmental performance, safety, efficiency, and sustainability perspective. However, users' perceptions of safety have yet to develop at a level that will increase social acceptance, and technological maturity is not at a sufficient level are essential obstacles to adopting autonomous buses. This review provides reassurance of a comprehensive evaluation, presenting an integrated fuzzy decision-making model to evaluate 20 different accessible autonomous buses within the framework of four primary and 33 sub-sustainability criteria and to analyze their integration into urban transportation systems. The research aims to fill critical research gaps in the literature by examining autonomous buses from a sustainability perspective with their technical, social, economic, environmental, and operational dimensions.

The findings of this study are of paramount importance, emphasizing the need to integrate user-oriented and reliable autonomous vehicles into urban transportation systems to achieve sustainability goals. The study also reveals the critical effects of advanced decision-making procedures and their role in evaluating innovative driverless bus technologies. The results highlight the Proterra Catalyst (A9) as the autonomous bus with the highest sustainability performance, followed by the Mercedes-Benz Future Bus (A18) and the Mercedes-Benz eCitaro (A8). Conversely, King Long (A15 and A16) was found to have the lowest sustainability performance.

When the effects of sustainability criteria on autonomous vehicle selection were examined, the most critical and essential criteria were determined as SO1-Accessibility (0.0351), TO1-Reliability (0.0349), and SO3-User Satisfaction (0.0347), respectively. These results underscore the practical implications of user-oriented approaches and more reliable autonomous vehicle systems in terms of safety for urban transportation to become more sustainable.

Robustness tests conducted through variations in criteria weighting and alternative ranking methods, comparisons with other MCDM techniques, and Monte Carlo simulations affirm the validity, reliability, and stability of the proposed decision-making procedure and its results. Accordingly, it proves the importance of using an advanced

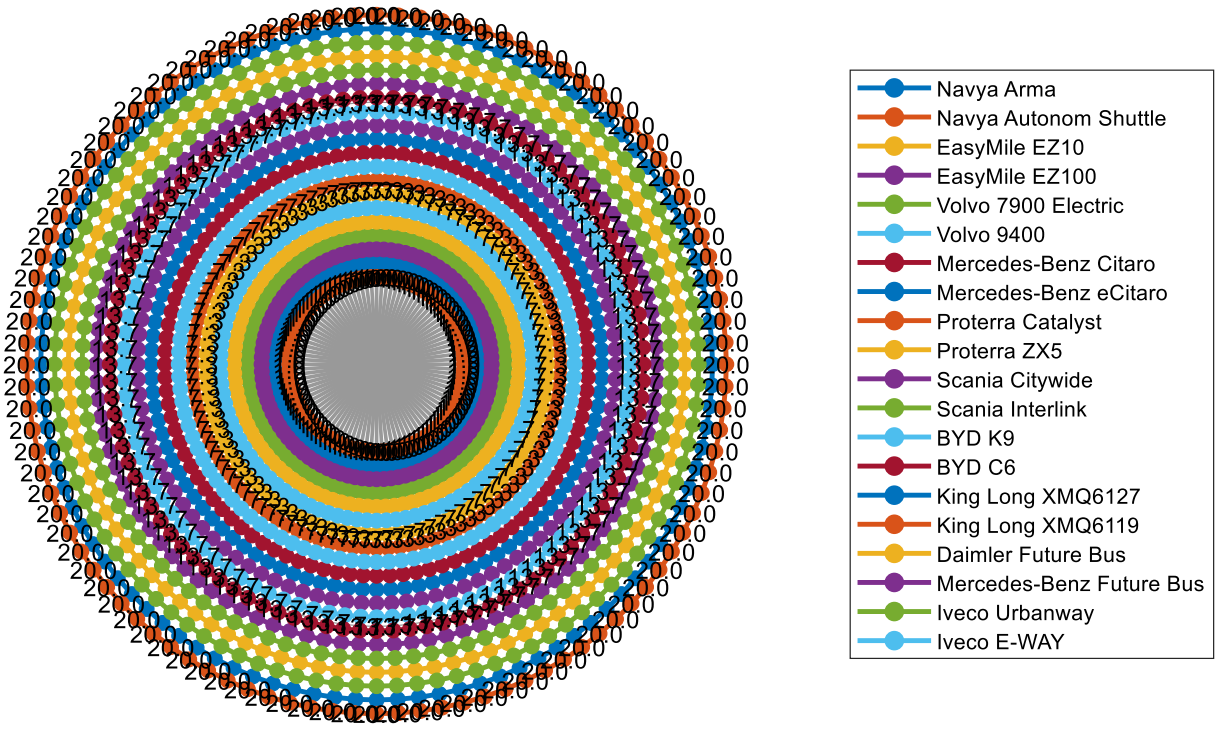


Fig. 14. Ranks of alternatives over different ϖ values ($\theta_1 = \theta_2 = 1$).

methodological framework for a balanced assessment of sustainability criteria in a complex and uncertain decision-making problem.

The results and findings obtained in the study provide critical and valuable insights with extremely critical policy and managerial implications for policymakers and decision-makers in the relevant industry. These insights pave the way for a future where urban transport is more efficient and user-friendly. In this context, decision-makers of urban transport authorities can configure autonomous bus fleets by considering the findings obtained in this study. By integrating autonomous buses into urban transportation systems, they can prioritize accessibility, reliability, and user satisfaction, ushering in a new era of public transport. At the same time, decision-makers can base their planning and implementation on this issue on the findings obtained in this study. It is a highly strategic and critical decision for practitioners to invest in deep learning-based traffic management systems with advanced distance sensing and artificial intelligence so that autonomous buses can be adapted to urban traffic in real time and integrated perfectly into urban transportation systems.

However, the study's findings suggest that policymakers should create inclusive legal regulations to eliminate barriers to autonomous bus technologies and legal regulations so that autonomous buses can be perfectly adapted to urban transportation systems. In addition, to manage the obstacles related to the start-up costs of autonomous buses, financial incentives, data discounts, subsidies, etc. Dissemination of practices and development of cooperation between public authorities, local authorities, and private enterprises are also among the study's recommendations. Competent urban transportation authorities must raise public awareness about autonomous buses and facilitate the adoption of these vehicles through various campaigns and informational applications. This responsibility is key to ensuring the successful integration of autonomous buses into urban transport systems.

Considering the contributions and innovations of this research, this study fills a significant and critical gap in the literature by examining the sustainability performances of autonomous buses in terms of economic, social, environmental, technical, and operational dimensions. The comprehensive nature of this study provides a potent, reliable, and innovative decision-making procedure that can address all dimensions of sustainability together and provide a comprehensive analysis. In this context, using a decision-making model consisting of the integration of fuzzy cluster-based Aczel-Alsina LMAW and DOBI methods provides a new methodological contribution to the literature on the comparison and evaluation of autonomous buses. This integrated decision-making model provides a comprehensive and balanced analysis by considering direct and indirect interactions and relationships between criteria and an easy-to-understand, applicable, and practical solution algorithm for decision-makers.

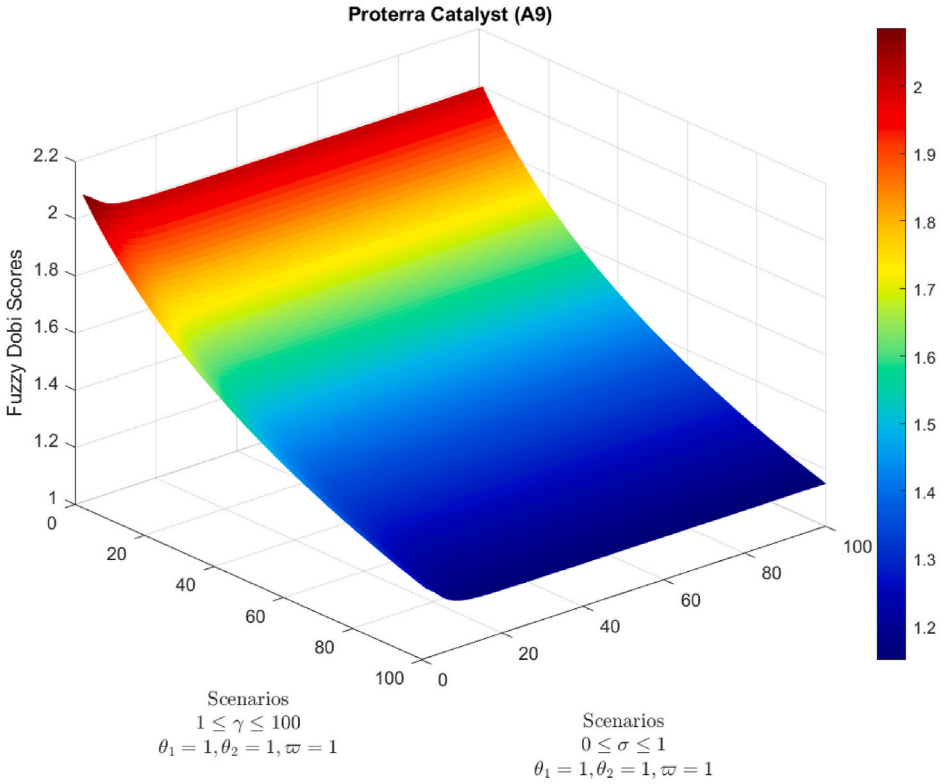
Furthermore, the inferences and findings presented in this study can serve as a roadmap for decision-makers. It guides them in the development and improvement processes that companies producing autonomous buses will undertake. The goal is to create vehicles that are not only technologically advanced but also user-friendly. By considering the results of this study and the insights it provides, autonomous vehicle manufacturers can design and produce vehicles that prioritize the needs and comfort of all stakeholders of urban transportation.

In conclusion, this review underlines the critical and transformative potential of autonomous (driverless) buses in addressing the problems encountered in urban transportation and achieving sustainability goals. The review bridges research and practice by examining autonomous bus alternatives with an innovative decision-making model within the framework of sustainability criteria in a multidimensional manner. The results and findings of the study underscore the urgency of addressing technological, economic, and societal challenges to successfully integrate autonomous buses into urban transportation systems for all stakeholders of urban transportation. In this context, a methodological framework that provides multidimensional and in-depth analysis is required for autonomous (driverless) buses to be the most basic and critical component of urban transportation systems in the future. The review significantly contributes to the relevant industry and literature, laying the groundwork for further research and innovative applications in this field.

7. Conclusions and future research trends

This paper presents a comprehensive methodological framework for

(a)



(b)

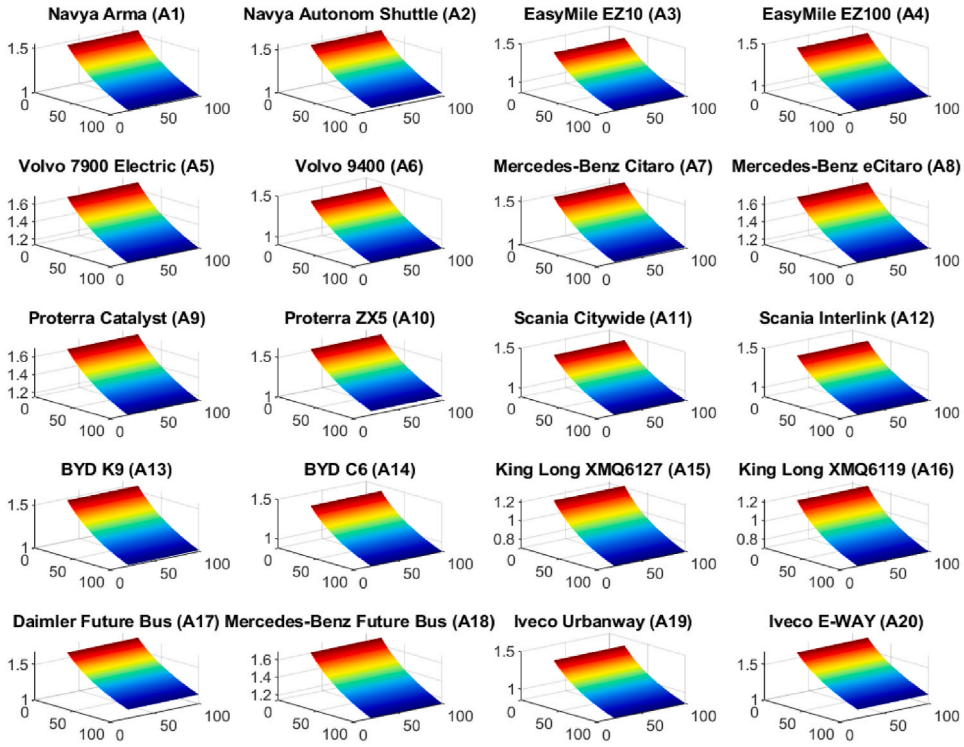


Fig. 15. Parameters γ and σ: Changing the integrated DOBI functions of alternatives (a) Alternative 9, and (b) All alternatives.

Table 14
Descriptive profile of the random weight sets.

	min	Max	mean	0.25	median	0.75	std	skewness	kurtosis
SO1	0.0001	0.0747	0.0302	0.0144	0.0299	0.0454	0.0179	0.0516	1.8778
SO2	0.0001	0.0738	0.0304	0.0158	0.0299	0.0450	0.0175	0.0597	1.9955
SO3	0.0001	0.0904	0.0301	0.0141	0.0295	0.0458	0.0185	0.1689	1.9801
SO4	0.0000	0.0747	0.0302	0.0149	0.0296	0.0445	0.0177	0.1472	2.0500
SO5	0.0000	0.0725	0.0310	0.0168	0.0315	0.0452	0.0172	−0.0353	1.9448
SO6	0.0000	0.0751	0.0298	0.0143	0.0299	0.0446	0.0176	0.0652	1.9122
SO7	0.0000	0.0746	0.0303	0.0148	0.0307	0.0444	0.0177	0.0726	1.9578
EC1	0.0000	0.0715	0.0300	0.0157	0.0297	0.0444	0.0169	0.0559	1.9123
EC2	0.0001	0.0714	0.0300	0.0146	0.0298	0.0449	0.0174	0.0394	1.9098
EC3	0.0001	0.0753	0.0302	0.0163	0.0299	0.0441	0.0174	0.0838	1.9829
EC4	0.0000	0.0753	0.0295	0.0138	0.0287	0.0446	0.0178	0.1527	1.9341
EC5	0.0000	0.0745	0.0309	0.0155	0.0316	0.0455	0.0179	0.0016	1.9582
EC6	0.0000	0.0803	0.0301	0.0147	0.0306	0.0451	0.0178	0.0492	1.9678
EC7	0.0001	0.0826	0.0304	0.0159	0.0307	0.0447	0.0172	0.0802	2.0587
EC8	0.0000	0.0741	0.0303	0.0155	0.0303	0.0445	0.0175	0.0636	1.9731
EN1	0.0000	0.0795	0.0302	0.0154	0.0302	0.0451	0.0176	0.0499	1.9855
EN2	0.0003	0.0757	0.0302	0.0147	0.0303	0.0452	0.0178	0.0806	1.9028
EN3	0.0002	0.0761	0.0305	0.0149	0.0309	0.0445	0.0179	0.0727	1.9598
EN4	0.0001	0.0722	0.0301	0.0154	0.0305	0.0446	0.0176	0.0886	1.9558
EN5	0.0000	0.0680	0.0306	0.0153	0.0304	0.0456	0.0176	0.0336	1.9318
EN6	0.0000	0.0753	0.0299	0.0145	0.0301	0.0444	0.0175	0.0698	1.9313
EN7	0.0000	0.0736	0.0306	0.0168	0.0302	0.0449	0.0172	0.0104	1.9608
EN8	0.0000	0.0765	0.0294	0.015	0.0289	0.0432	0.0172	0.1082	2.0053
EN9	0.0000	0.0778	0.0306	0.0167	0.0306	0.0447	0.0170	0.0784	2.0416
EN10	0.0001	0.0816	0.0302	0.0152	0.0304	0.0443	0.0178	0.1254	2.0734
TO1	0.0002	0.0772	0.0308	0.0163	0.0313	0.0449	0.0176	0.0493	2.0887
TO2	0.0003	0.0708	0.0315	0.0174	0.0312	0.0465	0.0176	−0.0085	1.9324
TO3	0.0000	0.0785	0.0311	0.0158	0.0312	0.0456	0.0174	0.0320	1.9309
TO4	0.0001	0.0728	0.0304	0.0151	0.0307	0.0453	0.0178	0.0440	1.9160
TO5	0.0000	0.0720	0.0300	0.0148	0.0300	0.0442	0.0176	0.0677	1.9156
TO6	0.0000	0.0732	0.0300	0.0151	0.0297	0.0449	0.0177	0.0925	1.9303
TO7	0.0001	0.0747	0.0302	0.0165	0.0304	0.0445	0.0174	0.0911	2.0234
TO8	0.0000	0.0744	0.0303	0.0144	0.0313	0.0457	0.0181	0.0058	1.8431

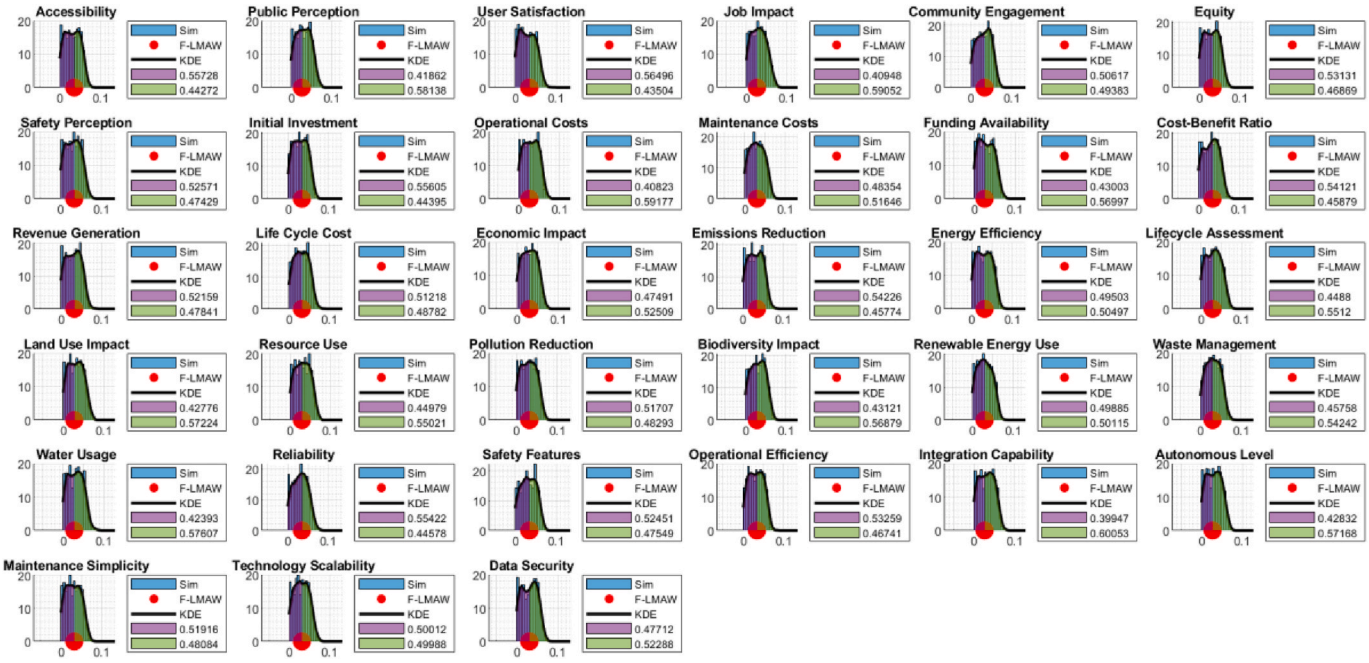


Fig. 16. Distribution of the random weights and fuzzy LMAW criteria weights.

analyzing the sustainability performance of driverless autonomous buses in urban transportation systems. The study evaluated the sustainability performances of 20 autonomous bus alternatives under 33 criteria, proposing an innovative decision-making procedure. The findings highlight the sustainable performance of the Proterra Catalyst (A9), Mercedes-Benz Future Bus (A18), and Mercedes-Benz eCitaro (A8) models. These results underscore the critical role of user-oriented, reliable, and accessible autonomous vehicle technologies in achieving sustainable urban transportation goals, providing valuable insights to the reader.

By eliminating the methodological evaluation deficiencies in the existing literature, this study's fuzzy cluster-based decision-making

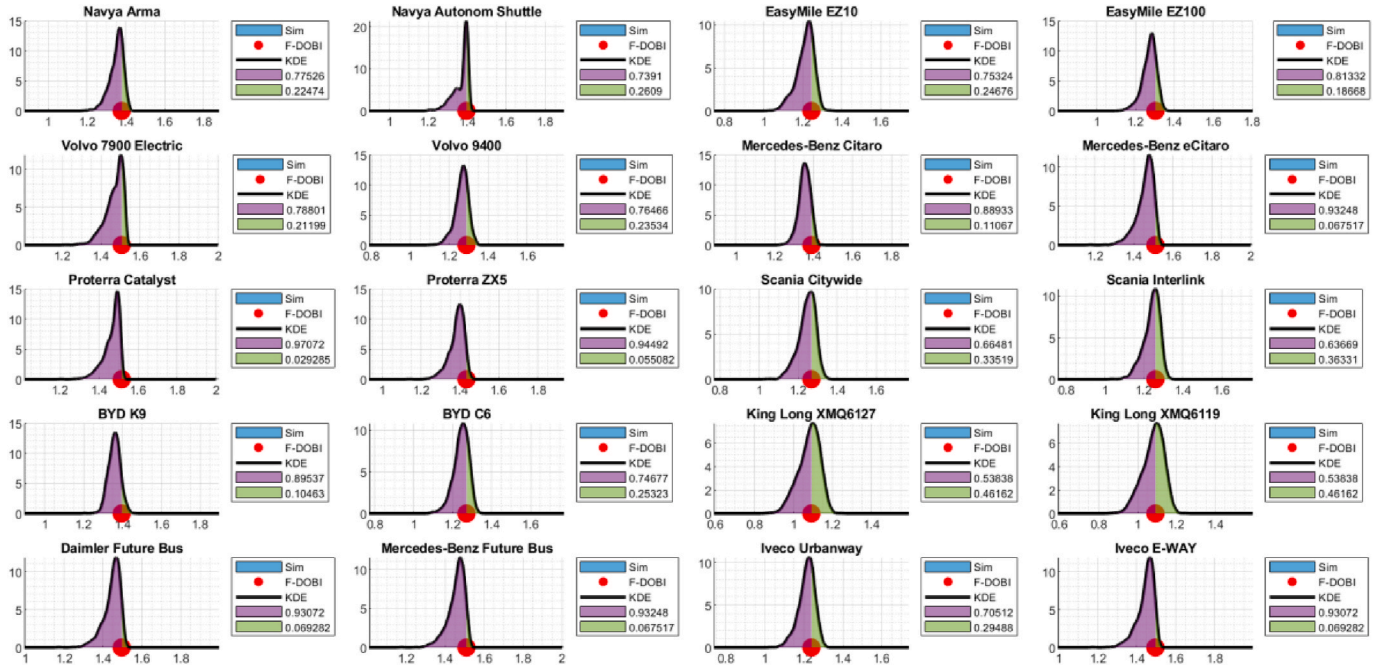


Fig. 17. Distribution of fuzzy DOBI scores under random weight sets.

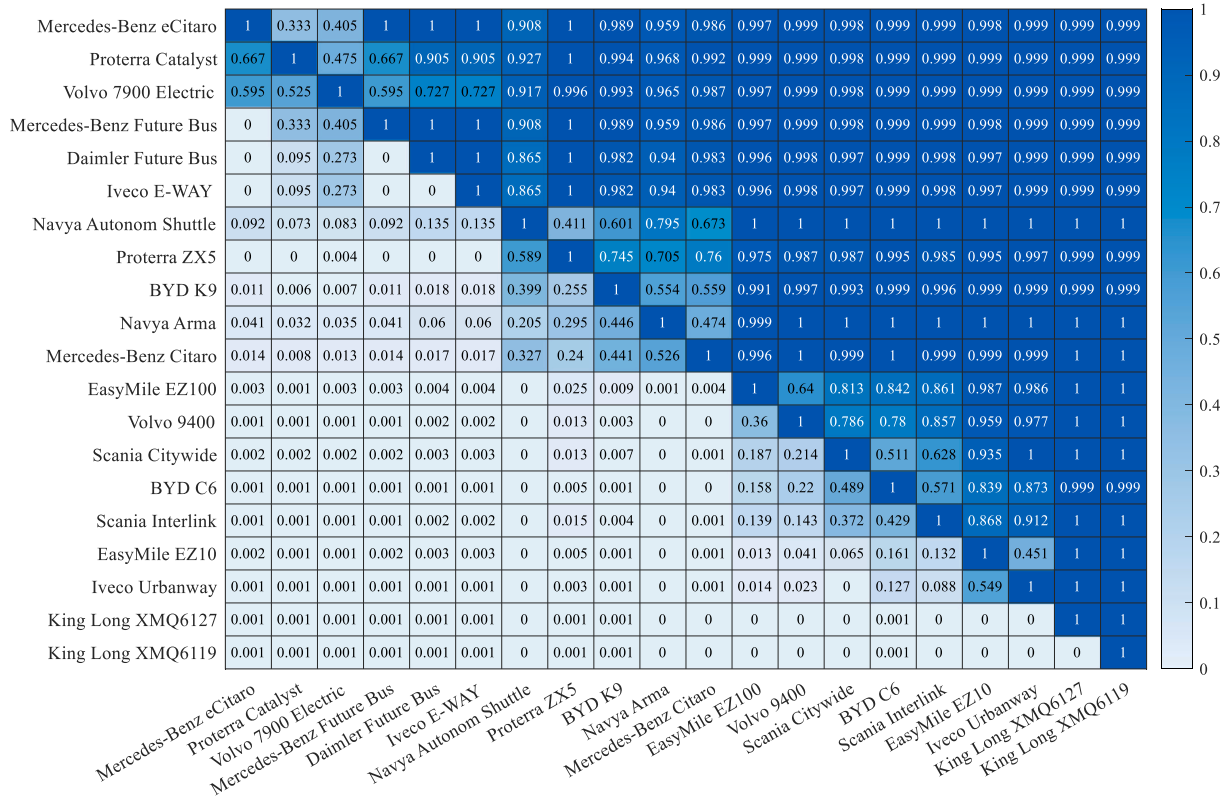


Fig. 18. How many times alternatives in a row is ranked higher than alternatives in a column.

procedure is proven by comprehensive robustness checks. The study's critical contributions to the literature and practical applications serve as a guiding guide for urban transportation authorities and policymakers, providing them with a clear path forward. The findings also underscore the potential for technological advancement and increased societal acceptance to facilitate faster and easier adoption of driverless autonomous buses, further supporting the reader in their decision-making process.

Although this study provides valuable inferences and insights and a comprehensive assessment, it is also subject to some limitations. First of all, the results obtained in the study are related to twenty alternatives and 33 sustainability criteria. The results may change if more advanced and dominant products are produced by new actors who may enter the autonomous vehicle market and more criteria are added. In addition,

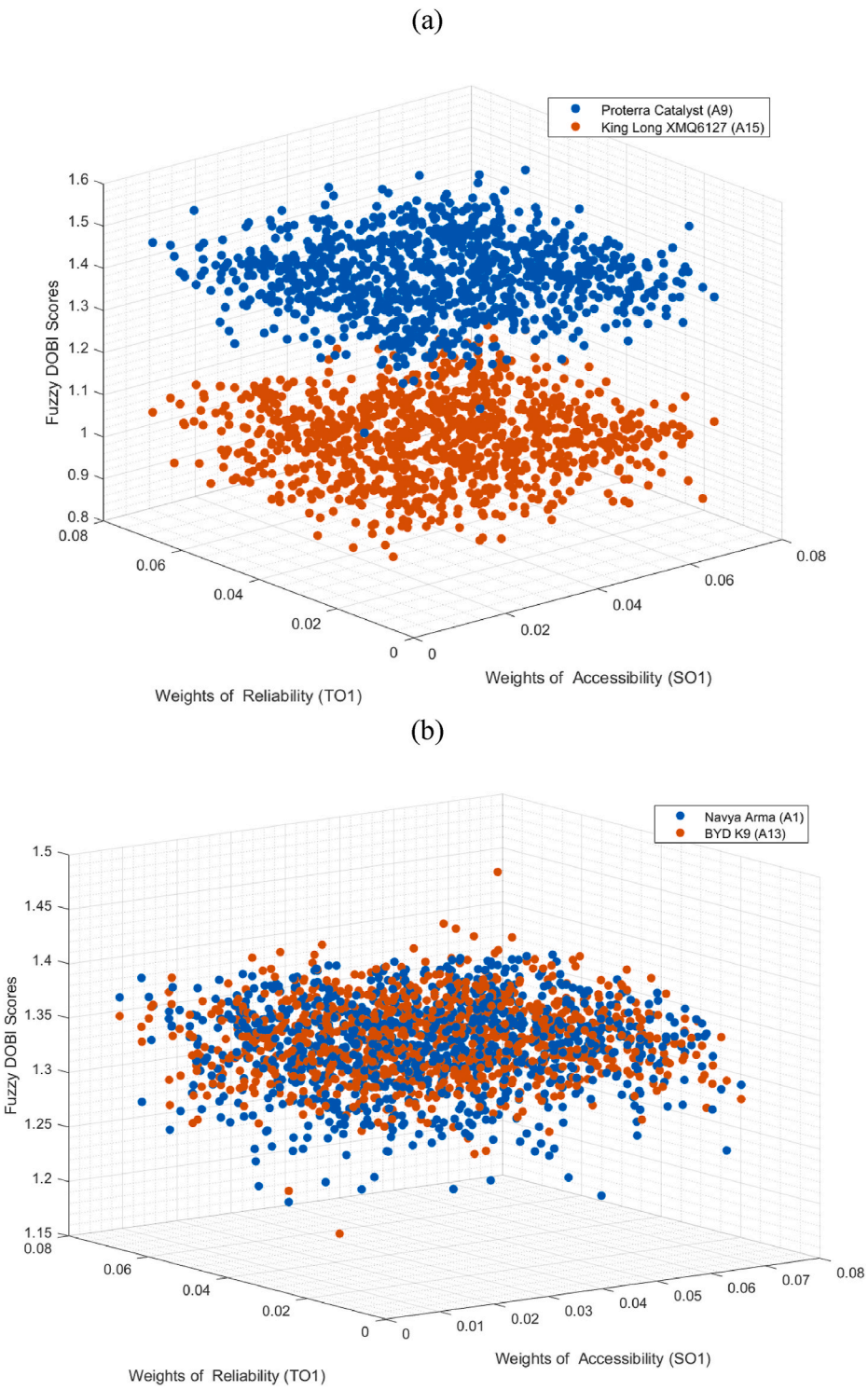


Fig. 19. Comparison of the alternative scores under different weight sets (a) Superiority of A9 over A15 (b) Situation of neither alternative A1 nor A13 being able to establish superiority over the other.

prototype vehicles were not included in the scope of the study because they have yet to be commercialized. However, their inclusion in future studies could significantly impact the results, as these vehicles may introduce new technological advancements and sustainability criteria. The number of autonomous bus alternatives may increase as these vehicles are completed and included in standard production processes.

Secondly, this study dealt with the sustainability performance of autonomous buses with a theoretical approach and did not include simulation-based or experimental applications in real traffic environments. The need for real-world testing is urgent, as it can significantly enhance the applicability of the proposed theoretical evaluation framework in real-world conditions.

To overcome these limitations in future studies, researchers can include a more up-to-date and expanded set of criteria and alternatives. This forward-looking approach can make the decision-making process more comprehensive and detailed, paving the way for further generalization of the results and advancing our understanding of autonomous vehicle sustainability.

Table 15
Ranking comparison with other techniques.

Alternatives	Alternative names	Fuzzy DOBI	Fuzzy MACONT	Fuzzy TOPSIS	Fuzzy MARCOS
A1	Navya Arma	11	11	11	14
A2	Navya Autonom Shuttle	8	8	10	18
A3	EasyMile EZ10	17	18	17	16
A4	EasyMile EZ100	12	12	12	11
A5	Volvo 7900 Electric	4	1	6	1
A6	Volvo 9400	13	13	13	12
A7	Citaro	10	10	8	9
A8	eCitaro	2	3	2	2
A9	Proterra Catalyst	1	2	1	6
A10	Proterra ZX5	7	7	7	7
A11	Scania Citywide	15	15	14	17
A12	Scania Interlink	16	16	15	15
A13	BYD K9	9	9	9	8
A14	BYD C6	14	14	16	10
A15	King Long XMQ6127	19	19	19	19
A16	King Long XMQ6119	20	20	20	20
A17	Daimler Future Bus	5	5	4	3
A18	Mercedes-Benz Future Bus	3	4	3	4
A19	Iveco Urbanway	18	17	18	13
A20	Iveco E-WAY	6	6	5	5

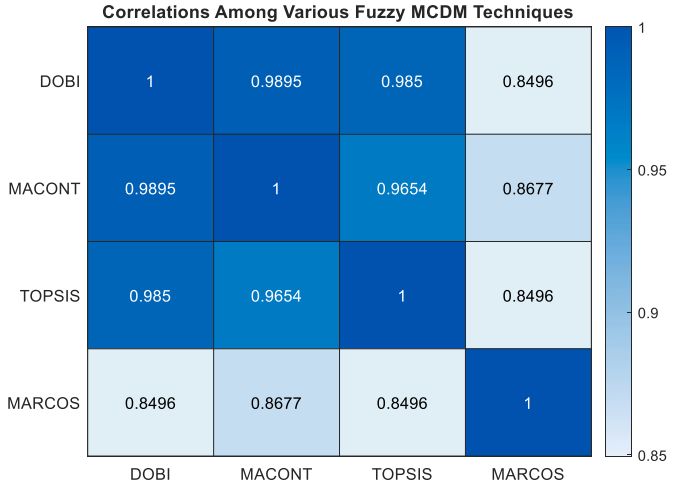


Fig. 20. Correlations among different fuzzy MCDM techniques.

Secondly, conducting experimental studies that test the performance of autonomous buses in actual traffic conditions or simulation environments can increase the results' accuracy and compatibility with real-life situations. Additionally, field studies that focus on user satisfaction and societal acceptance could make it possible to develop policy recommendations that could significantly influence the adoption of autonomous buses and their impact on society.

Finally, computer-based decision support systems can be created by developing the basic algorithm of the decision-making model proposed in this study in this form or by further developing it. These decision support systems can facilitate the implementation of the proposed decision-making model, making the model more user-friendly while ensuring that the results obtained are more consistent and reliable. Moreover, the use of dynamic modeling approaches to monitor changes in sustainability criteria over time holds great potential for a more reliable forecast of future requirements for urban transportation, instilling optimism for its future applications.

CRediT authorship contribution statement

Ömer Faruk Görçün: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Formal analysis, Conceptualization. **Mehmet Özçalıcı:** Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis, Data curation, Conceptualization. **Hasan Emin Gurler:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis, Data curation. **Dragan Pamucar:** Writing – review & editing, Writing – original draft, Software, Methodology, Data curation. **Vladimir Simic:** Writing – review & editing, Writing – original draft, Validation, Resources, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.engappai.2025.111034.

Data availability

Data will be made available on request.

References

Akbulut, K., Kici, A., Bezgin, N.O., 2017. Metrobus Isletmesi ile ilgili olarak meydana gelen kazaların ve etkilesimlerin irdelenmesi. Ulastirma Kongresi 219–226.
Akram, M., Zahid, K., Kahraman, C., 2024. A new ELECTRE-based decision-making framework with spherical fuzzy information for the implementation of autonomous vehicles project in Istanbul. Knowl. Base Syst. 283. https://doi.org/10.1016/j.knosys.2023.111207.
Ariza-Álvarez, A., López-Lambas, M.E., Soria-Lara, J.A., 2023. Analysing the acceptance of autonomous buses in real-life traffic environments: pilot project with tourists in Malaga, Spain. Transport. Plann. Technol. 46 (1). https://doi.org/10.1080/03081060.2022.2159407.
Babaei, P., Riahinia, N., Ebadati E, O.M., Azimi, A., 2024. Autonomous vehicles' object detection architectures ranking based on multi-criteria decision-making techniques. Int. J. Inf. Technol. 16 (4). https://doi.org/10.1007/s41870-023-01517-y.
Badi, I., Bouraima, M.B., Stević, Željko, Oloketuyi, E.A., Makinde, O.O., 2024. Optimizing vendor-managed inventory in multi-tier distribution systems. Spectrum of Operational Research 1 (1), 33–43. https://doi.org/10.31181/sor1120243.
Badi, I., Demir, G., Bouraima, M.B., Maraka, A.K., 2025. Vendor managed inventory in practice: efficient scheduling and delivery optimization. Spectrum of Decision Making and Applications 2 (1), 157–165. https://doi.org/10.31181/sdmap21202515.
Basuri, T., Gazi, K.H., Bhaduri, P., Das, S.G., Mondal, S.P., 2025. Decision-analytics-based sustainable location problem - neutrosophic CRITIC-COPRAS assessment model. Management Science Advances 2 (1), 19–58. https://doi.org/10.31181/msa2120257.
Biswas, A., Gazi, K.H., Sankar, P.M., Ghosh, A., 2025. A decision-making framework for sustainable highway restaurant site selection: AHP-TOPSIS approach based on the fuzzy numbers. Spectrum of Operational Research 2 (1), 1–26. https://doi.org/10.31181/sor2120256.
Biswas, A., Reon, M.A.O., Das, P., Tasneem, Z., Muyeen, S.M., Das, S.K., Badal, F.R., Sarker, S.K., Hassan, M.M., Abhi, S.H., Islam, M.R., Ali, M.F., Ahamed, M.H., Islam, M.M., 2022. State-of-the-Art review on recent advancements on lateral control of autonomous vehicles. In: IEEE Access, vol. 10. https://doi.org/10.1109/ACCESS.2022.3217213.
Božanić, D., Pamučar, D., Milić, A., Marinković, D., Komazec, N., 2022. Modification of the logarithm methodology of additive weights (LMAW) by a triangular fuzzy number and its application in multi-criteria decision making. Axioms 11 (3). https://doi.org/10.3390/axioms11030089.
Chougule, A., Chamola, V., Sam, A., Yu, F.R., Sikdar, B., 2024. A comprehensive review on limitations of autonomous driving and its impact on accidents and collisions. IEEE

Open Journal of Vehicular Technology 5. https://doi.org/10.1109/OJVT.2023.3335180.
Correa, G., Muñoz, P., Falaguerra, T., Rodriguez, C.R., 2017. Performance comparison of conventional, hybrid, hydrogen and electric urban buses using well to wheel analysis. Energy 141. https://doi.org/10.1016/j.energy.2017.09.066.
Cui, Y., Xu, S., Jiang, Y., Zhang, Z., 2023. Transportation infrastructure development and subjective socio-economic status: a quasi-experiment in establishing rural bus stations in China. Research in Transportation Business and Management 49. https://doi.org/10.1016/j.rtbm.2023.101006.
Deveci, M., Gokasar, I., Pamucar, D., Chen, Y., Coffman, D.M., 2023. Sustainable E-scooter parking operation in urban areas using fuzzy Dombi based RAFSI model. Sustain. Cities Soc. 91. https://doi.org/10.1016/j.scs.2023.104426.
Deveci, M., Pamucar, D., Gokasar, I., 2021. Fuzzy Power Heronian function based CoCoSo method for the advantage prioritization of autonomous vehicles in real-time traffic management. Sustain. Cities Soc. 69. https://doi.org/10.1016/j.scs.2021.102846.
Deveci, M., Pamucar, D., Gokasar, I., Koppen, M., Gupta, B.B., 2022b. Personal mobility in metaverse with autonomous vehicles using Q-rung orthopair fuzzy sets based OPA-RAFSI model. IEEE Trans. Intell. Transport. Syst. https://doi.org/10.1109/TITS.2022.3186294.
Deveci, M., Pamucar, D., Gokasar, I., Pedrycz, W., Wen, X., 2022a. Autonomous bus operation alternatives in urban areas using fuzzy dombi-bonferroni operator based decision making model. IEEE Trans. Intell. Transport. Syst. 1–10. https://doi.org/10.1109/TITS.2022.3202111.
Dwipayana, A.D., Pradana, A., Sasue, R.R.O., Adiyasa, I.W., 2024. Autonomous bus transit simulation in the Bali tourism development corporation (BTDC) area. Jurnal Teknologi Transportasi Dan Logistik 5 (1), 31–46. https://doi.org/10.52920/jttl.v5i1.258.
Eti, S., Dinçer, H., Yüksel, S., Gökalp, Y., 2025. A new fuzzy decision-making model for enhancing electric vehicle charging infrastructure. Spectrum of Decision Making and Applications 2 (1), 94–99. https://doi.org/10.31181/sdmap21202513.
Fidanoglu, A., Gokasar, I., Deveci, M., 2023. Integrating shared autonomous vehicles in Last-Mile public transportation. Sustain. Energy Technol. Assessments 57. https://doi.org/10.1016/j.seta.2023.103214.
Gokasar, I., Arisoy, A.A., Deveci, M., Mardani, A., 2024. Evaluation of a new real-life traffic management using a limited number of connected autonomous vehicles. IEEE Trans. Intell. Transport. Syst. 25 (3). https://doi.org/10.1109/TITS.2023.3285913.
Gokasar, I., Simic, V., Deveci, M., Senapati, T., 2023. Alternative prioritization of freeway incident management using autonomous vehicles in mixed traffic using a type-2 neutrosophic number based decision support system. Eng. Appl. Artif. Intell. 123. https://doi.org/10.1016/j.engappai.2023.106183.
Hasan, U., Whyte, A., AlJassmi, H., 2022. A microsimulation modelling approach to quantify environmental footprint of autonomous buses. Sustainability 14 (23), 15657.
Hasan, U., Whyte, A., AlJassmi, H., 2024. A multi-criteria decision-making framework for sustainable road transport systems: integrating stakeholder-cost-environment-energy for a highway case study in United Arab Emirates. J. Clean. Prod. 450. https://doi.org/10.1016/j.jclepro.2024.141831.
Huang, C.Y., Lin, Y.C., Lu, Y.F., Wang, L.C., Kuo, Y.T., Cheng, J.C., Sun, Y., 2020. An ANFIS based derivations of inference rules for users' adoptions of autonomous vehicles. 2020 International Conference on Fuzzy Theory and its Applications, IFUZZY 2020. https://doi.org/10.1109/iFUZZY50310.2020.9297811.
I.E.T.T., 2024. I.E.T.T Operating Report.
Imran, R., Ullah, K., 2025. Circular intuitionistic fuzzy edas approach: a new paradigm for decision-making in the automotive industry sector. Spectrum of Engineering and Management Sciences 3 (1), 76–92. https://doi.org/10.31181/sems31202537i.
Karami, A., Karami, H., Mehdizadeh, M., 2023. A world of fully autonomous mobility options: on long-distance travel mode choice. Technol. Forecast. Soc. Change 194. https://doi.org/10.1016/j.techfore.2023.122702.
Korkmaz, H., Fidanoglu, A., Ozcelik, S., Okumus, A., 2021. User acceptance of autonomous public transport systems: extended UTAUT2 model. Journal of Public Transportation 23 (1). https://doi.org/10.5038/2375-0901.23.1.5.
Kuo, Y.H., Leung, J.M.Y., Yan, Y., 2023. Public transport for smart cities: recent innovations and future challenges. Eur. J. Oper. Res. 306 (3). https://doi.org/10.1016/j.ejor.2022.06.057.
Launonen, P., Salonen, A.O., Liimatainen, H., 2021. Icy roads and urban environments. Passenger experiences in autonomous vehicles in Finland. Transport. Res. F Traffic Psychol. Behav. 80. https://doi.org/10.1016/j.trf.2021.03.015.
Lee, J., Mao, R., Pervez, A., 2024. Perceived risk of crime on driverless public bus and ride-pooling services in China. Travel Behaviour and Society 35. https://doi.org/10.1016/j.tbs.2023.100730.
Li, H., Jin, Z., Cui, H., Tu, H., 2023. An exploration of the preferences and mode choice behavior between autonomous demand-responsive transit and traditional buses. International Journal of Transportation Science and Technology. https://doi.org/10.1016/j.ijtst.2023.07.004.
López-Lambas, M.E., Alonso, A., 2019. The driverless bus: an analysis of public perceptions and acceptability. Sustainability 11 (18). https://doi.org/10.3390/su11184986.
Mandal, U., Seikh, M.R., 2023. Interval-valued spherical fuzzy MABAC method based on Dombi aggregation operators with unknown attribute weights to select plastic waste management process. Appl. Soft Comput. 145, 110516.
Mishra, A.R., Rani, P., Pamucar, D., Alshamrani, A.M., Alrasheedi, A.F., 2024. Intuitionistic fuzzy MACONT method for logistics 4.0 based circular economy interested regions assessment in the agri-food sector. Facta Univ. – Ser. Mech. Eng. https://doi.org/10.22190/FUME241001052M.
Moslem, S., Solieman, H., Oubahman, L., Duleba, S., Senapati, T., Pilla, F., 2023. Assessing public transport supply quality: a comparative analysis of analytical network process and analytical hierarchy process. Journal of Soft Computing and Decision Analytics 1 (1), 124–138. https://doi.org/10.31181/jscda11202311.
Mouratidis, K., Cobeña Serrano, V., 2021. Autonomous buses: intentions to use, passenger experiences, and suggestions for improvement. Transport. Res. F Traffic Psychol. Behav. 76. https://doi.org/10.1016/j.trf.2020.12.007.
Nanaki, E.A., Koroneos, C.J., Roset, J., Susca, T., Christensen, T.H., De Gregorio Hurtado, S., Rybka, A., Kopitovic, J., Heidrich, O., López-Jiménez, P.A., 2017. Environmental assessment of 9 European public bus transportation systems. Sustain. Cities Soc. 28. https://doi.org/10.1016/j.scs.2016.08.025.
Nastjuk, I., Herrenkind, B., Marrone, M., Brendel, A.B., Kolbe, L.M., 2020. What drives the acceptance of autonomous driving? An investigation of acceptance factors from an end-user's perspective. Technol. Forecast. Soc. Change 161. https://doi.org/10.1016/j.techfore.2020.120319.
Owczarzak, Ł., Zak, J., 2015. Design of passenger public transportation solutions based on autonomous vehicles and their multiple criteria comparison with traditional forms of passenger transportation. Transp. Res. Procedia 10. https://doi.org/10.1016/j.trpro.2015.09.001.
Papadima, G., Genitsaris, E., Karagiotas, I., Naniopoulos, A., Nalmpantis, D., 2020. Investigation of acceptance of driverless buses in the city of Trikala and optimization of the service using Conjoint Analysis. Util. Policy 62. https://doi.org/10.1016/j.jup.2019.100994.
Połom, M., Wiśniewski, P., 2021. Assessment of the emission of pollutants from public transport based on the example of diesel buses and trolleybuses in gdynia and sopot. Int. J. Environ. Res. Publ. Health 18 (16). https://doi.org/10.3390/ijerph18168379.
Rahnamay Bonab, S., Jafarzadeh Ghoushchi, S., Deveci, M., Haseli, G., 2023. Logistic autonomous vehicles assessment using decision support model under spherical fuzzy set integrated Choquet Integral approach. Expert Syst. Appl. 214. https://doi.org/10.1016/j.eswa.2022.119205.
Rosell, J., Allen, J., 2020. Test-riding the driverless bus: determinants of satisfaction and reuse intention in eight test-track locations. Transport. Res. Pol. Pract. 140. https://doi.org/10.1016/j.tra.2020.08.013.
Salonen, A.O., Haavisto, N., 2019. Towards autonomous transportation. Passengers' experiences, perceptions and feelings in a driverless shuttle bus in Finland. Sustainability 11 (3). https://doi.org/10.3390/su11030588.
Seikh, M.R., Chatterjee, P., 2025. Sustainable strategies for electric vehicle adoption: a confidence level-based interval-valued spherical fuzzy MEREC-VIKOR approach. Inf. Sci. 699, 121814.
Seikh, M.R., Mandal, U., 2023. Interval-valued Fermatean fuzzy Dombi aggregation operators and SWARA based PROMETHEE II method to bio-medical waste management. Expert Syst. Appl. 226, 120082.
Sitinjak, C., Tahir, Z., Toriman, M.E., Lyndon, N., Simic, V., Musselwhite, C., Simanullang, W.F., Hamzah, F.M., 2023. Assessing public acceptance of autonomous vehicles for smart and sustainable public transportation in urban areas: a case study of jakarta, Indonesia. Sustainability 15, 7445.
Tanveer, M., Kashmiri, F.A., Naeem, H., Yan, H., Qi, X., Abbas Rizvi, S.M., Wang, T., Lu, H., 2020. An assessment of age and gender characteristics of mixed traffic with autonomous and manual vehicles: a cellular automata approach. Sustainability 12 (7). https://doi.org/10.3390/su12072922.
Taş, M.A., Çakır, E., 2021. Ranking triangular and trapezoidal fuzzy MARCOS methods on road risk. In: 2021 IEEE 6th International Forum on Research and Technology for Society and Industry (RTSI), pp. 370–375. IEEE.
Tütüncü, K.A., Gül, N.N., Bölükbaş, U., Güneri, A.F., 2023. Integer linear programming approach for the personnel shuttles routing problem in yıldız campus in Istanbul. Journal of Soft Computing and Decision Analytics 1 (1), 303–316. https://doi.org/10.31181/jscda11202326.
Wicki, M., 2021. How do familiarity and fatal accidents affect acceptance of self-driving vehicles? Transport. Res. F Traffic Psychol. Behav. 83. https://doi.org/10.1016/j.trf.2021.11.004.
Yang, W., Pang, Y., 2020. New q-rung orthopair fuzzy Bonferroni mean Dombi operators and their application in multiple attribute decision making. IEEE Access 8, 50587–50610. https://doi.org/10.1109/ACCESS.2020.2979780.
Yang, W., Pang, Y., 2022. T-spherical fuzzy Bonferroni mean operators and their application in multiple attribute decision making. Mathematics 10 (6). https://doi.org/10.3390/math10060988.
Younis Al-Zibaree, H.K., Konur, M., 2023. Fuzzy analytic hierarchal process for sustainable public transport system. Journal of Operations Intelligence 1 (1), 1–10. https://doi.org/10.31181/jopi1120234.
Yu, L., Shao, X., Yan, X., 2017. Autonomous overtaking decision making of driverless bus based on deep Q-learning method. IEEE International Conference on Robotics and Biomimetics, ROBIO 2017, 2018-January. https://doi.org/10.1109/ROBIO.2017.8324756, 2017.
Zakeri, S., Konstantas, D., Sorooshian, S., Chatterjee, P., 2024. A novel ML-MCDM-based decision support system for evaluating autonomous vehicle integration scenarios in Geneva's public transportation. Artif. Intell. Rev. 57 (310).
Zarbakhshnia, N., Ma, Z., 2024. Critical success factors for the adoption of AVs in sustainable urban transportation. Transp. Policy 156, 62–76. https://doi.org/10.1016/j.tranpol.2024.07.00.